

PYTHON DEVELOPER'S HANDBOOK

A Practical Reference Manual



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Using This Handbook

This handbook is organized as a practical reference manual rather than a traditional textbook. Each section is designed to provide a concise explanation of a topic, followed by examples, best practices, and lessons learned through hands-on development.

The handbook follows a consistent format to make information easy to locate and understand.

Section Format

Most topics follow this structure:

- **Purpose** – Introduces the topic and its role in Python development.
- **Why It Matters** – Explains why the concept is important.
- **Quick Reference** – Summarizes commonly used syntax, functions, or commands.
- **Syntax** – Demonstrates the basic language structure.
- **Core Concepts** – Explains the primary ideas behind the topic.
- **Examples** – Provides practical demonstrations.
- **Common Beginner Errors** – Highlights frequent mistakes and how to avoid them.
- **Best Practices** – Recommends proven development techniques.
- **Developer's Note** – Shares practical lessons learned through real projects.
- **Related Topics** – References other sections of the handbook.
- **Key Takeaways** – Summarizes the most important concepts.

Not every section contains every heading, but the overall structure remains consistent throughout the handbook.

Code Examples

Python code examples are identified by the following marker:

<-> Python

This marker indicates that the text immediately following is a Python code example or command intended for demonstration or reference.

Diagrams

Workflow diagrams and mental models are included throughout the handbook to illustrate concepts that are more easily understood visually than through text alone.

Cross-References

Many topics include a **Related Topics** section that points to other chapters where a concept is explored in greater detail. These references are intended to reinforce how individual concepts work together throughout Python development.

A Living Reference

This handbook is intended to grow alongside my experience as a Python developer. As new projects are completed and new technologies are learned, additional topics, examples, and best practices will be added.

This handbook was developed while learning Python through hands-on projects. It reflects practical experience, lessons learned, and workflows refined through real development rather than serving as a replacement for official documentation or textbooks.

Some reference material has been adapted from publicly available documentation, textbooks, and other educational resources. Such material is included solely for educational and reference purposes. Original sources retain their respective copyrights, and no claim of authorship or ownership is made for that content.

Project Charter

Project Name:

Python Developer's Handbook

Purpose

To create a single, living reference manual that documents my understanding of Python programming, software development, and the projects I build.

The handbook is intended to be a practical reference written from experience rather than a replacement for textbooks or official documentation.

Objective

Capture the concepts, workflows, lessons learned, and best practices developed through hands-on programming.

Guiding Principles

- Write from understanding.
 - Explain concepts in our own words.
 - Use real projects as examples.
 - Build one section at a time.
 - Update continuously as skills grow.
-

Success Criteria

Years from now, I should be able to open this handbook, find an answer in minutes, and quickly return to productive development.

This handbook will grow alongside my experience as a developer and will never be considered complete.

1 PYTHON DEVELOPMENT ENVIRONMENT

1.1 Installing Python

Overview

Python is a high-level, interpreted programming language known for its readability, versatility, and ease of learning. It is used to develop desktop applications, web applications, automation scripts, data analysis tools, artificial intelligence systems, cybersecurity utilities, and scientific software.

Before Python programs can be developed or executed, the Python interpreter must be installed on the computer.

Installation

Python should always be downloaded from the official Python website:

<https://www.python.org>

Download the latest stable release that matches your operating system and run the installer.

During installation, enable the option:

Add Python to PATH

This allows Python to be executed from any Command Prompt or PowerShell window without requiring the full installation path.

After enabling this option, complete the installation using the default settings unless your project requires a custom configuration.

Why Install from Python.org?

Although Python is available from several sources, including the Microsoft Store, installing directly from the official Python website provides several advantages:

- Access to the latest stable Python release.
- Complete installation of the Python interpreter and standard library.
- Reliable compatibility with development tools such as Visual Studio Code and Python virtual environments.
- Greater control over future updates and installations.

Things to Remember

- Download Python from the official Python website.
- Enable Add Python to PATH during installation.
- Install the latest stable release unless a project requires a specific version.
- Verify the installation before creating your first project.

Developer's Note

Throughout this handbook, Python is installed directly from python.org and all examples assume the standard installation using the python command from PowerShell.

1.2 Verifying the Installation

Overview

After Python has been installed, it is important to verify that the installation completed successfully. Verification confirms that Python is available from the command line and is ready to execute programs.

This simple check also confirms that the **Add Python to PATH** option was configured correctly during installation.

Verify the Python Installation

Open **PowerShell** or **Command Prompt** and enter:

```
python --version
```

If Python is installed correctly, a response similar to the following will be displayed:

Python 3.13.0

The version number shown will depend on the version of Python currently installed.

Start the Python Interpreter

To start the interactive Python interpreter, enter:

```
python
```

If successful, the Python prompt appears:

```
Python 3.13.0 (tags/...)
Type "help", "copyright", "credits", or "license" for more information.
```

```
>>>
```

The `>>>` prompt indicates that Python is running and ready to accept commands.

Try a simple calculation:

```
>>> 2 + 3
5
```

Or display text:

```
>>> print("Hello, World!")
Hello, World!
```

Exit the Python Interpreter

To leave the interactive interpreter, enter:

```
exit()
```

or press:

Ctrl + Z

followed by **Enter** (Windows).

The PowerShell or Command Prompt prompt will return.

Common Problems

Problem	Possible Cause
'python' is not recognized...	Python is not installed or Add Python to PATH was not selected during installation.
Incorrect Python version	Multiple versions of Python may be installed on the system.
Command works in one terminal but not another	Restart the terminal after installation so it recognizes the updated PATH.

Things to Remember

Verify the installation before creating your first project.

- `python --version` displays the installed Python version.
- `python` starts the interactive interpreter.
- `exit()` closes the interpreter and returns to the command prompt.
- If the `python` command is not recognized, verify the installation and the system PATH settings.

Developer's Note

Throughout this handbook, Python commands are executed from **PowerShell**. If you install a newer version of Python in the future, rerun `python --version` to confirm that your terminal is using the expected version before starting a project.

1.3 Updating pip

Overview

pip is Python's package manager. It is used to install, update, and remove third-party Python packages such as Django, Plotly, NumPy, and Pandas.

Keeping pip up to date ensures compatibility with the latest packages and provides access to bug fixes, performance improvements, and new features.

Checking the Installed Version

To display the currently installed version of pip, open **PowerShell** and enter:

```
python -m pip --version
```

The command returns information similar to:

```
pip 26.1.2 from C:\Python313\Lib\site-packages\pip (python 3.13)
```

The exact version and installation path will vary depending on your system.

Updating pip

To upgrade pip to the latest available version, run:

```
python -m pip install --upgrade pip
```

If an update is available, pip downloads and installs the newer version automatically.

After the update completes, verify the installation by running:

```
python -m pip --version
```

Why Use python -m pip?

Although the command:

```
pip install
```

often works, the recommended practice is to use:

```
python -m pip
```

This ensures that the version of pip associated with the current Python interpreter is used. This is especially important when multiple versions of Python are installed on the same computer.

Things to Remember

- pip is Python's package manager.
 - Keep pip updated before installing project packages.
 - Use python -m pip instead of pip when possible.
 - Verify the installed version after updating.
 - Updating pip does not affect your existing Python projects.
-

Developer's Note

Throughout this handbook, package management is performed using python -m pip. This approach avoids ambiguity and ensures that packages are installed into the correct Python environment.

1.4 Creating a Project Folder

Overview

Before writing any code, create a dedicated folder for the project. Keeping each project in its own directory helps organize source code, documentation, virtual environments, and other project files.

A well-organized folder structure makes projects easier to maintain, share, and back up.

Create the Project Folder

Open **PowerShell** and navigate to the location where your Python projects will be stored.

For example:

```
cd F:\
```

Create a new project folder using the `mkdir` command:

```
mkdir Python_Study
```

Move into the new folder:

```
cd Python_Study
```

Create a folder for the project:

```
mkdir Learning_Log
```

Navigate into the project folder:

```
cd Learning_Log
```

The PowerShell prompt should now display the project directory, indicating that you are ready to create a virtual environment and begin development.

Why Create a Separate Project Folder?

Creating a dedicated folder for each project provides several benefits:

- Keeps project files organized.
 - Prevents unrelated files from becoming mixed together.
 - Makes projects easier to archive or share.
 - Simplifies Git repository management.
 - Provides a logical location for the project's virtual environment.
-

Things to Remember

- Create a separate folder for every Python project.
 - Give project folders meaningful names.
 - Navigate into the project folder before creating a virtual environment.
 - Keep all project-related files inside the project folder.
-

Developer's Note

Throughout this handbook, projects are stored inside a dedicated **Python_Study** directory. While the folder names used in the examples reflect my own development environment, any folder structure that is organized and consistent will work equally well.

1.5 Creating a Virtual Environment

Overview

A **virtual environment** is an isolated Python workspace created for a specific project. It allows each project to maintain its own versions of Python packages and dependencies without affecting the global Python installation or other projects.

Using a virtual environment is considered a Python best practice and should be done for virtually every project.

Why Use a Virtual Environment?

Virtual environments provide several important benefits:

- Isolate project dependencies from other projects.
- Prevent package version conflicts.
- Protect the system-wide Python installation.
- Make projects easier to share and reproduce.
- Ensure that everyone working on the project uses the same package versions.

For example, one project may require Django 5.x while another requires Django 6.x. Each project can have its own virtual environment without causing conflicts.

Creating a Virtual Environment

Navigate to the project's root directory in PowerShell, then run:

```
python -m venv ll_env
```

Where:

- `python` starts the Python interpreter.
- `-m` tells Python to run a module.
- `venv` is Python's built-in virtual environment module.
- `ll_env` is the name of the virtual environment folder.

The name `ll_env` is simply a convention used for the Learning Log project. The folder can be named anything, such as:

- `venv`
 - `.venv`
 - `project_env`
 - `my_environment`
-

1.6 Activating and Deactivating the Virtual Environment

Activating the Virtual Environment

Once the environment has been created, activate it before installing packages or running the project.

```
.\11_env\Scripts\Activate.ps1
```

When activated, the PowerShell prompt changes to display the environment name:

```
(11_env) PS F:\Python_Study\Python_Crash_Course\Django\learning_log>
```

This indicates that Python and pip commands now operate inside the virtual environment.

Deactivating the Virtual Environment

When finished working, leave the virtual environment by entering:

```
deactivate
```

The environment name disappears from the PowerShell prompt, indicating that the session has returned to the system's normal Python installation.

1.7 Installing Packages with pip

After activation, any packages installed with pip install are placed only inside this virtual environment.

Example:

```
pip install django
```

These packages are isolated from every other Python project on the computer.

Best Practices

- Create one virtual environment for each project.
- Activate the environment before running or modifying the project.
- Install packages only after the environment has been activated.
- Deactivate the environment when your work session is complete.
- Do not commit the virtual environment folder to Git. Instead, use a `.gitignore` file and track project dependencies with a `requirements.txt` file.

Developer's Note: Throughout the *Python Crash Course* projects, every Django project was developed inside its own virtual environment. This kept package installations isolated and prevented dependency conflicts between projects.

1.8 Managing Dependencies with requirements.txt

Overview

As a project grows, additional Python packages are often installed to support new features. A requirements.txt file records the project's package dependencies so the same development environment can be recreated on another computer.

Using a requirements.txt file is considered a Python best practice for projects that will be shared, deployed, or maintained over time.

Creating a requirements.txt File

After installing all required packages, create a dependency file by running:

```
python -m pip freeze > requirements.txt
```

This command records every package currently installed in the active virtual environment.

A typical requirements.txt file might look like:

```
Django==6.0.6  
asgiref==3.9.0  
sqlparse==0.5.3
```

Each entry records the package name and the exact version installed.

Installing Packages from requirements.txt

To recreate the same environment on another computer or after creating a new virtual environment, run:

```
python -m pip install -r requirements.txt
```

Python reads the file and installs every listed package automatically.

Why Use requirements.txt?

Using a dependency file provides several advantages:

- Documents every package required by the project.
- Makes projects easy to share with other developers.
- Ensures consistent package versions across different computers.
- Simplifies rebuilding a project's development environment.

Things to Remember

- Create a new requirements.txt file whenever project dependencies change.
- Keep the file under Git version control.
- Use `python -m pip freeze` to generate the file.
- Use `python -m pip install -r requirements.txt` to recreate the environment.

Developer's Note

Every Django project developed during *Python Crash Course* used a `requirements.txt` file to document package dependencies. Although small personal projects may not always require one, it quickly becomes an essential part of professional software development.

1.9 Common Virtual Environment Commands

Overview

During everyday development, a small number of commands are used repeatedly to manage virtual environments. Learning these commands makes setting up and maintaining Python projects much easier.

Frequently Used Commands

Command	Purpose
<code>python -m venv ll_env</code>	Create a new virtual environment.
<code>.\ll_env\Scripts\Activate.ps1</code>	Activate the virtual environment (PowerShell).
<code>deactivate</code>	Exit the virtual environment.
<code>python -m pip --version</code>	Display the installed version of pip.
<code>python -m pip install --upgrade pip</code>	Update pip to the latest version.
<code>python -m pip install package_name</code>	Install a package.
<code>python -m pip uninstall package_name</code>	Remove a package.
<code>python -m pip list</code>	Display all installed packages.
<code>python -m pip freeze</code>	Display installed packages in requirements.txt format.
<code>python -m pip install -r requirements.txt</code>	Install packages listed in a requirements.txt file.

Things to Remember

- Always activate the virtual environment before installing packages.
- Use `python -m pip` whenever possible.
- Keep package installations isolated within each project.
- Refer to this table whenever you need a quick command reference.

Developer's Note

Most day-to-day Python development relies on only a handful of commands. After working through several projects, these commands quickly become second nature.

1.10 Troubleshooting Virtual Environments

Overview

Most virtual environment problems are caused by an incorrect working directory, an inactive virtual environment, or an outdated package installation. Verifying a few key items usually resolves the issue quickly.

Common Problems and Solutions

Problem	Possible Cause	Solution
Virtual environment will not activate.	Incorrect folder or environment does not exist.	Verify the project folder and confirm the environment was created successfully.
<code>python</code> uses the wrong version.	Virtual environment is not active.	Activate the virtual environment before running Python commands.
Package cannot be imported.	Package has not been installed in the active environment.	Install the package using <code>python -m pip install package_name</code> .
<code>pip</code> installs packages globally.	Virtual environment is inactive.	Activate the virtual environment and install the package again.
<code>requirements.txt</code> is missing packages.	The file was created before the latest packages were installed.	Run <code>python -m pip freeze > requirements.txt</code> again.

Troubleshooting Checklist

Before assuming something is wrong, check the following:

- Am I in the correct project folder?
- Is the virtual environment active?
- Did I install the required package?
- Am I using the correct Python interpreter?
- Does `python --version` return the expected version?
- Does `python -m pip list` show the installed package?

Most problems can be solved by answering these questions before looking for more complex solutions.

Things to Remember

- Most issues are configuration problems rather than Python problems.
 - Activate the virtual environment before running project commands.
 - Verify your current directory before troubleshooting.
 - Read error messages carefully—they often identify the problem directly.
-

Developer's Note

Almost every virtual environment issue I encountered while working through *Python Crash Course* was caused by one of three things: I was in the wrong folder, I had forgotten to activate the virtual environment, or a required package had not yet been installed. Developing the habit of checking these first saved a great deal of time.

2 PYTHON LANGUAGE FUNDAMENTALS

2.1 Variables & Data Types

2.1.1 Variables

Purpose (Topic)

Variables are named locations in memory that store data while a program is running. They allow information to be saved, retrieved, and modified throughout the execution of a program.

Instead of repeatedly typing the same value throughout a program, a variable provides a meaningful name that can be referenced whenever the stored data is needed. Variables are one of the most fundamental concepts in programming and are used in virtually every Python application.

Concept Overview (Topic)

Think of a variable as a labeled storage container.

The **label** identifies the variable, while the **contents** represent the data stored inside it. The value stored in a variable can be replaced at any time during program execution.

For example:

- The variable `name` might store a person's name.
- The variable `age` might store an integer.
- The variable `temperature` might store a decimal value.

A program accesses the variable by its name rather than by the value itself.

Variable Naming Conventions (*Topic*)

Python variable names must follow specific rules:

- Must begin with a letter (A-Z or a-z) or an underscore (_).
- Cannot begin with a number.
- May contain letters, numbers, and underscores.
- Cannot contain spaces or special characters.
- Cannot use Python reserved keywords such as if, for, while, or class.
- Variable names are **case-sensitive**.

Examples:

✓ Valid

<-> Python

```
name = "Skip"
age = 70
project_name = "Learning Log"
_employee = "Alice"
```

✗ Invalid

<-> Python

```
2name = "Skip"
project-name = "Learning Log"
first name = "Skip"
class = "Python"
```

Creating Variables (*Topic*)

Variables are created automatically when a value is assigned using the assignment operator (=).

Python does not require variables to be declared before use.

Example:

<-> Python

```
name = "Skip"  
age = 70  
salary = 27.00
```

Python determines the appropriate data type based on the assigned value.

Assigning Values (*Topic*)

The assignment operator (=) stores a value in a variable.

<-> Python

```
project = "Python Developer Handbook"  
completed = False  
temperature = 98.6
```

A variable may be reassigned at any time.

<-> Python

```
count = 1  
count = 2  
count = 3
```

Each new assignment replaces the previous value.

Multiple Assignment (*Topic*)

Python allows multiple variables to be assigned in a single statement.

<-> Python

```
x, y, z = 10, 20, 30
```

The values are assigned in order from left to right.

Python also allows multiple variables to reference the same value.

<-> Python

```
width = height = 500
```

Examples (*Topic*)

Example 1

<-> Python

```
name = "Skip"
```

```
print(name)
```

Output

Skip

Example 2

<-> Python

```
age = 70
```

```
print(age)
```

Output

70

Example 3

<-> Python

```
city = "Houston"
```

```
print(f"{name} lives in {city}.")
```

Output

Skip lives in Houston.

Common Beginner Errors (*Topic*)

- Misspelling a variable name.
- Using spaces within a variable name.
- Attempting to use a variable before assigning a value.
- Accidentally overwriting an existing value.
- Confusing the assignment operator (=) with the equality operator (==).

Example:

<-> Python

```
if age = 70:
```

Incorrect.

Correct:

```
if age == 70:
```

Best Practices (*Topic*)

- Use descriptive variable names.
- Follow Python's snake_case naming convention.
- Keep names short but meaningful.
- Avoid single-letter names except in simple loops or mathematical expressions.
- Do not reuse variables for unrelated purposes.
- Treat variable names as documentation for your code.

Good examples:

<-> Python

customer_name
project_directory
total_cost
current_temperature

Poor examples:

<-> Python

x
abc
thing
temp1

Developer's Note (*Topic*)

One of the biggest improvements I noticed while working through *Python Crash Course* was that descriptive variable names made debugging much easier. Variables such as `topic`, `entry`, `highs`, `low_temps`, and `earthquake_data` immediately conveyed their purpose, making programs easier to understand months later. Investing a little extra effort in choosing meaningful names pays dividends as projects grow in size and complexity.

Key Takeaways (*Topic*)

- Variables store information used by a program.
- Variables are created when a value is assigned.
- Python automatically determines the variable's data type.
- Variable names should be descriptive and follow snake_case conventions.
- Variables can be updated or reassigned at any time.
- Clear variable names improve readability, debugging, and long-term maintenance.

2.1.2 Data Types

Purpose (Topic)

Every value stored in a Python variable has a data type. A data type defines what kind of information the variable contains and determines the operations that can be performed on it.

Understanding data types is essential because Python treats text, numbers, logical values, and collections differently. Choosing the appropriate data type helps create programs that are efficient, reliable, and easy to maintain.

Concept Overview (Topic)

A data type describes the nature of the information stored in memory.

When a value is assigned to a variable, Python automatically determines its data type based on the value being assigned.

For example:

- Text is stored as a **string** (`str`).
- Whole numbers are stored as **integers** (`int`).
- Decimal numbers are stored as **floating-point numbers** (`float`).
- Logical values are stored as **Booleans** (`bool`).

<-> Python

```
name = "Skip"           # String
age = 70                 # Integer
temperature = 98.6       # Float
retired = False          # Boolean
```

Understanding data types allows you to select the proper operations, avoid programming errors, and write code that behaves as expected.

Quick Reference *(Topic)*

Function / Type	Purpose	Example
<code>type()</code>	Display the data type of a variable or value.	<code>type(age)</code>
<code>str()</code>	Convert a value to a string.	<code>str(25)</code>
<code>int()</code>	Convert a value to an integer.	<code>int("25")</code>
<code>float()</code>	Convert a value to a floating-point number.	<code>float("3.14")</code>
<code>bool()</code>	Convert a value to a Boolean.	<code>bool(1)</code>

Built-in Data Types *(Topic)*

Python includes several built-in data types.

Data Type	Description	Example
<code>str</code>	Text	<code>"Python"</code>
<code>int</code>	Whole numbers	<code>25</code>
<code>float</code>	Decimal numbers	<code>98.6</code>
<code>bool</code>	True or False values	<code>True</code>
<code>list</code>	Ordered collection	<code>[1, 2, 3]</code>
<code>tuple</code>	Immutable ordered collection	<code>(1, 2, 3)</code>
<code>set</code>	Unordered collection of unique values	<code>{1, 2, 3}</code>
<code>dict</code>	Key-value pairs	<code>{"name": "Skip"}</code>
<code>NoneType</code>	Represents no value	<code>None</code>

Syntax (*Topic*)

Assigning different data types:

<-> Python

```
name = "Skip"  
age = 70  
height = 6.0  
retired = False
```

Checking the data type of a variable:

<-> Python

```
print(type(name))  
print(type(age))  
print(type(height))  
print(type(retired))
```

Core Concepts (*Topic*)

Strings (*str*)

Store text.

<-> Python

```
language = "Python"
```

Integers (*int*)

Store whole numbers.

<-> Python

```
age = 70
```

Floating-Point Numbers (float)

Store decimal values.

<-> Python

```
price = 19.95
```

Booleans (bool)

Represent logical values.

<-> Python

```
logged_in = True  
finished = False
```

Collections

Collections store multiple values in a single variable.

Examples include:

- Lists
- Tuples
- Dictionaries
- Sets

These collection types are covered in detail later in this chapter.

Mutable vs. Immutable (*Topic*)

Some Python objects can be modified after they are created, while others cannot.

Immutable Types

- Strings
- Integers
- Floats
- Tuples
- Booleans

Once created, their values cannot be changed.

Mutable Types

- Lists
- Dictionaries
- Sets

These objects can be modified without creating a new object.

Example:

<-> Python

```
colors = ["red", "blue"]  
colors.append("green")
```

The list now contains three values.

Examples (*Topic*)

Example 1

<-> Python

```
name = "Skip"
```

```
print(type(name))
```

Output

```
<class 'str'>
```

Example 2

<-> Python

```
age = 70
```

```
print(type(age))
```

Output

```
<class 'int'>
```

Example 3

<-> Python

```
numbers = [1, 2, 3]
```

```
print(type(numbers))
```

Output

```
<class 'list'>
```

Common Beginner Errors (*Topic*)

- Treating numbers as strings.
- Forgetting quotation marks around text.
- Performing mathematical operations on string values.
- Assuming all numbers are integers.
- Forgetting that input() always returns a string.

Incorrect:

<-> Python

```
age = "70"
```

```
print(age + 5)
```

Result:

`TypeError`

Correct:

<-> Python

```
age = int("70")
```

```
print(age + 5)
```

Output

`75`

Best Practices (*Topic*)

- Choose the appropriate data type for the information being stored.
 - Use descriptive variable names regardless of the data type.
 - Use `type()` when learning or debugging programs.
 - Convert data types only when necessary.
 - Keep data types consistent throughout your program.
-

Developer's Note (*Topic*)

While working through the **Python Crash Course** projects, I found that understanding data types became increasingly important as the projects grew in complexity. Reading CSV files, processing JSON data from APIs, creating visualizations with Plotly, and building Django applications all required converting data into the correct type before it could be processed. Many debugging sessions were resolved simply by verifying that a variable contained the expected data type.

Related Topics (*Topic*)

- **2.1 Variables** – Variables store values of specific data types.
 - **2.3 Type Conversion** – Converting values between data types.
 - **2.4 Strings** – Working with text data.
 - **2.6 Numbers** – Performing mathematical operations.
 - **2.8 Boolean Values** – Logical data used in program flow.
 - **Chapter 6 – Data Visualization** – Converting CSV data from strings into integers and floats.
 - **Chapter 7 – APIs** – Working with JSON data returned by web services.
 - **Chapter 8 – Django Web Development** – Defining model fields using specific data types.
-

Key Takeaways (*Topic*)

- Every value in Python has a data type.
- Python automatically determines a variable's data type when a value is assigned.
- The `type()` function identifies the data type of a variable or value.
- Some data types are mutable, while others are immutable.
- Selecting the appropriate data type improves readability, reliability, and program performance.
- Understanding data types is fundamental to writing effective Python applications.

2.1.3 Type Conversion

Purpose *(Topic)*

Type conversion is the process of changing a value from one data type to another. Since different data types support different operations, converting values allows Python to process data correctly and avoid type-related errors.

Type conversion is a common task in programming and is used whenever data is read from user input, files, databases, or APIs.

Why It Matters *(Topic)*

Python stores data according to its type. A number stored as text cannot be used directly in mathematical calculations, and a numeric value must often be converted to a string before it can be displayed in a message.

Real-world Python programs constantly perform type conversions. Whether reading temperatures from a CSV file, processing JSON data from an API, accepting keyboard input, or displaying formatted output, converting data to the correct type is an essential programming skill.

Quick Reference *(Topic)*

Function	Purpose	Example
<code>str()</code>	Convert a value to a string.	<code>str(25)</code>
<code>int()</code>	Convert a value to an integer.	<code>int("25")</code>
<code>float()</code>	Convert a value to a floating-point number.	<code>float("3.14")</code>
<code>bool()</code>	Convert a value to a Boolean.	<code>bool(1)</code>

Syntax (*Topic*)

Convert a string to an integer:

<-> Python

```
age = int("70")
```

Convert a string to a floating-point number:

<-> Python

```
temperature = float("98.6")
```

Convert a number to a string:

<-> Python

```
message = str(2026)
```

Convert a value to a Boolean:

<-> Python

```
active = bool(1)
```

Core Concepts (*Topic*)

Implicit Conversion

Implicit conversion occurs automatically when Python safely converts one data type to another during an operation.

Example:

<-> Python

```
result = 5 + 2.5
```

Output

```
7.5
```

Python automatically converts the integer `5` into the floating-point value `5.0` before performing the calculation.

Explicit Conversion

Explicit conversion occurs when the programmer intentionally converts one data type into another by using a conversion function.

Example:

<-> Python

```
age = "70"  
age = int(age)
```

The variable `age` is converted from a string to an integer.

Examples (*Topic*)

Example 1 – Converting User Input

<-> Python

```
age = input("Enter your age: ")  
age = int(age)  
  
print(age + 5)
```

Output

If 70 is entered:

75

Example 2 – Converting to a Float

<-> Python

```
price = "19.95"  
price = float(price)  
  
print(price)
```

Output

19.95

Example 3 – Converting a Number to Text

<-> Python

```
year = 2026
```

```
print("Current year: " + str(year))
```

Output

```
Current year: 2026
```

Example 4 – Boolean Conversion

<-> Python

```
print(bool(0))
```

```
print(bool(1))
```

```
print(bool(""))
```

```
print(bool("Python"))
```

Output

```
False
```

```
True
```

```
False
```

```
True
```

Common Beginner Errors (*Topic*)

- Forgetting that `input()` always returns a string.
- Attempting mathematical operations on string values.
- Converting text that does not contain a valid number.
- Assuming every value can be converted successfully.

Incorrect:

<-> Python

```
age = input("Age: ")
```

```
print(age + 5)
```

Result

`TypeError`

Correct:

<-> Python

```
age = int(input("Age: "))
```

```
print(age + 5)
```

Best Practices (*Topic*)

- Convert data as soon as it is read when appropriate.
 - Verify that input contains valid data before converting it.
 - Use explicit conversions when readability is important.
 - Avoid unnecessary conversions between data types.
 - Remember that `input()` always returns a string.
-

Developer's Note (*Topic*)

Type conversion became essential throughout the Python Crash Course projects. CSV files stored temperatures as strings that had to be converted into integers before plotting graphs. API responses frequently returned numbers as text within JSON objects, and user input in Django forms required validation before being processed. Understanding when and how to convert data types eliminated many runtime errors and made the programs far more reliable.

Related Topics (*Topic*)

- **2.1 Variables**
 - **2.2 Data Types**
 - **2.4 Strings**
 - **2.6 Numbers**
 - **2.8 Boolean Values**
 - **Chapter 6 – Data Visualization**
 - **Chapter 7 – APIs**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Type conversion changes a value from one data type to another.
- Python performs some conversions automatically, while others require explicit conversion.
- The most common conversion functions are `str()`, `int()`, `float()`, and `bool()`.
- User input always begins as a string and often requires conversion.
- Proper type conversion prevents many common programming errors.
- Understanding type conversion is essential when working with files, APIs, databases, and user input.

2.1.4 Strings

Purpose *(Topic)*

Strings are one of Python's most commonly used data types and are used to store and manipulate text. A string can contain letters, numbers, symbols, punctuation, and whitespace, making it suitable for everything from user names and file paths to messages, URLs, and data retrieved from APIs.

Nearly every Python program uses strings, making them a fundamental part of the language.

Why It Matters *(Topic)*

Most information that users interact with is text. Whether displaying messages on the screen, reading data from a file, accepting user input, or communicating with web services, Python programs constantly process strings.

Understanding how strings work allows developers to search, modify, compare, and format text efficiently while avoiding common programming errors.

Quick Reference *(Topic)*

Function / Method	Purpose	Example
<code>len()</code>	Returns the number of characters.	<code>len(name)</code>
<code>.upper()</code>	Converts text to uppercase.	<code>name.upper()</code>
<code>.lower()</code>	Converts text to lowercase.	<code>name.lower()</code>
<code>.title()</code>	Capitalizes each word.	<code>name.title()</code>
<code>.strip()</code>	Removes leading and trailing spaces.	<code>name.strip()</code>
<code>.replace()</code>	Replaces text.	<code>text.replace("old", "new")</code>
<code>.split()</code>	Splits text into a list.	<code>text.split()</code>
<code>.find()</code>	Finds the position of text.	<code>text.find("Python")</code>

Syntax (*Topic*)

Strings are enclosed in either single or double quotation marks.

<-> Python

```
name = "Skip"
```

```
language = 'Python'
```

Multi-line strings use triple quotation marks.

<-> Python

```
message = """Python is
```

```
easy to learn
```

```
and powerful."""
```

Core Concepts (*Topic*)

Creating Strings

Strings can contain letters, numbers, symbols, and spaces.

<-> Python

```
first_name = "Skip"
```

```
city = "Houston"
```

String Concatenation

Strings can be combined using the + operator.

<-> Python

```
first_name = "Skip"
```

```
last_name = "Voss"
```

```
full_name = first_name + " " + last_name
```

```
print(full_name)
```

Output

```
Skip Voss
```

Repeating Strings

The multiplication operator repeats a string.

<-> Python

```
print("=" * 20)
```

Output

```
=====
```

Accessing Individual Characters

Each character has an index beginning with zero.

<-> Python

```
language = "Python"
```

```
print(language[0])
```

```
print(language[3])
```

Output

```
P
```

```
h
```

String Slicing

A slice extracts part of a string.

<-> Python

```
language = "Python"
```

```
print(language[0:3])
```

```
print(language[3:])
```

Output

```
Pyt
```

```
hon
```

Common String Methods (*Topic*)

Changing Case

<-> Python

```
name = "skip voss"
```

```
print(name.title())
```

```
print(name.upper())
```

```
print(name.lower())
```

Output

```
Skip Voss
```

```
SKIP VOSS
```

```
skip voss
```

Removing Whitespace

<-> Python

```
name = "  Python  "
```

```
print(name.strip())
```

Output

```
Python
```

Replacing Text

<-> Python

```
sentence = "I like Java."
```

```
print(sentence.replace("Java", "Python"))
```

Output

```
I like Python.
```

Splitting Text

<-> Python

```
colors = "red,green,blue"
```

```
print(colors.split(","))
```

Output

```
['red', 'green', 'blue']
```

Escape Characters (*Topic*)

Escape characters allow special formatting within strings.

Escape Sequence	Description
\n	New line
\t	Horizontal tab
\\	Backslash
\"	Double quotation mark
\'	Single quotation mark

Example:

<-> Python

```
print("Python\nDeveloper")
```

```
print("Name:\tSkip")
```

Output

```
Python
```

```
Developer
```

```
Name:    Skip
```

Examples (*Topic*)

Example 1

<-> Python

```
language = "Python"
```

```
print(len(language))
```

Output

6

Example 2

<-> Python

```
message = "Hello World"
```

```
print(message.upper())
```

Output

HELLO WORLD

Example 3

<-> Python

```
website = "python.org"
```

```
print(website.find("org"))
```

Output

7

Common Beginner Errors (*Topic*)

- Forgetting quotation marks around text.
- Mixing single and double quotation marks incorrectly.
- Using + with numbers and strings without conversion.
- Accessing an index that does not exist.
- Forgetting that strings are immutable.

Incorrect:

<-> Python

```
age = 70
```

```
print("Age: " + age)
```

Correct:

<-> Python

```
age = 70
```

```
print("Age: " + str(age))
```

Best Practices (*Topic*)

- Use descriptive variable names.
 - Prefer f-strings for formatted output (covered in the next topic).
 - Use string methods instead of writing unnecessary code.
 - Strip whitespace from user input before processing.
 - Use meaningful text that improves program readability.
-

Developer's Note (*Topic*)

Strings appeared in every Python Crash Course project. They were used for file names, API URLs, chart titles, Git commit messages, Django templates, HTML content, and user input. Learning Python's built-in string methods greatly reduced the amount of code needed to manipulate text and made programs cleaner and easier to read.

Related Topics (*Topic*)

- **2.1 Variables**
 - **2.2 Data Types**
 - **2.3 Type Conversion**
 - **2.5 String Formatting (f-Strings)**
 - **2.6 Numbers**
 - **Chapter 7 – APIs**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Strings store textual information.
- Strings are created using single, double, or triple quotation marks.
- Python provides many built-in methods for manipulating strings.
- Strings support indexing and slicing.
- Strings are immutable, meaning their contents cannot be modified directly.
- Mastering string operations is essential because nearly every Python program processes text.

2.1.5 String Formatting (f-Strings)

Purpose (Topic)

String formatting combines text and variable values into a single, readable string. Python provides several methods for formatting strings, with **formatted string literals (f-strings)** being the preferred approach because they are simple, efficient, and easy to read.

String formatting is commonly used to display information to users, generate reports, create log entries, and build dynamic messages.

Why It Matters (Topic)

Programs rarely display fixed text. Instead, they display information that changes while the program runs, such as names, dates, calculations, temperatures, and user input.

String formatting allows variables and expressions to be inserted directly into text without manually converting values or concatenating multiple strings. Using f-strings produces cleaner, more maintainable code.

Quick Reference (Topic)

Syntax	Purpose	Example
<code>f""</code>	Create an f-string.	<code>f"Hello {name}"</code>
<code>{variable}</code>	Insert a variable.	<code>{age}</code>
<code>{expression}</code>	Evaluate an expression.	<code>{price * quantity}</code>
<code>{value:.2f}</code>	Format a float to two decimal places.	<code>{cost:.2f}</code>
<code>{value:,}</code>	Add thousands separators.	<code>{population:,}</code>

Syntax (Topic)

Creating a basic f-string:

```
<-> Python
```

```
name = "Skip"
```

```
print(f"Welcome, {name}!")
```

Output

```
Welcome, Skip!
```

Core Concepts (*Topic*)

Inserting Variables

Variables are placed inside curly braces {}.

<-> Python

```
first_name = "Skip"
```

```
last_name = "Voss"
```

```
print(f"{first_name} {last_name}")
```

Output

```
Skip Voss
```

Inserting Expressions

Python evaluates expressions inside the braces before displaying the result.

<-> Python

```
price = 20
```

```
quantity = 3
```

```
print(f"Total: ${price * quantity}")
```

Output

```
Total: $60
```

Formatting Decimal Values

Floating-point numbers can be displayed with a specific number of decimal places.

<-> Python

```
pi = 3.14159265
```

```
print(f"{pi:.2f}")
```

Output

```
3.14
```

Formatting Large Numbers

Thousands separators improve readability.

<-> Python

```
population = 8234567
```

```
print(f"{population:,}")
```

Output

```
8,234,567
```

Combining Text and Variables

Multiple variables can be displayed within a single string.

<-> Python

```
city = "Houston"
```

```
temperature = 98
```

```
print(f"The temperature in {city} is {temperature}°F.")
```

Output

```
The temperature in Houston is 98°F.
```

Comparison of Formatting Methods (*Topic*)

Method	Example	Recommended
Concatenation	"Age: " + str(age)	No
<code>format()</code>	"Age: {}".format(age)	Occasionally
f-String	f"Age: {age}"	Yes

Python continues to support older formatting methods, but f-strings are generally preferred because they are more readable and concise.

Examples (*Topic*)

Example 1

<-> Python

```
name = "Skip"
```

```
print(f"Hello, {name}!")
```

Output

```
Hello, Skip!
```

Example 2

<-> Python

```
length = 12
```

```
width = 8
```

```
print(f"Area = {length * width}")
```

Output

```
Area = 96
```

Example 3

<-> Python

```
balance = 1250.5
```

```
print(f"Current Balance: ${balance:.2f}")
```

Output

```
Current Balance: $1250.50
```

Example 4

<-> Python

```
first = "Python"
```

```
second = "Developer"
```

```
print(f"{first} {second}")
```

Output

```
Python Developer
```

Common Beginner Errors (*Topic*)

- Forgetting to place the f before the opening quotation mark.
- Omitting the curly braces around variables.
- Using quotation marks inside braces incorrectly.
- Attempting to concatenate numbers without conversion when not using f-strings.

Incorrect:

<-> Python

```
name = "Skip"
```

```
print("Hello {name}")
```

Output

```
Hello {name}
```

Correct:

<-> Python

```
name = "Skip"
```

```
print(f"Hello {name}")
```

Output

```
Hello Skip
```

Best Practices (*Topic*)

- Prefer f-strings for new Python code.
 - Use descriptive variable names.
 - Keep formatting expressions simple and readable.
 - Format numbers for readability when displaying financial or statistical data.
 - Avoid unnecessary string concatenation.
-

Developer's Note (*Topic*)

By the middle of the Python Crash Course projects, nearly every program had switched to using f-strings. They simplified printing debugging information, displaying chart titles, creating Git commit messages, showing game statistics in Alien Invasion, and presenting data in Django templates. Once comfortable with f-strings, they quickly became the preferred method for producing readable output.

Related Topics (*Topic*)

- **2.1 Variables**
 - **2.2 Data Types**
 - **2.3 Type Conversion**
 - **2.4 Strings**
 - **2.6 Numbers**
 - **Chapter 6 – Data Visualization**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- f-strings provide a simple and readable way to combine text and variable values.
- Variables and expressions are enclosed in curly braces `{}`.
- f-strings can format numbers, calculations, and large values.
- They are generally preferred over concatenation and the older `format()` method.
- Using f-strings produces cleaner, easier-to-maintain Python code.

2.1.6 Numbers

Purpose *(Topic)*

Numbers are used to represent quantitative values and perform mathematical calculations. Python supports several numeric data types that allow developers to work with whole numbers, decimal values, complex numbers, and a wide variety of mathematical operations.

Numbers are fundamental to programming and are used in calculations, data analysis, scientific computing, financial applications, games, and automation.

Why It Matters *(Topic)*

Nearly every Python program performs some form of mathematical operation. Whether calculating scores in a game, graphing weather data, processing financial information, or analyzing API data, understanding Python's numeric types and operators is essential for producing accurate results.

Selecting the correct numeric type improves precision, readability, and program performance.

Quick Reference *(Topic)*

Operator / Function	Purpose	Example
<code>+</code>	Addition	<code>a + b</code>
<code>-</code>	Subtraction	<code>a - b</code>
<code>*</code>	Multiplication	<code>a * b</code>
<code>/</code>	Division	<code>a / b</code>
<code>//</code>	Floor Division	<code>a // b</code>
<code>%</code>	Modulus (Remainder)	<code>a % b</code>
<code>**</code>	Exponentiation	<code>a ** b</code>
<code>abs()</code>	Absolute Value	<code>abs(-5)</code>
<code>round()</code>	Round a Number	<code>round(3.14159, 2)</code>

Syntax (*Topic*)

Assigning numeric values:

<-> Python

```
age = 70
```

```
temperature = 98.6
```

```
price = 19.95
```

Performing calculations:

<-> Python

```
total = 25 + 15
```

```
average = (90 + 85 + 95) / 3
```

Core Concepts (*Topic*)

Integers (int)

Integers represent whole numbers.

<-> Python

```
employees = 12
```

```
year = 2026
```

Examples include positive numbers, negative numbers, and zero.

Floating-Point Numbers (float)

Floats represent numbers containing decimal values.

<-> Python

```
pi = 3.14159
```

```
price = 19.95
```

```
temperature = 98.6
```

Complex Numbers (complex)

Python also supports complex numbers used in advanced mathematics and engineering.

<-> Python

```
value = 3 + 4j
```

Although important in scientific computing, complex numbers are used less frequently in general application development.

Arithmetic Operators (Topic)

Operator	Description	Example	Result
+	Addition	5 + 3	8
-	Subtraction	5 - 3	2
*	Multiplication	5 * 3	15
/	Division	5 / 2	2.5
//	Floor Division	5 // 2	2
%	Modulus	5 % 2	1
**	Exponentiation	5 ** 2	25

Order of Operations (Topic)

Python follows the standard mathematical order of operations.

1. Parentheses ()
2. Exponents **
3. Multiplication, Division, Floor Division, Modulus
4. Addition and Subtraction

Example:

<-> Python

```
result = (5 + 3) * 2  
print(result)
```

Output

16

Useful Numeric Functions (*Topic*)

Absolute Value

<-> Python

```
print(abs(-42))
```

Output

42

Rounding Numbers

<-> Python

```
pi = 3.14159265
```

```
print(round(pi, 2))
```

Output

3.14

Finding Maximum and Minimum Values

<-> Python

```
numbers = [12, 5, 18, 3]
```

```
print(max(numbers))
```

```
print(min(numbers))
```

Output

18

3

Examples (*Topic*)

Example 1

<-> Python

```
length = 15
```

```
width = 8
```

```
area = length * width
```

```
print(area)
```

Output

```
120
```

Example 2

<-> Python

```
price = 49.95
```

```
tax = 0.0825
```

```
total = price + (price * tax)
```

```
print(round(total, 2))
```

Output

```
54.07
```

Example 3

<-> Python

```
score = 87
```

```
print(score % 10)
```

Output

```
7
```

Example 4

<-> Python

```
number = 5
```

```
print(number ** 3)
```

Output

```
125
```

Common Beginner Errors (*Topic*)

- Confusing `/` with `//`.
- Forgetting operator precedence.
- Dividing by zero.
- Expecting floating-point calculations to always be exact.
- Mixing numeric values and strings without conversion.

Incorrect:

<-> Python

```
print("Age: " + 70)
```

Result

```
TypeError
```

Correct:

<-> Python

```
age = 70
```

```
print(f"Age: {age}")
```

Best Practices (*Topic*)

- Use integers for whole numbers whenever appropriate.
 - Use floating-point numbers for decimal values.
 - Use parentheses to improve readability in complex calculations.
 - Round values when presenting financial or statistical information.
 - Prefer f-strings when displaying numeric results.
-

Developer's Note (*Topic*)

Numbers appeared throughout the Python Crash Course projects. They were used to calculate game scores in Alien Invasion, plot temperature and earthquake data, compute chart scales, process API results, and perform calculations within Django applications. Understanding arithmetic operators and numeric functions made it much easier to interpret data and produce accurate visualizations.

Related Topics (*Topic*)

- **2.2 Data Types**
 - **2.3 Type Conversion**
 - **2.4 Strings**
 - **2.5 String Formatting (f-Strings)**
 - **2.7 Operators**
 - **Chapter 6 – Data Visualization**
 - **Chapter 7 – APIs**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Python supports integers, floating-point numbers, and complex numbers.
- Arithmetic operators perform mathematical calculations.
- Python follows the standard order of operations.
- Built-in functions such as `abs()`, `round()`, `max()`, and `min()` simplify common calculations.
- Choosing the correct numeric type improves program accuracy and readability.
- Numeric operations are used throughout Python programming, from simple scripts to scientific computing and web applications.

2.1.7 Operators

Purpose (*Topic*)

Operators are special symbols that instruct Python to perform specific operations on variables and values. They are used to perform mathematical calculations, compare values, assign data, and evaluate logical expressions.

Operators are one of the fundamental building blocks of programming and appear in nearly every Python program.

Why It Matters (*Topic*)

Every decision, calculation, and comparison in a program relies on operators. Whether determining if a user has entered the correct password, calculating the area of a rectangle, updating a game score, or deciding which code path to execute, operators make those actions possible.

Understanding the different categories of operators is essential for writing accurate, efficient, and readable Python code.

Quick Reference (*Topic*)

Operator	Purpose	Example
+	Addition	$a + b$
-	Subtraction	$a - b$
*	Multiplication	$a * b$
/	Division	a / b
//	Floor Division	$a // b$
%	Modulus	$a \% b$
**	Exponentiation	$a ** b$
=	Assignment	$x = 5$
==	Equal To	$x == 5$
!=	Not Equal To	$x != 5$
>	Greater Than	$x > 5$
<	Less Than	$x < 5$
>=	Greater Than or Equal To	$x >= 5$
<=	Less Than or Equal To	$x <= 5$
and	Logical AND	$a \text{ and } b$
or	Logical OR	$a \text{ or } b$
not	Logical NOT	$\text{not } a$

Syntax (Topic)

Arithmetic operation:

<-> Python

```
result = 15 + 10
```

Comparison operation:

<-> Python

```
age >= 18
```

Logical operation:

<-> Python

```
age >= 18 and citizen == True
```

Core Concepts (Topic)

Arithmetic Operators

Arithmetic operators perform mathematical calculations.

Operator	Description	Example	Result
+	Addition	5 + 2	7
-	Subtraction	5 - 2	3
*	Multiplication	5 * 2	10
/	Division	5 / 2	2.5
//	Floor Division	5 // 2	2
%	Modulus	5 % 2	1
**	Exponentiation	5 ** 2	25

Assignment Operators

Assignment operators store or update values.

Operator	Description	Example
=	Assign value	x = 5
+=	Add and assign	x += 2
-=	Subtract and assign	x -= 2
*=	Multiply and assign	x *= 2
/=	Divide and assign	x /= 2

Example:

<-> Python

```
score = 100
```

```
score += 25
```

```
print(score)
```

Output

```
125
```

Comparison Operators

Comparison operators compare two values and always return either True or False.

Operator	Description
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Example:

<-> Python

```
age = 70
```

```
print(age >= 65)
```

Output

```
True
```

Logical Operators

Logical operators combine multiple conditions.

Operator	Description
<code>and</code>	Both conditions must be true.
<code>or</code>	At least one condition must be true.
<code>not</code>	Reverses a Boolean value.

Example:

<-> Python

```
age = 70
```

```
member = True
```

```
print(age >= 65 and member)
```

Output

```
True
```

Identity Operators

Identity operators determine whether two variables reference the same object in memory.

Operator	Description
<code>is</code>	Objects are identical
<code>is not</code>	Objects are different

Example:

<-> Python

```
a = None
```

```
print(a is None)
```

Output

```
True
```

Membership Operators

Membership operators determine whether a value exists within a sequence or collection.

Operator	Description
<code>in</code>	Value exists in the sequence
<code>not in</code>	Value does not exist in the sequence

Example:

<-> Python

```
colors = ["red", "green", "blue"]
```

```
print("green" in colors)
```

Output

```
True
```

Examples (Topic)

Example 1 – Arithmetic

<-> Python

```
x = 10
```

```
y = 4
```

```
print(x + y)
```

```
print(x * y)
```

Output

```
14
```

```
40
```

Example 2 – Comparison

<-> Python

```
temperature = 85
```

```
print(temperature > 90)
```

Output

```
False
```

Example 3 – Logical

<-> Python

```
logged_in = True
```

```
admin = False
```

```
print(logged_in and admin)
```

Output

```
False
```

Example 4 – Membership

<-> Python

```
language = "Python"
```

```
print("P" in language)
```

Output

```
True
```

Operator Precedence (*Topic*)

When an expression contains multiple operators, Python follows a predefined order of evaluation.

The general order is:

1. Parentheses `()`
2. Exponentiation `**`
3. Multiplication, Division, Floor Division, Modulus
4. Addition and Subtraction
5. Comparison Operators
6. Logical Operators (`not`, `and`, `or`)

When in doubt, use parentheses to make expressions easier to read.

Example:

<-> Python

```
result = (5 + 3) * 2
```

```
print(result)
```

Output

```
16
```

Common Beginner Errors (*Topic*)

- Confusing the assignment operator (=) with the equality operator (==).
- Forgetting operator precedence.
- Using is instead of == when comparing values.
- Dividing by zero.
- Forgetting that comparison operators return Boolean values.

Incorrect:

<-> Python

```
if age = 70:
```

Correct:

<-> Python

```
if age == 70:
```

Best Practices (*Topic*)

- Use parentheses to improve readability.
 - Keep expressions simple whenever possible.
 - Use == when comparing values.
 - Reserve is for comparisons with objects such as [None](#).
 - Break long logical expressions into smaller, easier-to-read statements.
-

Developer's Note (*Topic*)

Operators were used throughout every project in the Python Crash Course. Arithmetic operators calculated game scores and graph values, comparison operators controlled program flow, logical operators validated user input, and membership operators searched collections of data. Learning the purpose of each operator made the code easier to read and greatly simplified debugging.

Related Topics (*Topic*)

- **2.2 Data Types**
- **2.3 Type Conversion**
- **2.6 Numbers**
- **2.8 Boolean Values**
- **2.9 Conditional Statements**
- **2.10 Loops**
- **2.11 Defining Functions**
- **Chapter 8 – Django Web Development**

Key Takeaways (*Topic*)

- Operators perform calculations, comparisons, assignments, and logical evaluations.
- Python provides arithmetic, assignment, comparison, logical, identity, and membership operators.
- Comparison operators always return a Boolean value.
- Logical operators combine multiple conditions to make decisions.
- Understanding operator precedence helps prevent unexpected results.
- Operators are used throughout every Python program and form the basis of calculations and decision-making.

2.2 Program Flow

2.2.1 Boolean Values

Purpose (Topic)

Boolean values represent one of two possible states: **True** or **False**. They are used to make decisions within a program and control the flow of execution based on whether a condition is met.

Booleans form the foundation of conditional statements, loops, user authentication, data validation, and countless other programming tasks.

Why It Matters (Topic)

Programs constantly evaluate conditions before taking action. Questions such as:

- Is the user logged in?
- Is the password correct?
- Has the file been found?
- Is the temperature above freezing?
- Has the game ended?

all produce either **True** or **False**.

Without Boolean values, programs would be unable to make decisions or respond intelligently to changing conditions.

Quick Reference *(Topic)*

Function / Operator	Purpose	Example
<code>True</code>	Boolean true value	<code>logged_in = True</code>
<code>False</code>	Boolean false value	<code>finished = False</code>
<code>bool()</code>	Convert a value to Boolean	<code>bool(1)</code>
<code>==</code>	Equality comparison	<code>age == 70</code>
<code>!=</code>	Not equal	<code>age != 70</code>
<code>></code>	Greater than	<code>score > 90</code>
<code><</code>	Less than	<code>score < 90</code>
<code>>=</code>	Greater than or equal	<code>age >= 18</code>
<code><=</code>	Less than or equal	<code>age <= 65</code>

Syntax *(Topic)*

Assigning Boolean values:

<-> Python

```
logged_in = True
```

```
finished = False
```

Using a Boolean expression:

<-> Python

```
age = 70
```

```
print(age >= 65)
```

Output

```
True
```

Core Concepts (*Topic*)

The Boolean Data Type

Python has a built-in Boolean data type named bool.

A Boolean variable can contain only one of two values:

- True
- False

Example:

<-> Python

```
available = True  
completed = False
```

Comparison Expressions

Most Boolean values are produced by comparing two values.

<-> Python

```
age = 70  
  
print(age > 65)  
print(age == 70)  
print(age < 18)
```

Output

```
True  
True  
False
```

Boolean Variables

Instead of repeatedly evaluating the same condition, a Boolean result can be stored in a variable.

<-> Python

```
age = 70

is_senior = age >= 65

print(is_senior)
```

Output

True

This makes programs easier to read and maintain.

Truthy and Falsy Values

Python considers some values to be **False** even though they are not the Boolean value False.

Examples of falsy values include:

- False
- None
- 0
- 0.0
- "" (empty string)
- [] (empty list)
- {} (empty dictionary)
- () (empty tuple)
- set() (empty set)

Everything else is generally considered **truthy**.

Example:

<-> Python

```
print(bool(""))  
print(bool("Python"))  
print(bool(0))  
print(bool(42))
```

Output

```
False  
True  
False  
True
```

Examples (*Topic*)

Example 1 – Simple Comparison

<-> Python

```
temperature = 85  
  
print(temperature > 90)
```

Output

```
False
```

Example 2 – Boolean Variable

<-> Python

```
logged_in = True  
  
print(logged_in)
```

Output

```
True
```

Example 3 – Comparing Strings

<-> Python

```
username = "Skip"
```

```
print(username == "Skip")
```

Output

```
True
```

Example 4 – Using bool()

<-> Python

```
print(bool([]))
```

```
print(bool([1, 2, 3]))
```

Output

```
False
```

```
True
```

Common Beginner Errors (*Topic*)

- Confusing = (assignment) with == (comparison).
- Assuming every non-zero value is automatically True without understanding truthy and falsy values.
- Comparing strings with incorrect capitalization.
- Forgetting that Boolean values are case-sensitive (True, False).

Incorrect:

<-> Python

```
if logged_in = True:
```

Correct:

<-> Python

```
if logged_in == True:
```

Or, even better:

<-> Python

```
if logged_in:
```

Best Practices (*Topic*)

- Give Boolean variables meaningful names.
- Use names that read naturally in conditions.
- Prefer `if logged_in:` instead of `if logged_in == True:`.
- Understand which values Python considers truthy and falsy.
- Keep Boolean expressions simple and easy to understand.

Good examples:

<-> Python

`is_logged_in`
`has_permission`
`is_complete`
`file_exists`

Poor examples:

<-> Python

`flag`
`x`
`status1`

Developer's Note (*Topic*)

Boolean values became increasingly important throughout the Python Crash Course projects. They controlled game states in Alien Invasion, determined whether users were authenticated in Django, validated input, checked for API responses, and controlled loops and conditional logic. Choosing descriptive Boolean variable names made the code much easier to read because conditions could almost be read as plain English.

Related Topics (*Topic*)

- **2.2 Data Types**
 - **2.3 Type Conversion**
 - **2.7 Operators**
 - **2.9 Conditional Statements**
 - **2.10 Loops**
 - **2.11 Defining Functions**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Boolean values can only be **True** or **False**.
- Boolean expressions evaluate conditions and control program flow.
- Comparison operators return Boolean values.
- Python recognizes truthy and falsy values when evaluating conditions.
- Meaningful Boolean variable names improve code readability.
- Boolean logic is the foundation of decision-making in every Python program.

2.2.2 Conditional Statements

Purpose (Topic)

Conditional statements allow a program to make decisions by executing different blocks of code based on whether a condition evaluates to **True** or **False**.

Conditional logic enables programs to respond intelligently to user input, process data, validate information, and control the flow of execution.

Why It Matters (Topic)

Without conditional statements, every program would execute the same instructions regardless of circumstances. Conditional logic allows a program to react dynamically by asking questions such as:

- Is the user logged in?
- Has the correct password been entered?
- Is the temperature above freezing?
- Does the file exist?
- Has the game ended?

These decisions allow software to behave differently depending on the current situation.

Quick Reference (Topic)

Statement	Purpose	Example
<code>if</code>	Execute code when a condition is true.	<code>if age >= 18:</code>
<code>elif</code>	Test another condition if the first is false.	<code>elif score >= 80:</code>
<code>else</code>	Execute code when all previous conditions are false.	<code>else:</code>
<code>and</code>	Both conditions must be true.	<code>age > 18 and member</code>
<code>or</code>	Either condition may be true.	<code>admin or manager</code>
<code>not</code>	Reverse a Boolean value.	<code>not logged_in</code>

Syntax (*Topic*)

Basic if statement:

<-> Python

```
age = 70
```

```
if age >= 65:  
    print("Senior Citizen")
```

Output

```
Senior Citizen
```

Core Concepts (*Topic*)

The if Statement

The if statement executes a block of code only when its condition evaluates to True.

<-> Python

```
temperature = 95
```

```
if temperature > 90:  
    print("It's a hot day.")
```

Output

```
It's a hot day.
```

The if...else Statement

When the condition is false, the else block executes.

<-> Python

```
age = 16

if age >= 18:
    print("Adult")
else:
    print("Minor")
```

Output

Minor

The if...elif...else Statement

Use elif when testing multiple conditions.

<-> Python

```
score = 87

if score >= 90:
    print("Grade A")
elif score >= 80:
    print("Grade B")
elif score >= 70:
    print("Grade C")
else:
    print("Needs Improvement")
```

Output

Grade B

Nested Conditional Statements

One conditional statement may appear inside another.

<-> Python

```
logged_in = True
```

```
admin = False
```

```
if logged_in:
```

```
    if admin:
```

```
        print("Administrator Access")
```

```
    else:
```

```
        print("Standard User")
```

Output

Standard User

Nested conditionals should be used carefully to avoid making code difficult to read.

Logical Operators in Conditions

Multiple conditions can be combined using logical operators.

<-> Python

```
age = 70
```

```
member = True
```

```
if age >= 65 and member:
```

```
    print("Senior Member Discount")
```

Output

Senior Member Discount

Examples (*Topic*)

Example 1 – Checking a Password

<-> Python

```
password = "Python123"

if password == "Python123":
    print("Access Granted")
else:
    print("Access Denied")
```

Output

Access Granted

Example 2 – Positive or Negative Number

<-> Python

```
number = -15

if number >= 0:
    print("Positive")
else:
    print("Negative")
```

Output

Negative

Example 3 – Comparing Strings

<-> Python

```
language = "Python"
```

```
if language == "Python":  
    print("Welcome Developer!")
```

Output

```
Welcome Developer!
```

Example 4 – Multiple Conditions

<-> Python

```
age = 70
```

```
veteran = True
```

```
if age >= 65 and veteran:  
    print("Eligible for both discounts.")
```

Output

```
Eligible for both discounts.
```

Indentation (*Topic*)

Python uses indentation to define code blocks.

Every statement belonging to an `if`, `elif`, or `else` block must be indented consistently.

Correct:

<-> Python

```
if True:
    print("Correct")
```

Incorrect:

<-> Python

```
if True:
print("Incorrect")
```

Improper indentation results in an `IndentationError`.

Common Beginner Errors (*Topic*)

- Forgetting the colon (`:`) after an `if`, `elif`, or `else` statement.
- Incorrect indentation.
- Using `=` instead of `==`.
- Writing multiple independent `if` statements when `elif` is more appropriate.
- Creating deeply nested conditionals that are difficult to read.

Incorrect:

<-> Python

```
if age = 70:
    print("Senior")
```

Correct:

<-> Python

```
if age == 70:
    print("Senior")
```

Best Practices (*Topic*)

- Keep conditional expressions simple.
- Use descriptive Boolean variable names.
- Prefer `elif` over multiple independent `if` statements when testing related conditions.
- Limit nesting whenever possible.
- Write conditions that read naturally in plain English.

Example:

<-> Python

```
if is_logged_in:
```

is generally more readable than:

<-> Python

```
if is_logged_in == True:
```

Developer's Note (*Topic*)

Conditional statements appeared in every project completed during the Python Crash Course. They controlled ship movement and collision detection in Alien Invasion, validated user authentication in Django, checked API responses before processing data, determined chart colors, and handled user input throughout the Learning Log application. Writing clear conditional logic made debugging significantly easier because program decisions became easier to follow.

Related Topics (*Topic*)

- **2.7 Operators**
 - **2.8 Boolean Values**
 - **2.10 Loops**
 - **2.11 Defining Functions**
 - **2.24 Exception Handling**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Conditional statements allow programs to make decisions.
- The primary conditional statements are `if`, `elif`, and `else`.
- Conditions evaluate to either `True` or `False`.
- Proper indentation is required for conditional blocks.
- Logical operators allow multiple conditions to be combined.
- Clear, well-structured conditional statements improve readability, maintainability, and debugging.

2.2.3 Loops

Purpose (*Topic*)

Loops allow a program to repeatedly execute a block of code until a specified condition has been met or until every item in a collection has been processed. Rather than writing the same code multiple times, loops automate repetitive tasks, making programs shorter, more efficient, and easier to maintain.

Loops are one of the most powerful control structures in Python and are used extensively in data processing, file handling, games, web applications, and automation.

Why It Matters (*Topic*)

Many programming tasks involve repetition. Examples include:

- Reading every line in a file.
- Processing each item in a list.
- Displaying records from a database.
- Updating game objects every frame.
- Repeating user prompts until valid input is received.

Without loops, developers would need to duplicate code repeatedly, making programs difficult to maintain and prone to errors.

Quick Reference (*Topic*)

Statement	Purpose	Example
<code>for</code>	Iterate through a sequence.	<code>for item in items:</code>
<code>while</code>	Repeat while a condition is true.	<code>while active:</code>
<code>range()</code>	Generate a sequence of numbers.	<code>range(5)</code>
<code>break</code>	Exit a loop immediately.	<code>break</code>
<code>continue</code>	Skip to the next iteration.	<code>continue</code>
<code>pass</code>	Placeholder statement.	<code>pass</code>

Syntax (*Topic*)

Basic `for` loop:

<-> Python

```
for number in range(5):  
    print(number)
```

Output

```
0  
1  
2  
3  
4
```

Basic `while` loop:

<-> Python

```
count = 1
```

```
while count <= 5:  
    print(count)  
    count += 1
```

Output

```
1  
2  
3  
4  
5
```

Core Concepts (*Topic*)

The `for` Loop

A `for` loop processes every item in a sequence.

<-> Python

```
colors = ["red", "green", "blue"]
```

```
for color in colors:  
    print(color)
```

Output

```
red  
green  
blue
```

The loop automatically stops after processing the last item.

Using `range()`

The `range()` function generates a sequence of numbers.

<-> Python

```
for number in range(1, 6):  
    print(number)
```

Output

```
1  
2  
3  
4  
5
```

`range(start, stop)` includes the starting value but excludes the stopping value.

The while Loop

A `while` loop continues executing as long as its condition remains `True`.

<-> Python

```
count = 1

while count <= 3:
    print(count)
    count += 1
```

Output

```
1
2
3
```

Unlike a for loop, the programmer is responsible for updating the loop condition.

Nested Loops

A loop may contain another loop.

<-> Python

```
for row in range(2):
    for column in range(3):
        print(row, column)
```

Output

```
0 0
0 1
0 2
1 0
1 1
1 2
```

Nested loops are commonly used when processing tables, grids, game boards, and matrices.

Loop Control Statements

break

Immediately exits the loop.

<-> Python

```
for number in range(10):
```

```
    if number == 5:
```

```
        break
```

```
    print(number)
```

Output

0

1

2

3

4

continue

Skips the remainder of the current iteration and moves to the next.

<-> Python

```
for number in range(5):
```

```
    if number == 2:
```

```
        continue
```

```
    print(number)
```

Output

0

1

3

4

`pass`

Acts as a placeholder when no action is required.

<-> Python

```
for number in range(5):  
    pass
```

Examples (*Topic*)

Example 1 – Counting

<-> Python

```
for number in range(1, 11):  
    print(number)
```

Output

```
1  
2  
3  
4  
5  
6  
7  
8  
9  
10
```

Example 2 – Processing a List

<-> Python

```
languages = ["Python", "Java", "C++"]
```

```
for language in languages:  
    print(language)
```

Output

Python

Java

C++

Example 3 – User-Controlled Loop

<-> Python

```
active = True
```

```
while active:  
    print("Running...")  
    active = False
```

Output

Running...

Example 4 – Finding an Item

<-> Python

```
numbers = [4, 8, 15, 16, 23, 42]
```

```
for number in numbers:
```

```
    if number == 23:
        print("Found!")
        break
```

Output

Found!

Infinite Loops (*Topic*)

An infinite loop occurs when the loop's terminating condition is never met.

Incorrect:

<-> Python

```
count = 1
```

```
while count <= 5:
    print(count)
```

This loop never ends because count is never updated.

Correct:

<-> Python

```
count = 1
```

```
while count <= 5:
    print(count)
    count += 1
```

Always ensure that a while loop eventually becomes False.

Common Beginner Errors (*Topic*)

- Forgetting to update the loop variable in a while loop.
 - Incorrect indentation.
 - Confusing `range(5)` with `range(1, 5)`.
 - Accidentally creating infinite loops.
 - Modifying a collection while iterating over it.
-

Best Practices (*Topic*)

- Use a `for` loop when processing a collection.
 - Use a `while` loop when the number of iterations is unknown.
 - Keep loop bodies concise and easy to read.
 - Avoid deeply nested loops whenever possible.
 - Use meaningful variable names such as `student`, `entry`, or `temperature` instead of generic names like `x`.
-

Developer's Note (*Topic*)

Loops were used extensively throughout the Python Crash Course projects. The Alien Invasion game loop continuously updated the display and game state, weather and earthquake visualizations processed thousands of data points using `for` loops, API responses were parsed item by item, and Django frequently iterated through query results to display topics and entries. Learning to recognize when a task required repetition quickly became one of the most valuable programming skills developed during these projects.

Related Topics (*Topic*)

- **2.8 Boolean Values**
 - **2.9 Conditional Statements**
 - **2.11 Defining Functions**
 - **2.21 Reading Files**
 - **Chapter 6 – Data Visualization**
 - **Chapter 7 – APIs**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Loops repeat a block of code efficiently.
- Use a `for` loop when processing sequences and collections.
- Use a `while` loop when repetition depends on a condition.
- The `range()` function generates sequences of numbers for iteration.
- The `break`, `continue`, and `pass` statements provide additional loop control.
- Properly designed loops eliminate repetitive code and are essential for automation, data processing, and application development.

2.3 Functions

2.3.1 Defining Functions

Purpose (Topic)

Functions are reusable blocks of code designed to perform a specific task. Instead of writing the same code multiple times, a function allows that code to be written once and called whenever it is needed.

Functions improve organization, reduce duplication, simplify maintenance, and make programs easier to understand.

Why It Matters (Topic)

As programs become larger, repeating the same code in multiple locations makes them difficult to maintain. If the repeated code needs to change, every occurrence must be updated.

Functions solve this problem by placing reusable code into a single location. This approach follows the **DRY (Don't Repeat Yourself)** principle and allows developers to build modular, maintainable applications.

Quick Reference (Topic)

Statement / Function	Purpose	Example
<code>def</code>	Define a function.	<code>def greet():</code>
<code>return</code>	Return a value from a function.	<code>return total</code>
Function Call	Execute a function.	<code>greet()</code>
Parameters	Receive information.	<code>def greet(name):</code>
Arguments	Values passed to a function.	<code>greet("Skip")</code>

Syntax (*Topic*)

Defining a simple function:

<-> Python

```
def greet():  
    print("Welcome to Python!")
```

Calling the function:

<-> Python

```
greet()
```

Output

```
Welcome to Python!
```

Core Concepts (*Topic*)

Defining a Function

Functions are created using the def keyword followed by a function name and parentheses.

<-> Python

```
def display_message():  
    print("Learning Python one step at a time.")
```

The function does nothing until it is called.

Calling a Function

A function is executed by writing its name followed by parentheses.

<-> Python

```
display_message()
```

Output

```
Learning Python one step at a time.
```

Parameters

Parameters allow a function to receive information from the calling program.

<-> Python

```
def greet(name):  
    print(f"Hello, {name}!")
```

Calling the function:

<-> Python

```
greet("Skip")
```

Output

```
Hello, Skip!
```

Multiple Parameters

Functions may receive multiple pieces of information.

<-> Python

```
def introduce(name, city):  
    print(f"{name} lives in {city}.")
```

Calling the function:

<-> Python

```
introduce("Skip", "Houston")
```

Output

```
Skip lives in Houston.
```

Returning Values

Some functions produce information that should be returned to the calling program.

<-> Python

```
def square(number):  
    return number ** 2
```

Calling the function:

<-> Python

```
result = square(5)
```

```
print(result)
```

Output

25

Function Anatomy (*Topic*)

A function consists of several important parts.

<-> Python

```
def calculate_area(length, width):  
    area = length * width  
    return area
```

Component	Description
def	Begins the function definition.
calculate_area	Function name.
length, width	Parameters.
area = ...	Function body.
return area	Returns the calculated result.

Examples (*Topic*)

Example 1 – Greeting Function

<-> Python

```
def greet():  
    print("Welcome!")  
  
greet()
```

Output

Welcome!

Example 2 – Personalized Greeting

<-> Python

```
def greet(name):  
    print(f"Hello, {name}!")  
  
greet("Skip")
```

Output

Hello, Skip!

Example 3 – Returning a Calculation

<-> Python

```
def multiply(a, b):  
    return a * b  
  
print(multiply(6, 7))
```

Output

42

Example 4 – Reusing a Function

<-> Python

```
def display_title(title):  
    print(f"=== {title} ===")  
  
display_title("Python")  
display_title("Django")  
display_title("Git")
```

Output

```
=== Python ===  
=== Django ===  
=== Git ===
```

Common Beginner Errors (*Topic*)

- Forgetting the parentheses when calling a function.
- Forgetting the colon (:) after the function definition.
- Incorrect indentation.
- Using return outside of a function.
- Calling a function before it has been defined.
- Forgetting to pass required arguments.

Incorrect:

<-> Python

```
def greet()  
    print("Hello")
```

Correct:

<-> Python

```
def greet():  
    print("Hello")
```

Best Practices *(Topic)*

- Give functions descriptive names that describe their purpose.
- Keep each function focused on one task.
- Reuse functions instead of duplicating code.
- Keep functions short and easy to understand.
- Document complex functions with comments or docstrings.
- Follow Python's snake_case naming convention for function names.

Examples of good function names:

<-> Python

`calculate_total()`

`display_menu()`

`load_data()`

`save_file()`

`validate_input()`

Developer's Note *(Topic)*

Functions became one of the biggest turning points in the Python Crash Course. The Alien Invasion project demonstrated how dividing a game into small functions made thousands of lines of code manageable. Functions such as `check_events()`, `update_screen()`, and `fire_bullet()` each handled one responsibility, making the program easier to debug, test, and expand. The same design pattern carried forward into data visualization projects, API programs, and the Django Learning Log application.

Related Topics *(Topic)*

- **2.9 Conditional Statements**
 - **2.10 Loops**
 - **2.12 Parameters & Arguments**
 - **2.13 Return Values**
 - **2.14 Variable Scope**
 - **2.16 Classes**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Functions are reusable blocks of code that perform specific tasks.
- Functions are defined using the `def` keyword.
- A function must be called to execute its code.
- Parameters receive information, while arguments provide the values.
- Functions can return results to the calling program.
- Well-designed functions make programs easier to read, maintain, test, and expand.

2.3.2 Parameters & Arguments

Purpose *(Topic)*

Parameters and arguments allow functions to receive information from other parts of a program. They make functions flexible by enabling the same function to work with different values each time it is called.

Rather than creating separate functions for every possible situation, developers write one function that accepts parameters and processes whatever arguments are provided.

Why It Matters *(Topic)*

Functions become far more useful when they can work with different data. A greeting function, for example, should be able to greet any user—not just one specific person.

Parameters and arguments make code reusable, reduce duplication, and support modular programming. Nearly every function in Python accepts one or more arguments.

Quick Reference *(Topic)*

Term	Purpose	Example
Parameter	Variable defined in a function.	<code>def greet(name):</code>
Argument	Value passed to a function.	<code>greet("Skip")</code>
Positional Argument	Passed by position.	<code>greet("Skip")</code>
Keyword Argument	Passed by parameter name.	<code>greet(name="Skip")</code>
Default Parameter	Uses a default value if none is supplied.	<code>def greet(name="Guest")</code>
<code>*args</code>	Accepts multiple positional arguments.	<code>def total(*numbers):</code>
<code>**kwargs</code>	Accepts multiple keyword arguments.	<code>def profile(**data):</code>

Syntax (*Topic*)

Function with one parameter:

<-> Python

```
def greet(name):  
    print(f"Hello, {name}!")
```

Calling the function:

<-> Python

```
greet("Skip")
```

Output

```
Hello, Skip!
```

Core Concepts (*Topic*)

Parameters

Parameters are variables defined in a function's declaration. They act as placeholders for the values that will be supplied when the function is called.

<-> Python

```
def display_city(city):  
    print(city)
```

In this example, city is the parameter.

Arguments

Arguments are the actual values passed to a function.

<-> Python

```
display_city("Houston")
```

In this example, "Houston" is the argument.

Positional Arguments

Arguments are matched to parameters according to their position.

<-> Python

```
def introduce(name, city):  
    print(f"{name} lives in {city}.")
```

```
introduce("Skip", "Houston")
```

Output

```
Skip lives in Houston.
```

Keyword Arguments

Arguments may also be passed using the parameter names.

<-> Python

```
introduce(city="Houston", name="Skip")
```

Output

```
Skip lives in Houston.
```

Using keyword arguments makes function calls easier to read and eliminates concerns about argument order.

Default Parameter Values

Functions can define default values for parameters.

<-> Python

```
def greet(name="Guest"):
    print(f"Hello, {name}!")
```

Calling without an argument:

<-> Python

```
greet()
```

Output

```
Hello, Guest!
```

Calling with an argument:

<-> Python

```
greet("Skip")
```

Output

```
Hello, Skip!
```

Variable-Length Arguments (*args)

The *args syntax allows a function to accept any number of positional arguments.

<-> Python

```
def total(*numbers):
    print(sum(numbers))
```

```
total(5, 10, 15)
```

Output

```
30
```

Keyword Variable-Length Arguments (**kwargs)

The **kwargs syntax accepts any number of keyword arguments.

<-> Python

```
def profile(**person):  
    print(person)
```

```
profile(name="Skip", city="Houston")
```

Output

```
{'name': 'Skip', 'city': 'Houston'}
```

Examples (*Topic*)

Example 1 – One Parameter

<-> Python

```
def square(number):  
    print(number ** 2)
```

```
square(5)
```

Output

```
25
```

Example 2 – Two Parameters

<-> Python

```
def area(length, width):  
    print(length * width)
```

```
area(8, 10)
```

Output

```
80
```

Example 3 – Keyword Arguments

<-> Python

```
def employee(name, department):  
    print(f"{name} works in {department}.")  
  
employee(department="Engineering", name="Skip")
```

Output

Skip works in Engineering.

Example 4 – Default Value

<-> Python

```
def connect(server="localhost"):  
    print(f"Connecting to {server}")  
  
connect()  
connect("Production")
```

Output

Connecting to localhost
Connecting to Production

Common Beginner Errors (*Topic*)

- Confusing parameters with arguments.
- Supplying arguments in the wrong order.
- Forgetting required arguments.
- Misspelling keyword argument names.
- Defining mutable objects (such as lists) as default parameter values.

Incorrect:

<-> Python

```
def greet(name):  
    print(name)
```

```
greet()
```

Result

```
TypeError: greet() missing 1 required positional argument
```

Correct:

<-> Python

```
greet("Skip")
```

Best Practices (*Topic*)

- Use descriptive parameter names.
 - Keep the number of parameters reasonable.
 - Use keyword arguments when they improve readability.
 - Provide sensible default values when appropriate.
 - Reserve `*args` and `**kwargs` for situations where flexibility is required.
-

Developer's Note (*Topic*)

Parameters became increasingly important throughout the Python Crash Course projects. Nearly every custom function accepted data, whether it was screen objects in Alien Invasion, file names in data visualization projects, JSON responses from APIs, or request objects in Django views. Understanding how parameters and arguments worked made it much easier to read library documentation because almost every Python function follows these same principles.

Related Topics (*Topic*)

- **2.11 Defining Functions**
 - **2.13 Return Values**
 - **2.14 Variable Scope**
 - **2.16 Classes**
 - **Chapter 6 – Data Visualization**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- Parameters define the information a function expects.
- Arguments provide the actual values passed to a function.
- Python supports positional, keyword, and default arguments.
- The `*args` and `**kwargs` syntax allows functions to accept a variable number of arguments.
- Well-designed parameters make functions more flexible, reusable, and easier to understand.
- Understanding parameters and arguments is essential for writing modular, professional Python applications.

2.3.3 Return Values

Purpose (*Topic*)

A return value is the result produced by a function and sent back to the part of the program that called it. Instead of simply displaying information, a function can calculate, process, or retrieve data and return that result for later use.

Return values make functions far more versatile by allowing their results to be stored, reused, combined with other calculations, or passed to additional functions.

Why It Matters (*Topic*)

Many functions perform calculations or retrieve information that other parts of a program need to use. Rather than printing a result directly, returning the value allows the calling program to decide what to do with it.

Returning values promotes modular programming, reduces duplicated code, and allows functions to become reusable building blocks for larger applications.

Quick Reference (*Topic*)

Statement	Purpose	Example
return	Send a value back to the caller.	return total
return	Exit a function immediately.	return
Multiple Returns	Return more than one value.	return x, y
None	Returned automatically if no value is specified.	return None

Syntax (*Topic*)

Returning a value from a function:

<-> Python

```
def square(number):  
    return number ** 2
```

Using the returned value:

<-> Python

```
result = square(5)
```

```
print(result)
```

Output

25

Core Concepts (*Topic*)

Returning a Single Value

The most common use of return is to send a single value back to the calling program.

<-> Python

```
def multiply(a, b):  
    return a * b
```

Calling the function:

<-> Python

```
answer = multiply(6, 7)
```

```
print(answer)
```

Output

42

Returning Multiple Values

Python allows multiple values to be returned at once.

<-> Python

```
def coordinates():  
    return 25, 50
```

Calling the function:

<-> Python

```
x, y = coordinates()  
  
print(x)  
print(y)
```

Output

```
25  
50
```

Python actually returns the values as a tuple.

Returning Early

A function may exit immediately when a condition is met.

<-> Python

```
def check_age(age):  
  
    if age < 0:  
        return "Invalid Age"  
  
    return "Valid Age"
```

Output

```
Valid Age
```

Returning early can simplify complex logic and improve readability.

Returning None

If a function does not explicitly return a value, Python automatically returns None.

<-> Python

```
def greet():  
    print("Hello")
```

Calling the function:

<-> Python

```
result = greet()
```

```
print(result)
```

Output

```
Hello
```

```
None
```

Examples (*Topic*)

Example 1 – Calculating an Area

<-> Python

```
def area(length, width):  
    return length * width
```

```
print(area(10, 8))
```

Output

```
80
```

Example 2 – Returning Text

<-> Python

```
def full_name(first, last):  
    return f"{first} {last}"  
  
print(full_name("Skip", "Voss"))
```

Output

Skip Voss

Example 3 – Returning a Boolean

<-> Python

```
def is_even(number):  
    return number % 2 == 0  
  
print(is_even(8))
```

Output

True

Example 4 – Returning Multiple Values

<-> Python

```
def statistics():  
  
    minimum = 5  
    maximum = 25  
  
    return minimum, maximum  
  
low, high = statistics()  
  
print(low)  
print(high)
```

Output

5
25

Common Beginner Errors (*Topic*)

- Confusing print() with return.
- Forgetting to include a return statement.
- Writing code after a return statement that never executes.
- Attempting to use a returned value when the function returns None.

Incorrect:

<-> Python

```
def add(a, b):  
    print(a + b)
```

```
result = add(5, 3)
```

```
print(result)
```

Output

8

None

Correct:

<-> Python

```
def add(a, b):  
    return a + b
```

```
result = add(5, 3)
```

```
print(result)
```

Output

8

Best Practices (*Topic*)

- Return values instead of printing them whenever the result may be reused.
 - Keep return statements simple and easy to understand.
 - Use meaningful variable names to store returned values.
 - Return early when it improves readability.
 - Ensure every execution path returns the expected type of value.
-

Developer's Note (*Topic*)

Understanding return values was a significant step in writing more professional Python code. Throughout the Python Crash Course projects, functions frequently returned calculated values, configuration settings, API responses, Django query results, and game state information. Separating computation from display made programs far more modular and easier to test because one function could calculate a result while another handled presentation.

Related Topics (*Topic*)

- **2.11 Defining Functions**
 - **2.12 Parameters & Arguments**
 - **2.14 Variable Scope**
 - **2.16 Classes**
 - **Chapter 6 – Data Visualization**
 - **Chapter 8 – Django Web Development**
-

Key Takeaways (*Topic*)

- The return statement sends information back to the calling program.
- Functions may return one value, multiple values, or no explicit value.
- Python automatically returns None if no return statement is provided.
- Returning values makes functions reusable and more flexible than simply printing results.
- Separating calculation from output leads to cleaner, more maintainable programs.
- Return values are fundamental to modular programming and are used throughout professional Python development.

2.3.4 Variable Scope

What Is Variable Scope?

Variable scope defines **where a variable exists and where it can be accessed** within a Python program.

Not every variable is available everywhere. A variable created in one part of a program may not exist in another part. Understanding scope helps prevent errors, avoids accidental data changes, and makes programs easier to maintain.

Python primarily uses two scopes that every developer should understand:

- **Global Scope** – Variables accessible throughout the program.
- **Local Scope** – Variables that exist only inside a function.

Global Variables

A global variable is created outside of any function. It remains available throughout the program unless its value is changed or deleted.

Example:

<-> Python

```
message = "Welcome to Python"
```

```
def greet():  
    print(message)
```

```
greet()  
print(message)
```

Output:

```
Welcome to Python  
Welcome to Python
```

The function can read the global variable because it exists outside the function and is visible throughout the module.

Local Variables

A local variable is created inside a function. It exists only while that function is running.

Example:

<-> Python

```
def greet():  
    message = "Hello, World!"  
    print(message)
```

```
greet()
```

Output:

```
Hello, World!
```

Attempting to access the variable outside the function produces an error.

<-> Python

```
def greet():  
    message = "Hello"
```

```
greet()  
print(message)
```

Output:

```
NameError: name 'message' is not defined
```

Once the function finishes executing, the local variable no longer exists.

Local Variables Override Global Variables

If a local variable has the same name as a global variable, the local variable temporarily takes precedence inside the function.

Example:

<-> Python

```
message = "Global"
```

```
def greet():  
    message = "Local"  
    print(message)
```

```
greet()  
print(message)
```

Output:

```
Local
```

```
Global
```

The local variable affects only the function. The global variable remains unchanged.

Modifying Global Variables

Functions can read global variables without any special syntax.

However, modifying a global variable requires the `global` keyword.

Example:

<-> Python

```
count = 0
```

```
def increment():  
    global count  
    count += 1
```

```
increment()  
print(count)
```

Output:

```
1
```

Without the `global` keyword, Python assumes `count` is a new local variable and raises an error.

Why Excessive Global Variables Are Discouraged

Although global variables are useful in certain situations, relying on them too heavily can create difficult-to-maintain programs.

Problems include:

- Functions become dependent on outside data.
- Bugs become harder to locate.
- Changes in one part of the program can unexpectedly affect another.
- Testing individual functions becomes more difficult.

A good programming practice is to pass values into functions as parameters and return results rather than modifying global variables directly.

The LEGB Rule

Python searches for variables using the **LEGB Rule**, which defines the order in which names are resolved.

Scope	Description
L – Local	Variables inside the current function.
E – Enclosing	Variables in any enclosing (nested) functions.
G – Global	Variables defined at the module level.
B – Built-in	Python's built-in names such as <code>print()</code> , <code>len()</code> , and <code>range()</code> .

Python searches these scopes in order until it finds the requested variable.

Example of the LEGB Rule

<-> Python

```
message = "Global"
```

```
def outer():
```

```
    message = "Enclosing"
```

```
    def inner():
```

```
        message = "Local"
```

```
        print(message)
```

```
    inner()
```

```
outer()
```

Output:

```
Local
```

The `inner()` function finds its own local variable first, so Python never searches the enclosing or global scopes.

Best Practices

- Keep variables as local as possible.
 - Pass information into functions using parameters.
 - Return values instead of modifying globals whenever practical.
 - Use meaningful variable names.
 - Reserve global variables for constants or data shared intentionally across the application.
-

Key Takeaways

- Scope determines where a variable can be accessed.
- Global variables exist throughout the module.
- Local variables exist only within the function where they are created.
- Local variables override global variables with the same name.
- The `global` keyword is required to modify a global variable from inside a function.
- Python resolves variable names using the **LEGB** search order.
- Limiting variable scope leads to cleaner, safer, and more maintainable code.

2.3.5 Lambda Functions

What Are Lambda Functions?

A **lambda function** is a small, anonymous function created using Python's lambda keyword.

Unlike a regular function created with def, a lambda function:

- Has no function name.
- Contains only a single expression.
- Automatically returns the result of that expression.
- Is commonly used for short, simple operations.

The general syntax is:

<-> Python

```
lambda parameters: expression
```

The expression is evaluated and its result is returned automatically.

A Simple Lambda Function

The following lambda function adds two numbers.

<-> Python

```
add = lambda x, y: x + y
```

```
print(add(5, 3))
```

Output:

8

This is equivalent to writing:

<-> Python

```
def add(x, y):  
    return x + y
```

```
print(add(5, 3))
```

Both functions produce the same result.

Lambda with One Parameter

Lambda functions can accept any number of parameters.

<-> Python

```
square = lambda x: x ** 2
```

```
print(square(7))
```

Output:

```
49
```

Lambda with Multiple Parameters

<-> Python

```
multiply = lambda a, b, c: a * b * c
```

```
print(multiply(2, 3, 4))
```

Output:

```
24
```

Using Lambda with sorted()

One of the most common uses of lambda functions is providing a custom sorting key.

<-> Python

```
students = [  
    ("Alice", 88),  
    ("Bob", 75),  
    ("Charlie", 95),  
]  
  
students.sort(key=lambda student: student[1])  
  
print(students)
```

Output:

```
[('Bob', 75), ('Alice', 88), ('Charlie', 95)]
```

The lambda function tells Python to sort using the second item (the score) in each tuple.

Using Lambda with map()

The map() function applies a function to every item in an iterable.

<-> Python

```
numbers = [1, 2, 3, 4, 5]  
  
squares = list(map(lambda x: x ** 2, numbers))  
  
print(squares)
```

Output:

```
[1, 4, 9, 16, 25]
```

Each number is squared automatically.

Using Lambda with filter()

The filter() function selects items that satisfy a condition.

<-> Python

```
numbers = [1, 2, 3, 4, 5, 6]
```

```
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))
```

```
print(even_numbers)
```

Output:

```
[2, 4, 6]
```

Only the even numbers remain.

Using Lambda with reduce()

The reduce() function repeatedly combines values into a single result. It is part of Python's functools module.

<-> Python

```
from functools import reduce
```

```
numbers = [1, 2, 3, 4]
```

```
product = reduce(lambda x, y: x * y, numbers)
```

```
print(product)
```

Output:

```
24
```

The multiplication occurs like this:

$((1 \times 2) \times 3) \times 4$

When to Use Lambda Functions

Lambda functions are ideal when:

- The function is short and simple.
- The function is used only once.
- A function must be passed as an argument.
- Working with `map()`, `filter()`, `sorted()`, or `reduce()`.

Example:

<-> Python

```
names = ["Alice", "Bob", "Charlie"]
```

```
sorted_names = sorted(names, key=lambda name: len(name))
```

```
print(sorted_names)
```

Output:

```
['Bob', 'Alice', 'Charlie']
```

The list is sorted by the length of each name.

When Not to Use Lambda Functions

Avoid lambda functions when:

- The logic requires multiple statements.
- The function becomes difficult to read.
- The function will be reused throughout the program.
- A descriptive function name improves readability.

In these cases, use a standard function created with `def`.

Lambda vs. Regular Functions

Feature	Lambda	def Function
Named	No	Yes
Multiple statements	No	Yes
Automatically returns value	Yes	No (return required)
Best for	Short, simple operations	Larger reusable functions

Best Practices

- Keep lambda expressions short and readable.
- Use them only when they improve clarity.
- Prefer regular functions for complex logic.
- Avoid deeply nested lambda expressions.
- Use descriptive `def` functions whenever readability is more important than brevity.

Key Takeaways

- Lambda functions are anonymous, single-expression functions.
- They automatically return the result of their expression.
- They are commonly used with `sorted()`, `map()`, `filter()`, and `reduce()`.
- Lambda functions are excellent for short, one-time operations.
- Regular functions created with `def` remain the better choice for complex or reusable logic.

2.4 Classes & Object-Oriented Programming (OOP)

2.4.1 Classes

What Is a Class?

A **class** is a blueprint or template used to create objects in Python.

Think of a class as the design for something rather than the thing itself. Just as an architect creates blueprints for a house before any houses are built, a programmer creates a class before creating objects.

A class defines:

- What data an object will store (attributes).
- What actions an object can perform (methods).
- How similar objects should behave.

Classes are one of the fundamental concepts of **Object-Oriented Programming (OOP)**.

Why Use Classes?

As programs become larger, organizing related data and functions becomes increasingly important.

Instead of keeping separate variables and functions scattered throughout a program, a class groups them together into a single logical unit.

Benefits include:

- Better organization.
- Code reuse.
- Easier maintenance.
- Improved readability.
- More scalable applications.

Nearly every modern Python framework, including Django, makes extensive use of classes.

Real-World Analogy

Consider a car manufacturer.

A blueprint specifies that every car has:

- A make
- A model
- A color
- An engine
- A method to start
- A method to stop

The blueprint is the **class**.

Each individual car built from that blueprint is an **object**.

Every car shares the same design but contains its own unique information.

Creating a Class

Classes are created using the class keyword.

<-> Python

```
class Dog:
    """A simple class representing a dog."""
    pass
```

This creates an empty class named Dog.

The pass statement simply tells Python that the class currently contains no code.

Naming Classes

Python follows a naming convention known as **PascalCase** for class names.

Examples:

Good:

<-> Python

```
class Employee:  
    pass
```

```
class StudentRecord:  
    pass
```

```
class BankAccount:  
    pass
```

Avoid:

<-> Python

```
class employee:  
    pass
```

```
class bank_account:  
    pass
```

Although these names are valid, they do not follow the standard Python style guidelines (PEP 8).

Adding Data to a Class

Classes become useful when they define attributes and behavior.

Python uses a special method named `__init__()` to initialize new objects.

<-> Python

```
class Dog:
```

```
    def __init__(self, name, age):  
        self.name = name  
        self.age = age
```

When a new object is created, Python automatically calls the `__init__()` method.

It initializes the object's data.

Understanding self

The first parameter of every instance method is traditionally named `self`.

`self` refers to the object currently being created or used.

For example:

<-> Python

```
self.name = name
```

means:

"Store the value of `name` inside this specific object's `name` attribute."

Every object has its own copy of these attributes.

Adding Methods

A class can also define methods.

Methods are simply functions that belong to a class.

<-> Python

```
class Dog:

    def __init__(self, name, age):
        self.name = name
        self.age = age

    def sit(self):
        print(f"{self.name} is now sitting.")

    def roll_over(self):
        print(f"{self.name} rolled over!")
```

The methods use the object's stored information through self.

Creating an Object

Creating an object is called **instantiation**.

<-> Python

```
my_dog = Dog("Buddy", 4)

print(my_dog.name)
print(my_dog.age)
```

Output:

```
Buddy
4
```

Python creates a new Dog object and stores the provided values in that object's attributes.

Calling Methods

Methods are called using dot notation.

<-> Python

```
my_dog = Dog("Buddy", 4)
```

```
my_dog.sit()
```

```
my_dog.roll_over()
```

Output:

```
Buddy is now sitting.
```

```
Buddy rolled over!
```

Creating Multiple Objects

One class can create many different objects.

<-> Python

```
dog1 = Dog("Buddy", 4)
```

```
dog2 = Dog("Lucy", 2)
```

```
print(dog1.name)
```

```
print(dog2.name)
```

Output:

```
Buddy
```

```
Lucy
```

Each object stores its own data while sharing the same class definition.

Classes vs. Objects

Class	Object
Blueprint	Individual instance
Defines attributes and methods	Contains actual data
Created once	Created many times
Template	Real working item

Think of it this way:

Class → Blueprint

Object → House built from the blueprint

Where Classes Are Used

Classes are everywhere in Python.

Examples include:

- Django models
- File objects
- Database connections
- Network sockets
- GUI applications
- Games
- APIs
- Data analysis libraries
- Custom business applications

Many of the Python libraries you have already used create objects behind the scenes.

Best Practices

- Use PascalCase for class names.
- Keep each class focused on a single responsibility.
- Give attributes meaningful names.
- Create methods that represent actions the object performs.
- Store related data together inside the class.

Key Takeaways

- A class is a blueprint used to create objects.
- Objects are individual instances of a class.
- Classes combine data (attributes) and behavior (methods).
- The `__init__()` method initializes new objects.
- The `self` parameter refers to the current object.
- Objects are created through instantiation.
- Classes form the foundation of Object-Oriented Programming and are widely used throughout Python, Django, and many third-party libraries.

2.4.2 Objects

What Is an Object?

An **object** is an individual instance of a class.

If a class is the blueprint, an object is the finished product created from that blueprint.

A class defines what an object should look like and what it can do, while an object contains its own unique data.

Classes Create Objects

A class by itself does nothing until an object is created from it.

Consider the following class:

<-> Python

```
class Dog:

    def __init__(self, name, age):
        self.name = name
        self.age = age
```

This defines what every `Dog` object should contain.

To actually use the class, we create an object.

<-> Python

```
my_dog = Dog("Buddy", 4)
```

The variable `my_dog` now refers to a new object created from the `Dog` class.

Every Object Is Unique

Even though objects come from the same class, each object stores its own data.

<-> Python

```
dog1 = Dog("Buddy", 4)
```

```
dog2 = Dog("Lucy", 2)
```

```
dog3 = Dog("Max", 6)
```

Each object has its own attributes.

<-> Python

```
print(dog1.name)
```

```
print(dog2.name)
```

```
print(dog3.name)
```

Output:

```
Buddy
```

```
Lucy
```

```
Max
```

Although all three are Dog objects, each stores different information.

Objects Have Attributes

Attributes are variables that belong to an object.

Using our example:

<-> Python

```
dog = Dog("Buddy", 4)
```

```
print(dog.name)
```

```
print(dog.age)
```

Output:

```
Buddy
```

```
4
```

The values are accessed using **dot notation**.

```
object.attribute
```

Objects Have Methods

Objects also have methods.

Methods are functions that belong to the object.

<-> Python

```
class Dog:

    def __init__(self, name, age):
        self.name = name
        self.age = age

    def bark(self):
        print("Woof!")
```

```
dog = Dog("Buddy", 4)
```

```
dog.bark()
```

Output:

Woof!

Methods are also accessed using dot notation.

```
object.method()
```

Objects Maintain Their Own State

An object's **state** is the collection of values stored in its attributes.

For example:

<-> Python

```
dog = Dog("Buddy", 4)
```

```
print(dog.name)
```

```
print(dog.age)
```

The object's state is:

Name: Buddy

Age: 4

If another object is created:

<-> Python

```
another_dog = Dog("Lucy", 2)
```

its state becomes:

Name: Lucy

Age: 2

Each object manages its own state independently.

Multiple Objects Can Use the Same Methods

Although each object stores different data, they all share the methods defined in the class.

<-> Python

```
class Dog:

    def __init__(self, name):
        self.name = name

    def introduce(self):
        print(f"My name is {self.name}.")
```

Now create two objects.

<-> Python

```
dog1 = Dog("Buddy")
dog2 = Dog("Lucy")

dog1.introduce()
dog2.introduce()
```

Output:

```
My name is Buddy.
My name is Lucy.
```

The same method behaves differently because each object has different data.

Objects Live in Memory

When an object is created, Python allocates memory for it.

<-> Python

```
dog = Dog("Buddy", 4)
```

The object `dog` contains two attributes:

- “`name` with the value “Buddy” “
- “`age` with the value 4 “

The variable `dog` refers to this object in memory. Whenever you access `dog.name` or `dog.age`, Python retrieves the corresponding attribute from that object.

Everything Is an Object in Python

One of Python's defining characteristics is that **almost everything is an object**.

Examples include:

<-> Python

```
number = 10
```

```
text = "Python"
```

```
numbers = [1, 2, 3]
```

Each of these is an object created from a class.

You can verify this using the `type()` function.

<-> Python

```
print(type(10))
```

```
print(type("Python"))
```

```
print(type([1, 2, 3]))
```

Output:

```
<class 'int'>
```

```
<class 'str'>
```

```
<class 'list'>
```

This demonstrates that integers, strings, and lists are all objects created from their respective classes.

Objects in Real Programs

As you continue learning Python, you'll work with objects constantly.

Examples include:

- File objects
- Database objects
- Django model objects
- HTTP request and response objects
- Plotly figure objects
- Pandas DataFrame objects
- Matplotlib figure objects

Even if you don't create the class yourself, you'll often work with objects created by Python or third-party libraries.

Best Practices

- Create descriptive class names.
- Give objects meaningful variable names.
- Store only related data within an object.
- Use methods to perform actions involving the object's data.
- Think of an object as representing one real-world entity or concept.

Key Takeaways

- An object is an instance of a class.
- Each object stores its own unique data.
- Attributes describe an object's state.
- Methods define an object's behavior.
- Objects are accessed using dot notation.
- Variables store references to objects in memory.
- Nearly everything in Python is an object, making object-oriented programming a fundamental part of the language.

2.4.3 Inheritance

What Is Inheritance?

Inheritance is one of the fundamental principles of **Object-Oriented Programming (OOP)**. It allows one class to inherit the attributes and methods of another class.

Rather than writing the same code multiple times, a new class can build upon an existing class by reusing its functionality and adding its own unique features.

The original class is called the **parent class** (or **base class**), and the new class is called the **child class** (or **derived class**).

Why Use Inheritance?

Inheritance promotes **code reuse** and makes programs easier to maintain.

Benefits include:

- Eliminates duplicate code.
- Makes programs easier to expand.
- Keeps related classes organized.
- Allows child classes to inherit existing functionality while adding new behavior.

Inheritance is widely used throughout Python and frameworks such as Django.

Parent and Child Classes

Suppose we create a general Animal class.

<-> Python

```
class Animal:

    def __init__(self, name):
        self.name = name

    def speak(self):
        print(f"{self.name} makes a sound.")
```

Now we can create a more specialized class that inherits from Animal.

<-> Python

```
class Dog(Animal):
    pass
```

The Dog class automatically inherits everything from Animal.

Creating an Object from the Child Class

Although the Dog class contains no code of its own, it still inherits the parent class's constructor and methods.

<-> Python

```
dog = Dog("Buddy")
```

```
print(dog.name)
```

```
dog.speak()
```

Output:

```
Buddy
```

```
Buddy makes a sound.
```

The Dog object behaves as though those attributes and methods were defined inside the Dog class.

Extending a Parent Class

A child class can add its own methods without changing the parent class.

<-> Python

```
class Animal:

    def __init__(self, name):
        self.name = name

    def speak(self):
        print(f"{self.name} makes a sound.")

class Dog(Animal):

    def fetch(self):
        print(f"{self.name} is fetching the ball.")
```

Now the Dog class has both inherited and new behavior.

<-> Python

```
dog = Dog("Buddy")

dog.speak()
dog.fetch()
```

Output:

```
Buddy makes a sound.
Buddy is fetching the ball.
```

Overriding Methods

A child class can replace an inherited method with its own version. This is known as **method overriding**.

<-> Python

```
class Animal:

    def __init__(self, name):
        self.name = name

    def speak(self):
        print(f"{self.name} makes a sound.")

class Dog(Animal):

    def speak(self):
        print(f"{self.name} barks.")
```

Now:

<-> Python

```
dog = Dog("Buddy")

dog.speak()
```

Output:

Buddy barks.

The child class's version of `speak()` replaces the parent's version.

Calling the Parent Constructor with super()

When a child class defines its own `__init__()` method, it often needs to initialize the parent class as well.

Python provides the `super()` function for this purpose.

<-> Python

```
class Animal:

    def __init__(self, name):
        self.name = name


class Dog(Animal):

    def __init__(self, name, breed):
        super().__init__(name)
        self.breed = breed
```

Creating an object:

<-> Python

```
dog = Dog("Buddy", "Golden Retriever")

print(dog.name)
print(dog.breed)
```

Output:

```
Buddy
Golden Retriever
```

The call to `super().__init__(name)` initializes the inherited name attribute before adding the new breed attribute.

Real-World Example

Consider a company.

An Employee class might contain information common to every employee:

- Name
- Employee ID
- Department

A Manager class could inherit from Employee while adding:

- Number of direct reports
- Budget responsibility
- Approval authority

The manager still has all the characteristics of an employee but also possesses additional responsibilities.

Inheritance in Django

Django makes extensive use of inheritance.

For example:

- Generic class-based views inherit from base view classes.
- User models inherit common authentication behavior.
- Custom classes often extend Django's built-in classes rather than starting from scratch.

Understanding inheritance will make many Django concepts much easier to understand.

Best Practices

- Use inheritance only when there is a true **"is-a"** relationship.
 - Keep parent classes general and reusable.
 - Keep child classes focused on specialized behavior.
 - Override methods only when different behavior is required.
 - Use `super()` when extending a parent's initialization.
-

Key Takeaways

- Inheritance allows one class to build upon another.
- The original class is the **parent** (base) class.
- The new class is the **child** (derived) class.
- Child classes inherit attributes and methods from their parent.
- Child classes can add new methods or override inherited ones.
- The `super()` function allows a child class to reuse the parent's constructor.
- Inheritance reduces duplicated code and supports building organized, maintainable object-oriented programs.

2.4.4 Polymorphism

What Is Polymorphism?

Polymorphism is one of the core principles of **Object-Oriented Programming (OOP)**. The word comes from Greek and means "**many forms**."

In programming, polymorphism means that **different objects can respond to the same method call in different ways**.

In other words, multiple classes can define a method with the same name, but each class provides its own implementation.

This allows you to write flexible code that works with many different object types without needing to know the specific class being used.

Why Use Polymorphism?

Polymorphism allows programs to treat different objects in a consistent manner while still allowing each object to perform its own specialized behavior.

Benefits include:

- Reduces conditional statements.
 - Makes code easier to extend.
 - Improves flexibility.
 - Encourages reusable designs.
 - Simplifies maintenance.
-

A Simple Example

Suppose we have two different classes.

<-> Python

```
class Dog:

    def speak(self):
        print("Woof!")
```

```
class Cat:

    def speak(self):
        print("Meow!")
```

Although both classes have a method named `speak()`, each class performs a different action.

Using the Objects

<-> Python

```
dog = Dog()
cat = Cat()
```

```
dog.speak()
cat.speak()
```

Output:

Woof!

Meow!

The same method name produces different results depending on the object calling it.

This is polymorphism.

Polymorphism with Inheritance

Polymorphism is commonly used together with inheritance.

<-> Python

```
class Animal:

    def speak(self):
        print("Some generic animal sound.")

class Dog(Animal):

    def speak(self):
        print("Woof!")

class Cat(Animal):

    def speak(self):
        print("Meow!")
```

Both child classes override the parent's speak() method.

Treating Different Objects the Same

One of the greatest strengths of polymorphism is that different objects can be processed using the same code.

<-> Python

```
animals = [Dog(), Cat()]
```

```
for animal in animals:  
    animal.speak()
```

Output:

Woof!

Meow!

Notice that the loop never checks whether an object is a Dog or a Cat. It simply calls `speak()`, and each object knows how to respond.

Without Polymorphism

Without polymorphism, the program might look like this:

<-> Python

```
if isinstance(animal, Dog):  
    animal.speak()
```

```
elif isinstance(animal, Cat):  
    animal.speak()
```

As additional animal types are added, the code becomes longer and more difficult to maintain. With polymorphism, no changes are needed.

Real-World Analogy

Imagine a television remote.

Regardless of the television's manufacturer, you press the **Power** button.

Different televisions respond differently:

- Samsung turns on.
- Sony turns on.
- LG turns on.

The button is the same.

The behavior depends on the device receiving the command.

Polymorphism works in much the same way.

Polymorphism Throughout Python

Python uses polymorphism extensively.

Consider the built-in `len()` function.

<-> Python

```
print(len("Python"))  
print(len([1, 2, 3, 4]))  
print(len({"a": 1, "b": 2}))
```

Output:

```
6  
4  
2
```

The same `len()` function works with strings, lists, dictionaries, tuples, and many other object types.

Each object knows how to calculate its own length.

Another Common Example

The + operator also demonstrates polymorphism.

<-> Python

```
print(5 + 3)
```

```
print("Hello " + "World")
```

```
print([1, 2] + [3, 4])
```

Output:

```
8
```

```
Hello World
```

```
[1, 2, 3, 4]
```

Although the same operator is used, Python performs different actions depending on the object types involved.

Duck Typing

Python often relies on a concept called **duck typing**.

A common saying is:

"If it walks like a duck and quacks like a duck, it's probably a duck."

In Python, if an object has the required methods, Python generally doesn't care what class the object belongs to.

For example:

<-> Python

```
class Duck:

    def speak(self):
        print("Quack!")
```

```
class Person:

    def speak(self):
        print("Hello!")
```

```
def make_it_speak(thing):
    thing.speak()
```

```
make_it_speak(Duck())
make_it_speak(Person())
```

Output:

```
Quack!
Hello!
```

The function doesn't check the object's type.

It simply assumes the object has a `speak()` method.

This is one of Python's defining characteristics.

Best Practices

- Use polymorphism to simplify code.
- Design methods with consistent names across related classes.
- Avoid unnecessary type checking when objects can provide the required behavior.
- Let each class handle its own implementation.

Key Takeaways

- Polymorphism means "**many forms.**"
- Different objects can respond differently to the same method call.
- It is commonly used together with inheritance.
- Polymorphism reduces conditional logic and improves flexibility.
- Python's built-in functions and operators make extensive use of polymorphism.
- Duck typing allows Python to focus on an object's behavior rather than its specific class.

2.4.5 Special Methods (`__init__`, `__str__`, etc.)

What Are Special Methods?

Special methods are predefined methods in Python that begin and end with double underscores (`__`). They are sometimes called **magic methods** or **dunder methods** (short for "double underscore").

These methods allow your objects to integrate naturally with Python's built-in functions and operators.

For example:

- `__init__()` initializes a new object.
- `__str__()` defines how an object is displayed as a string.
- `__len__()` defines what `len()` returns.
- `__add__()` defines how the `+` operator behaves.

Although there are many special methods available, a few are used far more frequently than the others.

The `__init__()` Method

The `__init__()` method is known as the **constructor**.

It is automatically called whenever a new object is created.

<-> Python

```
class Dog:
```

```
    def __init__(self, name, age):
        self.name = name
        self.age = age
```

```
dog = Dog("Buddy", 4)
```

Python automatically calls:

<-> Python

```
Dog.__init__(dog, "Buddy", 4)
```

You never call `__init__()` directly when creating an object.

Its purpose is to initialize the object's attributes.

The `__str__()` Method

The `__str__()` method controls what is displayed when an object is printed.

Without `__str__()`:

<-> Python

```
class Dog:
```

```
    def __init__(self, name):
        self.name = name
```

```
dog = Dog("Buddy")
```

```
print(dog)
```

Output:

```
<__main__.Dog object at 0x1045af850>
```

Python displays the object's memory location, which isn't very useful.

Now add a `__str__()` method.

<-> Python

```
class Dog:
```

```
    def __init__(self, name):
        self.name = name
```

```
    def __str__(self):
        return f"Dog: {self.name}"
```

```
dog = Dog("Buddy")
```

```
print(dog)
```

Output:

```
Dog: Buddy
```

The object now provides a much more meaningful description.

The `__repr__()` Method

The `__repr__()` method provides the **official string representation** of an object.

It is primarily intended for developers and debugging.

<-> Python

```
class Dog:

    def __init__(self, name):
        self.name = name

    def __repr__(self):
        return f"Dog('{self.name}')
```

Now:

<-> Python

```
dog = Dog("Buddy")
```

```
print(repr(dog))
```

Output:

```
Dog('Buddy')
```

Many developers write `__repr__()` so that it clearly shows how the object was created.

The `__len__()` Method

The `len()` function works because many Python objects implement the `__len__()` method.

You can define it for your own classes.

<-> Python

```
class Team:

    def __init__(self, members):
        self.members = members

    def __len__(self):
        return len(self.members)
```

```
team = Team(["Alice", "Bob", "Charlie"])
```

```
print(len(team))
```

Output:

3

Python automatically calls `team.__len__()` behind the scenes.

The `__add__()` Method

Special methods can also define how operators behave.

<-> Python

```
class Counter:

    def __init__(self, value):
        self.value = value

    def __add__(self, other):
        return Counter(self.value + other.value)

counter1 = Counter(10)
counter2 = Counter(5)

result = counter1 + counter2

print(result.value)
```

Output:

15

The + operator now knows how to combine two Counter objects.

Other Common Special Methods

Method	Purpose
<code>__init__()</code>	Initializes a new object
<code>__str__()</code>	Defines the user-friendly string representation
<code>__repr__()</code>	Defines the developer-oriented representation
<code>__len__()</code>	Returns the object's length
<code>__add__()</code>	Implements the + operator
<code>__eq__()</code>	Implements the == operator
<code>__lt__()</code>	Implements the < operator
<code>__getitem__()</code>	Supports indexing (object[index])
<code>__iter__()</code>	Makes an object iterable

You won't use all of these every day, but you'll encounter them frequently when working with larger Python projects and third-party libraries.

Why Special Methods Matter

Special methods allow your custom classes to behave like Python's built-in types.

For example, because lists implement these methods, you can write:

<-> Python

```
numbers = [1, 2, 3]
```

```
print(len(numbers))
```

```
print(numbers[0])
```

The list object understands how to respond because its class defines methods such as `__len__()` and `__getitem__()`.

When you define these methods in your own classes, your objects become more intuitive and easier to use.

Special Methods in Django

Django uses special methods extensively.

For example, Django models often define a `__str__()` method so that meaningful information appears in the Django Admin interface.

A simple example might look like this:

<-> Python

```
class Topic(models.Model):

    text = models.CharField(max_length=200)

    def __str__(self):
        return self.text
```

Without `__str__()`, the Django Admin would display a generic object reference instead of the topic's name.

Best Practices

- Always use `__init__()` to initialize object attributes.
- Implement `__str__()` whenever a readable string representation is helpful.
- Use `__repr__()` to aid debugging and development.
- Only implement other special methods when they provide meaningful behavior.
- Avoid redefining operators unless the new behavior is clear and intuitive.

Key Takeaways

- Special methods begin and end with double underscores (`__`).
- They allow custom objects to work naturally with Python's built-in functions and operators.
- `__init__()` initializes new objects.
- `__str__()` defines how an object is displayed to users.
- `__repr__()` provides a developer-focused representation.
- Methods such as `__len__()` and `__add__()` allow custom classes to support built-in operations.
- Special methods make your classes behave like native Python objects.

2.5 Files

2.5.1 Reading Files

Why Read Files?

Many Python programs need to read information stored outside the program itself. Rather than hard-coding data into your code, you can store it in a file and read it whenever the program runs.

Common examples include:

- Configuration files
- Text documents
- Log files
- CSV files
- JSON files
- Data exported from databases

Reading files allows your programs to work with changing data without modifying the source code.

Opening a File

Python uses the built-in `open()` function to open a file.

The general syntax is:

<-> Python

```
file = open("filename.txt", "r")
```

The "r" indicates **read mode**. Since reading is the default mode, it can usually be omitted.

Reading an Entire File

The simplest way to read a file is with the `read()` method.

Suppose a file named **message.txt** contains:

Hello, World!

Welcome to Python.

You can read the entire file like this:

<-> Python

```
with open("message.txt") as file:  
    contents = file.read()
```

```
print(contents)
```

Output:

Hello, World!

Welcome to Python.

The entire file is loaded into the variable `contents`.

Why Use with?

Notice the statement:

<-> Python

```
with open("message.txt") as file:
```

The `with` statement automatically closes the file when you're finished reading it.

Without `with`, you would need to close the file manually.

<-> Python

```
file = open("message.txt")
```

```
contents = file.read()
```

```
file.close()
```

Although this works, using `with` is considered best practice because it guarantees the file is closed properly, even if an error occurs.

Reading One Line at a Time

Sometimes a file is too large to read all at once.

The `readline()` method reads one line at a time.

Suppose **colors.txt** contains:

Red

Green

Blue

<-> Python

```
with open("colors.txt") as file:
```

```
    print(file.readline())
```

```
    print(file.readline())
```

```
    print(file.readline())
```

Output:

Red

Green

Blue

Each call to `readline()` reads the next line in the file.

Looping Through a File

A more common approach is to loop through the file.

<-> Python

```
with open("colors.txt") as file:
```

```
    for line in file:
```

```
        print(line.rstrip())
```

Output:

Red

Green

Blue

The `rstrip()` method removes the newline character (`\n`) stored at the end of each line.

Reading All Lines into a List

The `readlines()` method reads every line and stores them in a list.

<-> Python

```
with open("colors.txt") as file:
    lines = file.readlines()

print(lines)
```

Output:

```
['Red\n', 'Green\n', 'Blue\n']
```

Each line includes the newline character (`\n`).

To display the lines cleanly:

<-> Python

```
for line in lines:
    print(line.strip())
```

Output:

```
Red
```

```
Green
```

```
Blue
```

File Paths

Python can open files using either a **relative path** or an **absolute path**.

A relative path refers to the current working directory.

<-> Python

```
with open("notes.txt") as file:
    contents = file.read()
```

An absolute path specifies the file's complete location.

<-> Python

```
with open(r"F:\Python_Study\notes.txt") as file:
    contents = file.read()
```

The `r` before the string creates a **raw string**, allowing Windows paths to be written without escaping backslashes.

Checking Whether a File Exists

Attempting to open a file that doesn't exist raises an error.

<-> Python

```
with open("missing_file.txt") as file:
    contents = file.read()
```

Output:

```
FileNotFoundError: [Errno 2] No such file or directory: 'missing_file.txt'
```

Later in this handbook, you'll learn how to handle this exception gracefully using `try` and `except`.

Best Practices

- Use the `with` statement whenever opening files.
 - Prefer relative paths when working within a project.
 - Use descriptive file names.
 - Read large files one line at a time when practical.
 - Close files automatically by using `with` rather than calling `close()` manually.
-

Key Takeaways

- Python reads files using the `open()` function.
- The `with` statement automatically closes files when finished.
- `read()` reads the entire file.
- `readline()` reads one line at a time.
- Looping through a file is an efficient way to process its contents.
- `readlines()` stores every line in a list.
- Files can be accessed using either relative or absolute paths.
- Trying to open a file that doesn't exist raises a `FileNotFoundError`.

2.5.2 Writing Files

Why Write Files?

In addition to reading data, Python programs often need to **save information** for future use.

Writing to files allows a program to:

- Save user input.
- Create log files.
- Generate reports.
- Export data.
- Store program settings.
- Save results for later processing.

Python uses the same `open()` function for writing files, but with different file modes.

File Modes

The second argument to `open()` specifies how the file should be opened.

Mode	Description
"r"	Read an existing file (default).
"w"	Write to a file. Creates a new file or overwrites an existing one.
"a"	Append data to the end of an existing file. Creates the file if it doesn't exist.
"x"	Create a new file. Raises an error if the file already exists.

Writing to a New File

Use **write mode ("w")** to create a file and write text.

<-> Python

```
with open("notes.txt", "w") as file:  
    file.write("Learning Python is fun!")
```

If **notes.txt** does not exist, Python creates it.

If it already exists, **all previous contents are erased** before the new text is written.

Writing Multiple Lines

The `write()` method writes exactly what you provide.

To create multiple lines, include newline characters (`\n`).

<-> Python

```
with open("notes.txt", "w") as file:
    file.write("Learning Python\n")
    file.write("Learning Django\n")
    file.write("Building Projects\n")
```

The resulting file contains:

```
Learning Python
Learning Django
Building Projects
```

Without `\n`, all of the text would appear on a single line.

Appending to a File

Use **append mode ("a")** when you want to preserve the existing contents of a file.

Suppose **notes.txt** already contains:

```
Learning Python
Learning Django
```

Now append another line.

<-> Python

```
with open("notes.txt", "a") as file:
    file.write("Building Projects\n")
```

The updated file becomes:

```
Learning Python
Learning Django
Building Projects
```

Unlike "w" mode, append mode never removes the existing contents.

Creating a New File Only

Sometimes you want to ensure that an existing file is not accidentally overwritten.

Use **exclusive creation mode ("x")**.

<-> Python

```
with open("report.txt", "x") as file:  
    file.write("Monthly Report")
```

If **report.txt** already exists, Python raises a `FileExistsError`.

Writing User Input to a File

Files are often used to save information entered by a user.

<-> Python

```
name = input("Enter your name: ")
```

```
with open("users.txt", "a") as file:  
    file.write(name + "\n")
```

Each name entered is added to the end of the file.

Writing Numbers

The write() method only accepts strings.

Attempting to write a number directly results in an error.

Incorrect:

<-> Python

```
age = 25
```

```
with open("age.txt", "w") as file:  
    file.write(age)
```

Output:

TypeError: write() argument must be str, not int

Correct:

<-> Python

```
age = 25
```

```
with open("age.txt", "w") as file:  
    file.write(str(age))
```

The str() function converts the integer into a string before writing it.

Combining Reading and Writing

Programs often read data, modify it, and write it back.

A simple example:

<-> Python

```
with open("notes.txt") as file:
    contents = file.read()

updated_contents = contents.upper()

with open("notes_upper.txt", "w") as file:
    file.write(updated_contents)
```

This program:

1. Reads the original file.
2. Converts the contents to uppercase.
3. Writes the modified text to a new file.

Common Mistakes

Be careful when using write mode ("w").

This code:

<-> Python

```
with open("notes.txt", "w") as file:
    file.write("New text")
```

Completely replaces the previous contents of **notes.txt**.

If preserving existing data is important, use append mode ("a") instead.

Best Practices

- Use the with statement when writing files.
- Use "w" only when you intend to replace the file's contents.
- Use "a" when adding new information.
- Convert non-string values using `str()` before writing.
- Use meaningful file names.
- Include newline characters (`\n`) when writing multiple lines.

Key Takeaways

- Python writes files using the `open()` function.
- `"w"` creates a new file or overwrites an existing one.
- `"a"` adds data to the end of an existing file.
- `"x"` creates a new file and prevents accidental overwriting.
- The `write()` method writes strings to a file.
- Use `str()` to convert numbers before writing them.
- Always use the `with` statement to ensure files are closed properly.

2.5.3 Working with File Paths

What Is a File Path?

A **file path** tells Python where a file or folder is located on your computer.

Whenever you read from or write to a file, Python must know its location.

For example:

<-> Python

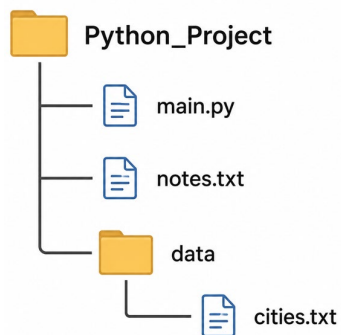
```
with open("notes.txt") as file:
    contents = file.read()
```

In this example, Python looks for **notes.txt** in the program's current working directory.

Relative Paths

A **relative path** specifies a file's location relative to the current working directory.

Suppose your project has the following structure:



To open **notes.txt**:

<-> Python

```
with open("notes.txt") as file:
    contents = file.read()
```

To open **cities.txt** inside the **data** folder:

<-> Python

```
with open("data/cities.txt") as file:
    contents = file.read()
```

Relative paths make projects portable because they do not depend on where the project is stored on a particular computer.

Absolute Paths

An **absolute path** specifies the complete location of a file.

Windows example:

<-> Python

```
with open(r"F:\Python_Study\Python_Crash_Course\data\cities.txt") as file:  
    contents = file.read()
```

The `r` creates a **raw string**, allowing backslashes to be written normally.

An absolute path always starts at the root of the drive.

Why Prefer Relative Paths?

Suppose you move your project from:

`F:\Python_Study\`

to:

`D:\Projects\`

If you used absolute paths, every path in your program would need to be updated.

If you used relative paths, **no code changes are required**, as long as the project folder structure remains the same.

For this reason, relative paths are generally preferred for project files.

The pathlib Module

Modern Python programs often use the pathlib module instead of manually building file paths. pathlib provides an object-oriented way to work with files and folders.

Import the Path class:

<-> Python

```
from pathlib import Path
```

Create a path object:

<-> Python

```
file_path = Path("notes.txt")
```

Read the file:

<-> Python

```
contents = file_path.read_text()
```

```
print(contents)
```

pathlib automatically handles many operating system differences and often produces cleaner, more readable code.

Joining Paths

Rather than manually typing slashes, pathlib allows paths to be joined using the / operator.

<-> Python

```
from pathlib import Path
```

```
file_path = Path("data") / "cities.txt"
```

```
print(file_path)
```

Output (Windows):

```
data\cities.txt
```

This approach is both readable and portable.

Checking Whether a File Exists

Before opening a file, you may want to verify that it exists.

Using pathlib:

<-> Python

```
from pathlib import Path
```

```
file_path = Path("notes.txt")
```

```
print(file_path.exists())
```

Output:

```
True
```

If the file is missing:

```
False
```

Determining the Current Working Directory

Sometimes it is useful to know where Python is currently looking for files.

<-> Python

```
from pathlib import Path
```

```
print(Path.cwd())
```

Example Output:

```
F:\Python_Study\Python_Crash_Course
```

This is the default location Python uses for relative paths.

Creating Directories

pathlib can also create directories.

<-> Python

```
from pathlib import Path
```

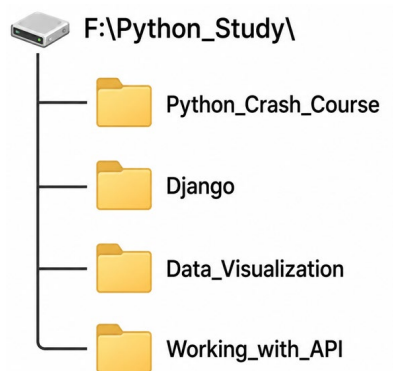
```
folder = Path("reports")
```

```
folder.mkdir(exist_ok=True)
```

The `exist_ok=True` argument prevents an error if the directory already exists.

Working with Your Projects

Throughout your Python studies, you've organized projects into folders such as:



Most of your programs should use **relative paths** within each project.

This makes it easy to move, back up, or share the project without modifying the code.

Best Practices

- Prefer relative paths for project files.
 - Use absolute paths only when necessary.
 - Use the `pathlib` module for new Python projects.
 - Check whether files exist before opening them when appropriate.
 - Keep related files together within the project folder.
-

Key Takeaways

- A file path tells Python where a file or folder is located.
- Relative paths are based on the current working directory.
- Absolute paths specify the complete location of a file.
- Relative paths make projects easier to move and share.
- The pathlib module provides a modern, object-oriented way to work with file paths.
- `Path.cwd()` returns the current working directory.
- `Path.exists()` checks whether a file or folder exists.
- `Path.mkdir()` creates new directories when needed.

2.6 Exceptions

2.6.1 Exception Handling

What Is an Exception?

An **exception** is an error that occurs while a program is running.

When Python encounters an error that it cannot handle, it raises an exception. If the exception is not handled, the program stops executing and displays an error message.

For example:

```
<-> Python  
print(10 / 0)
```

Output:

```
ZeroDivisionError: division by zero
```

Python stops the program because dividing by zero is not allowed.

Why Handle Exceptions?

Programs should be prepared for errors that may occur during normal use.

Examples include:

- A user enters invalid data.
- A file does not exist.
- A network connection is unavailable.
- A database cannot be reached.
- A calculation attempts to divide by zero.

Rather than crashing, a program can respond gracefully and continue running when appropriate.

The try and except Statements

Python handles exceptions using the `try` and `except` statements.

The general syntax is:

<-> Python

```
try:
```

```
    # Code that might cause an exception
```

```
except:
```

```
    # Code that runs if an exception occurs
```

If no exception occurs, the except block is skipped.

Handling Division by Zero

<-> Python

```
try:
```

```
    result = 10 / 0
```

```
except ZeroDivisionError:
```

```
    print("You cannot divide by zero.")
```

Output:

You cannot divide by zero.

Instead of crashing, the program displays a helpful message.

Handling Missing Files

One of the most common exceptions is attempting to open a file that does not exist.

<-> Python

```
try:
    with open("missing_file.txt") as file:
        contents = file.read()

except FileNotFoundError:
    print("The file could not be found.")
```

Output:

The file could not be found.

This example builds directly on what you learned in the **Reading Files** section.

Handling Multiple Exceptions

A try block may raise more than one type of exception.

Python allows multiple except blocks.

<-> Python

```
try:
    number = int(input("Enter a number: "))
    result = 100 / number

except ValueError:
    print("Please enter a valid integer.")

except ZeroDivisionError:
    print("Zero is not allowed.")
```

Each exception is handled independently.

The else Block

The `else` block executes only if **no exception occurs**.

<-> Python

```
try:
    number = int(input("Enter a number: "))

except ValueError:
    print("Invalid input.")

else:
    print(f"You entered {number}.")
```

If the user enters a valid integer, the else block executes.

The finally Block

The `finally` block always executes, regardless of whether an exception occurs.

It is commonly used to clean up resources.

<-> Python

```
try:
    file = open("notes.txt")

except FileNotFoundError:
    print("File not found.")

finally:
    print("Program finished.")
```

Output (if the file exists):

Program finished.

Output (if the file is missing):

File not found.

Program finished.

No matter what happens, the finally block always runs.

Complete Example

<-> Python

```
try:
    number = int(input("Enter a number: "))
    result = 100 / number

except ValueError:
    print("Please enter a valid integer.")

except ZeroDivisionError:
    print("You cannot divide by zero.")

else:
    print(f"Result: {result}")

finally:
    print("Program complete.")
```

This example demonstrates all four parts of Python's exception handling.

Common Exceptions

Exception	Cause
ZeroDivisionError	Division by zero
ValueError	Invalid value or conversion
TypeError	Incorrect data type
IndexError	Invalid list index
KeyError	Dictionary key not found
AttributeError	Attribute or method does not exist
FileNotFoundError	File cannot be located
NameError	Variable does not exist

Exception Handling in Django

Django handles many exceptions internally.

Examples include:

- Missing pages (**404 Not Found**)
- Permission errors (**403 Forbidden**)
- Database lookup failures
- Form validation errors

Understanding Python exceptions makes it easier to understand how Django reports and handles errors.

Best Practices

- Catch only the exceptions you expect.
 - Display meaningful error messages.
 - Avoid using a bare `except:` unless absolutely necessary.
 - Keep the code inside the `try` block as small as possible.
 - Use `finally` for cleanup operations such as closing files or releasing resources.
-

Key Takeaways

- An exception is a runtime error.
- Python raises exceptions when something unexpected occurs.
- The `try` block contains code that may fail.
- The `except` block handles specific exceptions.
- The `else` block runs only if no exception occurs.
- The `finally` block always executes.
- Proper exception handling makes programs more reliable, user-friendly, and easier to maintain.

2.6.2 Raising Exceptions

What Does It Mean to Raise an Exception?

So far, you've learned how to **catch** exceptions using try and except.

Sometimes, however, your own program needs to report that something has gone wrong.

This is done by **raising an exception**.

Python provides the raise statement for this purpose.

When an exception is raised, Python immediately stops the current operation and looks for an appropriate except block to handle it.

Why Raise Exceptions?

Programs often need to verify that data is valid before continuing.

Examples include:

- A number must be positive.
- A password must meet certain requirements.
- A user's age must be at least 18.
- A list cannot be empty.
- A file must exist before processing.

If these conditions are not met, raising an exception clearly communicates that something is wrong.

Basic Syntax

The general syntax is:

<-> Python

```
raise ExceptionType("Error message")
```

The exception type tells Python what kind of error occurred, and the message helps explain the problem.

Raising a ValueError

Suppose a program requires a positive number.

<-> Python

```
number = -5
```

```
if number < 0:  
    raise ValueError("The number must be positive.")
```

Output:

ValueError: The number must be positive.

Python stops the program because the value is not valid.

Raising an Exception Inside a Function

Exceptions are commonly raised inside functions.

<-> Python

```
def calculate_square_root(number):  
  
    if number < 0:  
        raise ValueError("Square roots require a non-negative number.")  
  
    return number ** 0.5
```

```
print(calculate_square_root(25))
```

Output:

5.0

If an invalid value is passed:

<-> Python

```
print(calculate_square_root(-9))
```

Output:

ValueError: Square roots require a non-negative number.

The function prevents invalid calculations by checking the input first.

Raising and Handling an Exception

An exception raised with raise can be handled just like any other exception.

<-> Python

try:

```
    age = -1
```

```
    if age < 0:
```

```
        raise ValueError("Age cannot be negative.")
```

```
except ValueError as error:
```

```
    print(error)
```

Output:

Age cannot be negative.

Instead of crashing, the program catches the exception and displays the error message.

Choosing the Right Exception

Python provides many built-in exception types.

Some common choices include:

Exception	Typical Use
ValueError	Invalid value
TypeError	Incorrect data type
FileNotFoundError	Missing file
IndexError	Invalid list index
KeyError	Missing dictionary key
RuntimeError	General runtime problem

Selecting the most appropriate exception makes programs easier to understand and debug.

Validating User Input

Raising exceptions is often part of input validation.

<-> Python

```
username = input("Enter your username: ")

if username == "":
    raise ValueError("Username cannot be empty.")
```

This prevents the program from continuing with invalid data.

When to Raise Exceptions

Raise an exception when:

- A function receives invalid input.
- A required resource is missing.
- Continuing would produce incorrect results.
- The caller should be informed of a problem.

Do **not** raise exceptions for normal program flow. Exceptions should represent unexpected or invalid conditions.

Exception Propagation

If an exception is not handled where it occurs, it moves up the call stack until Python finds an appropriate `except` block.

If no handler is found, the program terminates and displays the error.

This process is called **exception propagation**.

Best Practices

- Raise the most specific exception available.
 - Include clear, helpful error messages.
 - Validate input before processing it.
 - Catch exceptions only when you can handle them meaningfully.
 - Avoid using generic exceptions when a more specific one is appropriate.
-

Key Takeaways

- The `raise` statement allows your program to generate exceptions.
- Raising an exception immediately interrupts normal program execution.
- Built-in exceptions such as `ValueError` and `TypeError` should be used whenever appropriate.
- Exceptions raised by your code can be caught using `try` and `except`.
- Raising exceptions helps detect problems early and prevents invalid data from producing incorrect results.
- Well-chosen exceptions make programs easier to debug and maintain.

2.6.3 Creating Custom Exceptions (*optional / advanced*)

What Are Custom Exceptions?

Python provides many built-in exceptions such as:

- `ValueError`
- `TypeError`
- `FileNotFoundError`
- `IndexError`

These exceptions cover most situations.

However, when writing larger applications, you may encounter situations where none of the built-in exceptions clearly describe the problem.

In these cases, you can create your own exception classes.

These are called **custom exceptions**.

Why Create Custom Exceptions?

Custom exceptions make programs easier to understand because they describe **exactly** what went wrong.

For example, imagine writing a banking application.

Instead of raising a generic `ValueError`, you could raise:

- `InsufficientFundsError`
- `InvalidAccountError`
- `WithdrawalLimitExceededError`

The exception itself immediately communicates the problem.

Creating a Simple Custom Exception

Custom exceptions are simply classes that inherit from Python's built-in `Exception` class.

<-> Python

```
class NegativeAgeError(Exception):  
    """Raised when a negative age is entered."""  
    pass
```

Although this class contains no code, it is now a valid exception type.

Raising a Custom Exception

Once created, it can be raised just like any built-in exception.

<-> Python

```
class NegativeAgeError(Exception):  
    pass
```

```
age = -5
```

```
if age < 0:  
    raise NegativeAgeError("Age cannot be negative.")
```

Output:

```
NegativeAgeError: Age cannot be negative.
```

Catching a Custom Exception

Custom exceptions are handled using try and except.

<-> Python

```
class NegativeAgeError(Exception):  
    pass
```

```
try:
```

```
    age = -5
```

```
    if age < 0:  
        raise NegativeAgeError("Age cannot be negative.")
```

```
except NegativeAgeError as error:  
    print(error)
```

Output:

```
Age cannot be negative.
```

Creating a More Meaningful Exception

Suppose you're developing a banking application.

<-> Python

```
class InsufficientFundsError(Exception):  
    """Raised when an account balance is too low."""  
    pass
```

Use it within a function.

<-> Python

```
class InsufficientFundsError(Exception):  
    pass
```

```
balance = 100
```

```
withdrawal = 250
```

```
if withdrawal > balance:
```

```
    raise InsufficientFundsError("Insufficient account balance.")
```

Output:

```
InsufficientFundsError: Insufficient account balance.
```

The exception clearly communicates the problem without requiring additional explanation.

Custom Exceptions Can Store Information

Like any class, a custom exception can store additional information.

<-> Python

```
class TemperatureError(Exception):  
  
    def __init__(self, temperature):  
        self.temperature = temperature  
        super().__init__(f"Invalid temperature: {temperature}")
```

Now:

<-> Python

```
raise TemperatureError(-500)
```

Output:

```
TemperatureError: Invalid temperature: -500
```

When Should You Create Custom Exceptions?

Most Python programs never need custom exceptions.

They become useful when:

- Developing large applications.
- Creating reusable libraries.
- Building frameworks.
- Designing APIs.
- Representing business-specific rules.

For smaller scripts, built-in exceptions are usually sufficient.

Django and Custom Exceptions

Django itself defines many custom exceptions.

Examples include:

- `ValidationError`
- `PermissionDenied`
- `ObjectDoesNotExist`
- `Http404`

These specialized exceptions allow Django to respond appropriately to different situations.

As you work with larger frameworks, you'll encounter custom exceptions frequently.

Best Practices

- Create custom exceptions only when built-in exceptions are not descriptive enough.
 - Inherit from Python's `Exception` class.
 - Give exceptions meaningful names ending with `Error`.
 - Provide helpful error messages.
 - Keep custom exceptions focused on a single type of problem.
-

Key Takeaways

- Custom exceptions are user-defined exception classes.
- They inherit from Python's built-in `Exception` class.
- They make large programs easier to understand and maintain.
- Custom exceptions can be raised and handled like built-in exceptions.
- Most beginner and intermediate programs can rely on Python's built-in exceptions.
- Custom exceptions become more valuable as applications grow in size and complexity.

2.7 Modules & Packages

2.7.1 Importing Modules

What Is a Module?

A **module** is a Python file (.py) that contains reusable code.

Instead of writing the same functions or classes in every program, you can place them in a module and import them whenever needed.

For example:

- `math.py`
- `random.py`
- `pathlib.py`
- `datetime.py`

Every one of these is a Python module.

Why Use Modules?

Modules help organize code and promote reuse.

Benefits include:

- Eliminates duplicate code.
- Keeps programs organized.
- Makes code easier to maintain.
- Allows developers to share reusable functionality.
- Provides access to Python's extensive Standard Library.

As your programs grow, separating code into modules becomes an essential programming practice.

Importing an Entire Module

Use the import statement to load an entire module.

<-> Python

```
import math
```

```
print(math.sqrt(25))
```

```
print(math.pi)
```

Output:

```
5.0
```

```
3.141592653589793
```

Notice that the module name (math) precedes the function or variable.

Importing Specific Items

If you only need a few items from a module, you can import them directly.

<-> Python

```
from math import sqrt, pi
```

```
print(sqrt(25))
```

```
print(pi)
```

Output:

```
5.0
```

```
3.141592653589793
```

The module name is no longer required because the imported names are available directly.

Importing with an Alias

Long module names can be shortened using the `as` keyword.

<-> Python

```
import math as m
```

```
print(m.sqrt(81))
```

Output:

```
9.0
```

Aliases are commonly used with modules that have long names.

For example:

<-> Python

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

```
import numpy as np
```

These aliases are widely used throughout the Python community.

Importing Everything (*Generally Not Recommended*)

Python allows every public name in a module to be imported.

<-> Python

```
from math import *
```

```
print(sqrt(16))
```

```
print(pi)
```

Although this works, it is generally discouraged.

Importing everything can:

- Make code harder to read.
- Cause name conflicts.
- Make it unclear where functions originated.

Most professional Python developers avoid this approach.

Using the random Module

The random module generates random values.

<-> Python

```
import random
```

```
number = random.randint(1, 10)
```

```
print(number)
```

Example Output:

7

Each execution may produce a different number.

Using the datetime Module

The datetime module works with dates and times.

<-> Python

```
from datetime import datetime
```

```
today = datetime.now()
```

```
print(today)
```

Example Output:

2026-07-09 10:42:15.123456

Your output will reflect the current date and time.

Finding Available Functions

You can inspect a module using the dir() function.

<-> Python

```
import math
```

```
print(dir(math))
```

Example Output:

```
['acos', 'asin', 'atan', 'ceil', 'cos', 'floor', 'pi', 'pow', 'sqrt', ...]
```

This displays the names available within the module.

Module Documentation

Most modules include built-in documentation.

Use the `help()` function to view it.

<-> Python

```
import math
```

```
help(math)
```

Python displays information about the module, including its functions, constants, and usage.

Commonly Used Standard Library Modules

Module	Purpose
math	Mathematical functions and constants
random	Random numbers and selections
datetime	Dates and times
pathlib	File and directory paths
os	Operating system functions
json	Reading and writing JSON data
csv	Reading and writing CSV files

You'll encounter many of these modules throughout your Python projects.

Best Practices

- Import only the modules or functions you need.
 - Use aliases only when they improve readability.
 - Avoid `from module import *`.
 - Group all import statements at the top of the file.
 - Organize imports according to PEP 8:
 1. Standard Library modules
 2. Third-party modules
 3. Your own project modules
-

Key Takeaways

- A module is a Python file containing reusable code.
- The `import` statement loads a module.
- Use `from module import item` to import specific functions or variables.
- The `as` keyword creates convenient aliases.
- Avoid importing everything from a module with `*`.
- Python's Standard Library provides many powerful modules ready to use.
- Organizing code into modules makes programs cleaner, more reusable, and easier to maintain.

2.7.2 Creating Modules

What Is a Custom Module?

A **custom module** is simply a Python file that you create to hold reusable code.

Instead of placing all of your code into a single file, you can organize related functions, classes, and variables into separate modules.

As your programs grow, creating your own modules makes your code:

- Easier to read.
- Easier to maintain.
- Easier to reuse.
- Better organized.

A Simple Module

Suppose you create a file named **greetings.py**.

<-> Python

```
def say_hello(name):  
    print(f"Hello, {name}!")  
  
def say_goodbye(name):  
    print(f"Goodbye, {name}!")
```

This file is now a Python module.

Using the Module

Suppose another file named **main.py** exists in the same folder.

<-> Python

```
import greetings  
  
greetings.say_hello("Alice")  
greetings.say_goodbye("Alice")
```

Output:

```
Hello, Alice!  
Goodbye, Alice!
```

Python imports the functions from **greetings.py** and makes them available in **main.py**.

Importing Specific Functions

Instead of importing the entire module, you can import only the functions you need.

<-> Python

```
from greetings import say_hello
```

```
say_hello("Bob")
```

Output:

```
Hello, Bob!
```

Only say_hello() is imported.

Importing with an Alias

Modules can also be imported using an alias.

<-> Python

```
import greetings as greet
```

```
greet.say_hello("Charlie")
```

Output:

```
Hello, Charlie!
```

Aliases can make code shorter and easier to read.

Organizing Large Programs

As programs become larger, it is common to separate different responsibilities into different modules.

Example:

Project

`calculator.py`

`database.py`

`reports.py`

`main.py`

Each module has a specific purpose:

- `calculator.py` performs calculations.
- `database.py` handles data storage.
- `reports.py` generates reports.
- `main.py` controls the overall program.

This organization makes large projects much easier to maintain.

Modules Can Contain Classes

Modules are not limited to functions.

They can also contain classes.

For example, **vehicles.py**:

<-> Python

```
class Car:

    def __init__(self, make, model):
        self.make = make
        self.model = model

    def describe(self):
        print(f"{self.make} {self.model}")
```

Now use the class in **main.py**.

<-> Python

```
from vehicles import Car

my_car = Car("Ford", "Mustang")

my_car.describe()
```

Output:

Ford Mustang

This allows related classes to remain together in their own module.

The `__name__` Variable

Every Python module has a built-in variable named `__name__`.

When a file is run directly:

<-> Python

```
print(__name__)
```

Output:

```
__main__
```

When the same file is imported into another program, `__name__` becomes the module's name.

This behavior allows a module to distinguish between being **executed directly** and **imported by another program**.

Using `if __name__ == "__main__":`

A common Python pattern is:

<-> Python

```
def greet():  
    print("Hello!")
```

```
if __name__ == "__main__":  
    greet()
```

If the file is executed directly:

```
Hello!
```

If the file is imported into another program, the `greet()` function is **not** executed automatically.

This makes modules reusable without producing unwanted output.

Organizing Your Own Projects

As you've progressed through your Python studies, you've already begun organizing your work into separate project folders, such as:

- **Alien_Invasion**
- **Downloading_Data**
- **Working_with_API**
- **Django**
- **Data_Visualization**

Within each project, creating separate modules for different responsibilities keeps the code clean and manageable.

For example, in the **Alien Invasion** project, classes such as [Ship](#), [Alien](#), [Bullet](#), and [Settings](#) each have their own module. This organization makes the program easier to understand, test, and maintain.

Best Practices

- Give modules descriptive names.
- Keep each module focused on a single responsibility.
- Place related functions and classes together.
- Avoid creating modules that become excessively large.
- Use `if __name__ == "__main__":` when appropriate to separate reusable code from test code.

Key Takeaways

- A custom module is a Python file containing reusable code.
- Modules can contain functions, classes, variables, and constants.
- Import modules using the `import` statement.
- The `__name__` variable identifies whether a file is being run directly or imported.
- The `if __name__ == "__main__":` pattern allows modules to be both reusable and executable.
- Organizing code into modules makes programs cleaner, easier to maintain, and more scalable.

2.7.3 Packages

What Is a Package?

As programs grow larger, even modules can become difficult to organize.

A **package** is a collection of related modules stored together inside a directory.

Think of it this way:

- A **module** is a single Python file.
- A **package** is a folder that contains one or more related modules.

Packages help organize larger applications into logical groups.

Why Use Packages?

Packages provide several benefits:

- Organize related modules.
- Reduce clutter in large projects.
- Make code easier to maintain.
- Prevent naming conflicts.
- Allow code to be reused across multiple projects.

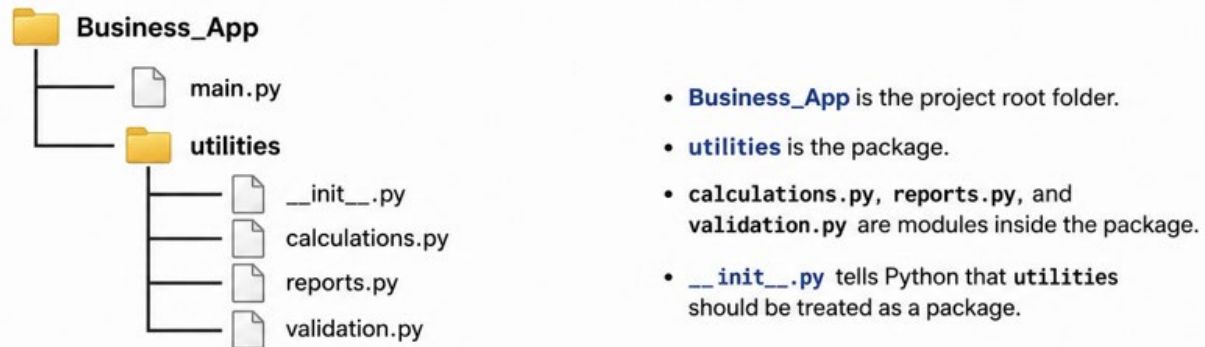
Large applications such as Django are built almost entirely from packages.

Example Package Structure

Suppose you're building a simple business application.

The project might be organized like this:

(A simple folder diagram works well here.)



In this example:

- **utilities** is the package.
- **calculations.py**, **reports.py**, and **validation.py** are modules inside the package.

The `__init__.py` File

Traditionally, a package contains a file named:

`__init__.py`

This file tells Python that the directory should be treated as a package.

Modern versions of Python can create packages without it in many situations, but you'll still encounter `__init__.py` frequently, especially in existing projects and third-party libraries.

The file may be:

- Completely empty.
 - Used for package initialization.
 - Used to control what is imported from the package.
-

Importing an Entire Module from a Package

Suppose **calculations.py** contains:

<-> Python

```
def add(a, b):  
    return a + b
```

You can import it like this:

<-> Python

```
from utilities import calculations
```

```
print(calculations.add(5, 3))
```

Output:

8

Importing a Specific Function

Instead of importing the entire module:

<-> Python

```
from utilities.calculations import add
```

```
print(add(5, 3))
```

Output:

8

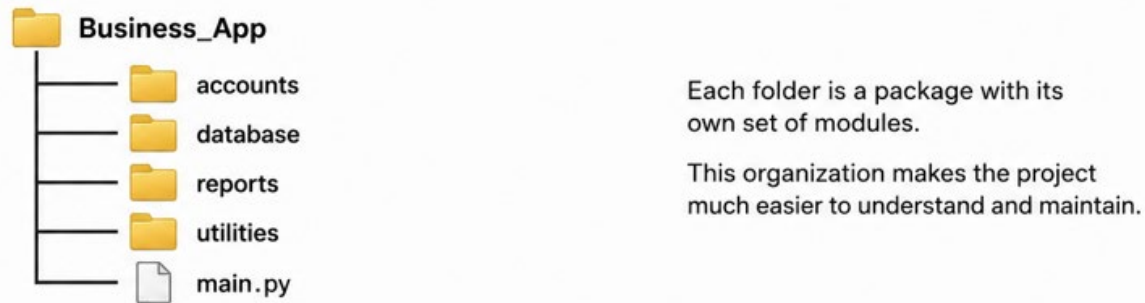
Only the `add()` function is imported.

Multiple Packages

Large projects often contain several packages.

A project might be organized like this:

(A simple folder diagram works well here.)



Each package has a specific responsibility.

This organization makes the project much easier to understand.

Packages in Your Python Projects

Although you may not have created many packages yourself yet, you've already used them extensively.

For example:

<-> Python

```
import matplotlib.pyplot as plt
```

Here:

- matplotlib is a package.
- pyplot is a module within that package.

Another example:

<-> Python

```
from pathlib import Path
```

Here:

- pathlib is a module from Python's Standard Library.
- Path is a class defined inside that module.

Packages and modules work together to organize Python code.

Packages in Django

Django provides an excellent example of packages.

When you created the **Learning Log** project, you worked with packages such as:

- `django.contrib`
- `django.urls`
- `django.db`
- `django.shortcuts`

Each package groups related functionality together.

Your own Django project also contains packages.

For example, each Django application (such as `learning_logs`) is organized as a package containing modules like:

- `models.py`
- `views.py`
- `admin.py`
- `urls.py`
- `forms.py`

This modular organization is one reason Django scales so well.

Packages vs. Modules

Module	Package
A single .py file	A directory containing related modules
Holds functions, classes, and variables	Organizes multiple modules
Imported directly	Contains one or more modules
Smallest reusable unit	Higher-level organizational structure

Best Practices

- Group related modules into packages.
 - Give packages descriptive names.
 - Keep each package focused on a single responsibility.
 - Avoid placing unrelated modules in the same package.
 - Organize projects so they remain easy to navigate as they grow.
-

Key Takeaways

- A package is a directory containing related Python modules.
- Packages organize large projects into logical sections.
- Modules are individual Python files within a package.
- The `__init__.py` file traditionally identifies a package.
- Packages help keep code organized, reusable, and maintainable.
- Django and many third-party libraries are organized as packages.

2.7.4 The Python Standard Library

What Is the Python Standard Library?

The **Python Standard Library** is a collection of modules that are included with every standard Python installation.

Rather than writing common functionality from scratch, you can import these modules and immediately begin using them in your programs.

The Standard Library is one of Python's greatest strengths because it provides powerful, well-tested tools for a wide variety of programming tasks.

Python's philosophy is often described as:

"Batteries Included."

This means that Python comes with many useful tools already built in.

Why Use the Standard Library?

The Standard Library allows you to:

- Save development time.
- Write less code.
- Use well-tested, reliable modules.
- Focus on solving problems rather than reinventing common functionality.

Instead of creating your own date library, file library, or random number generator, you simply import the appropriate module.

Importing a Standard Library Module

Standard Library modules are imported just like any other module.

<-> Python

```
import math
```

```
print(math.sqrt(64))
```

Output:

```
8.0
```

Python already includes the math module, so no installation is required.

Common Standard Library Modules

Some of the most frequently used modules include:

Module	Purpose
math	Mathematical functions and constants
random	Generate random numbers and make random selections
datetime	Work with dates and times
pathlib	Work with file and directory paths
os	Interact with the operating system
sys	Access Python interpreter information
json	Read and write JSON data
csv	Read and write CSV files
statistics	Perform common statistical calculations
collections	Specialized container data types

The math Module

The math module provides many mathematical functions.

<-> Python

```
import math
```

```
print(math.pi)
```

```
print(math.sqrt(49))
```

```
print(math.factorial(5))
```

Output:

```
3.141592653589793
```

```
7.0
```

```
120
```

The random Module

The random module generates random values.

<-> Python

```
import random
```

```
print(random.randint(1, 100))
```

Example Output:

```
42
```

Each execution may produce a different result.

The datetime Module

The datetime module works with dates and times.

<-> Python

```
from datetime import datetime
```

```
print(datetime.now())
```

Example Output:

```
2026-07-09 14:35:21.456789
```

The output reflects the current date and time on the computer running the program.

The pathlib Module

The pathlib module provides a modern way to work with files and directories.

<-> Python

```
from pathlib import Path
```

```
current_folder = Path.cwd()
```

```
print(current_folder)
```

Example Output:

```
F:\Python_Study\Python_Crash_Course
```

This module is generally preferred over manually building file paths.

The json Module

Many applications exchange information using the JSON format.

Python includes built-in support for JSON.

<-> Python

```
import json
```

```
person = {  
    "name": "Alice",  
    "age": 30  
}
```

```
json_data = json.dumps(person)
```

```
print(json_data)
```

Output:

```
{"name": "Alice", "age": 30}
```

The csv Module

The csv module makes it easy to work with comma-separated value (CSV) files.

<-> Python

```
import csv
```

```
with open("employees.csv") as file:
```

```
    reader = csv.reader(file)
```

```
    for row in reader:
```

```
        print(row)
```

This module is commonly used for importing and exporting spreadsheet data.

Finding Standard Library Modules

Python provides documentation for every Standard Library module.

You can view information directly in Python.

<-> Python

```
import math
```

```
help(math)
```

You can also list the contents of a module.

<-> Python

```
import math
```

```
print(dir(math))
```

These tools are valuable when learning unfamiliar modules.

Standard Library vs. Third-Party Libraries

It is important to distinguish between modules included with Python and those that must be installed separately.

Standard Library	Third-Party Library
Included with Python	Installed separately
Ready to use immediately	Usually installed with pip
Maintained as part of Python	Developed by the Python community or organizations

Examples:

Standard Library

- math
- random
- datetime
- json
- csv

Third-Party Libraries

- pandas
- numpy
- matplotlib
- plotly
- Django

Best Practices

- Check the Standard Library before writing functionality from scratch.
 - Become familiar with the most commonly used modules.
 - Read the official documentation when exploring a new module.
 - Import only the modules you need.
 - Prefer Standard Library solutions whenever they meet your requirements.
-

Key Takeaways

- The Python Standard Library is included with every Python installation.
- It contains hundreds of modules for common programming tasks.
- Standard Library modules require no additional installation.
- Modules such as math, random, datetime, pathlib, json, and csv are widely used.
- Third-party libraries, such as pandas and Django, must be installed separately.
- Learning the Standard Library allows you to write cleaner, more efficient Python programs with less code.

3 VISUAL STUDIO CODE

3.1 Explorer

Overview

The **Explorer** is VS Code's file manager. It displays your project's folders, files, and open editors, making it easy to navigate and organize your work.

When developing Python applications, you'll spend much of your time using the Explorer to create files, open projects, and manage your folder structure.

Key Features

- Open project folders
- Create files and folders
- Rename, move, and delete files
- Expand and collapse folders
- View open editors
- Display Git file status

Opening the Explorer

Open the Explorer by:

- Clicking the **Explorer** icon in the Activity Bar.
- Pressing **Ctrl + Shift + E**.

The Explorer displays the currently opened project and its folder structure.

Working with Files

Common tasks include:

- Creating new Python files (.py)
- Creating project folders
- Renaming files
- Moving files between folders
- Deleting files you no longer need

Keeping related files organized makes projects much easier to manage.

Project Organization

As projects grow, organize files into logical folders.

For example:

(PNG Folder Structure Here)

This approach keeps projects clean and makes navigation much easier.

Best Practices

- Open the **project folder**, not individual files.
 - Use descriptive file and folder names.
 - Keep related files together.
 - Collapse folders you're not actively using.
 - Remove unused files from your project.
-

Developer's Note

Throughout the **Python Crash Course**, we always opened the entire project folder (such as **Alien_Invasion**, **Downloading_Data**, or **Learning_Log**) rather than individual files. This allowed Git, IntelliSense, debugging, and the integrated terminal to work together seamlessly.

Key Takeaways

- The Explorer is VS Code's primary navigation tool.
- It displays your project's complete folder structure.
- Files and folders can be created and managed directly from the Explorer.
- A well-organized project is easier to navigate, debug, and maintain.

3.2 Terminal

Overview

The **Integrated Terminal** allows you to run commands without leaving VS Code. Whether you're executing Python scripts, managing virtual environments, installing packages, or using Git, the terminal keeps everything in one place.

Throughout your Python projects, the terminal became one of the most frequently used tools.

Key Features

- Run Python programs
 - Execute Git commands
 - Create and activate virtual environments
 - Install packages with pip
 - Navigate folders
 - Run Django development servers
-

Opening the Terminal

Open the Integrated Terminal by:

- Selecting **Terminal → New Terminal**
- Pressing **Ctrl + `** (backtick)

The terminal opens at the bottom of the VS Code window.

Common Commands

<-> Python

```
python hello.py
```

Run a Python program.

<-> Python

```
pip install requests
```

Install a package.

<-> Python

```
git status
```

View the current Git repository status.

<-> Python

```
cd Project_Name
```

Change to another directory.

Working with Virtual Environments

When using a virtual environment, activate it before running your project.

Windows PowerShell:

<-> Python

```
.\11_env\Scripts\Activate.ps1
```

Once activated, the environment name appears at the beginning of the terminal prompt.

Best Practices

- Open the terminal from within your project folder.
 - Activate your virtual environment before running Python projects.
 - Run git status frequently.
 - Read error messages carefully—they often tell you exactly what went wrong.
 - Keep the terminal open while developing.
-

Developer's Note

During **Learning Log**, we typically kept two terminals open:

- One running the Django development server.
- One for Git commands, package installation, and general project work.

This workflow allowed us to develop, test, and commit changes without interrupting the running server.

Key Takeaways

- The Integrated Terminal provides command-line access inside VS Code.
- Use it to run Python programs, Git commands, and package installations.
- Activate your virtual environment before working on a project.
- Keeping the terminal open improves efficiency during development.

3.3 Settings

Overview

VS Code provides a wide range of settings that allow you to customize the editor to match your workflow. Adjusting a few key settings can improve readability, automate repetitive tasks, and make development more efficient.

Most settings can be changed through the graphical interface or by editing the [settings.json](#) file.

Key Settings

Some of the most useful settings for Python development include:

- Auto Save
 - Editor Font Size
 - Terminal Font Size
 - Word Wrap
 - Color Theme
 - File Icon Theme
 - Tab Size
-

Opening Settings

Open Settings by:

- Selecting **File → Preferences → Settings**
- Pressing **Ctrl + ,** (Ctrl + Comma)

To edit the JSON configuration directly:

1. Open **Settings**.
 2. Click the **Open Settings (JSON)** icon in the upper-right corner.
-

Recommended Settings

For our Python development environment, we found these settings particularly useful:

Setting	Recommendation
Auto Save	<code>afterDelay</code>
Editor Font Size	18
Terminal Font Size	18
Word Wrap	On
Tab Size	4 spaces

Best Practices

- Increase the font size to reduce eye strain.
 - Enable Auto Save to avoid losing changes.
 - Leave the Python indentation at four spaces.
 - Customize VS Code to improve comfort and productivity.
 - Change only settings you understand.
-

Developer's Note

One of the first improvements we made was increasing both the editor and terminal font sizes. Combined with Auto Save, these small changes made coding sessions much more comfortable and allowed us to focus on writing code instead of managing files.

Key Takeaways

- VS Code is highly customizable.
- Most settings are available through the Settings window.
- `settings.json` provides direct access to configuration options.
- A few simple adjustments can significantly improve your development experience.
- Configure VS Code to fit your workflow, not someone else's.

3.4 Keyboard Shortcuts

Overview

Keyboard shortcuts are one of the easiest ways to improve productivity in VS Code. Instead of navigating menus, common tasks can be performed with a few keystrokes, allowing you to keep your hands on the keyboard and stay focused on coding.

Frequently Used Shortcuts

Shortcut	What it Does
Ctrl + Shift + P	Open the Command Palette
Ctrl + Shift + E	Open the Explorer
Ctrl + `	Open or close the Integrated Terminal
Ctrl + Shift + F	Search the current project
Shift + Alt + F	Format the current file
Ctrl + /	Comment or uncomment selected lines
Alt + ↑ / ↓	Move the current line or selected block
Shift + Alt + ↑ / ↓	Duplicate the current line or selected block
Ctrl + D	Select the next occurrence of the current word
Ctrl + Shift + K	Delete the current line
Ctrl + P	Quickly open a file by name
F2	Rename a symbol throughout the project

Best Practices

- Learn a few shortcuts at a time rather than trying to memorize them all.
- Use keyboard shortcuts for repetitive tasks to improve efficiency.
- Customize shortcuts if they better fit your workflow.
- Practice regularly until your most-used shortcuts become second nature.

Developer's Note

The shortcuts we used most often throughout the **Python Crash Course** were **Ctrl + `** to open the terminal, **Ctrl + /** to comment code, **Shift + Alt + F** to format files, and **Ctrl + Shift + E** to return to the Explorer. These quickly became part of our everyday workflow and significantly reduced the need to navigate menus.

Key Takeaways

- Keyboard shortcuts speed up common development tasks.
- Focus on learning the shortcuts you use every day.
- VS Code allows shortcuts to be customized.
- A small set of well-practiced shortcuts can noticeably improve productivity.

4 GIT & GitHub

4.1 Daily Workflow

Overview

Git provides a reliable way to track changes to your code, while GitHub stores your repositories online for backup, collaboration, and version control. Following a consistent daily workflow helps keep projects organized and ensures that every meaningful change is recorded.

Typical Workflow

1. Open your project in VS Code.
2. Activate the virtual environment (if required).
3. Write and test your code.
4. Check the repository status.

<-> Python

```
git status
```

5. Stage your changes.

<-> Python

```
git add .
```

6. Create a descriptive commit.

<-> Python

```
git commit -m "Describe the changes made"
```

7. Push your commits to GitHub (if using a remote repository).

<-> Python

```
git push
```

Best Practices

- Commit after completing a logical unit of work.
 - Write clear, descriptive commit messages.
 - Run git status before and after every commit.
 - Test your code before committing.
 - Commit frequently rather than saving many unrelated changes in one commit.
-

Developer's Note

Throughout the **Python Crash Course**, we developed the habit of checking git status constantly. Nearly every work session ended with a clean working tree before shutting down for the day. This routine made it easy to resume work the following morning and provided confidence that every completed feature had been safely committed.

Key Takeaways

- Follow the same Git workflow for every project.
- Commit small, meaningful changes.
- Check git status frequently.
- Use descriptive commit messages.
- End each work session with a clean repository.

4.2 Common Commands

Overview

Git provides a relatively small set of commands that you'll use repeatedly during everyday development. While Git includes hundreds of commands and options, mastering a handful of core commands is enough to manage most personal projects.

Frequently Used Commands

Command	Purpose
<code>git status</code>	Display the current repository status
<code>git add .</code>	Stage all modified and new files
<code>git add filename.py</code>	Stage a specific file
<code>git commit -m "message"</code>	Commit staged changes
<code>git log --oneline</code>	Display a condensed commit history
<code>git diff</code>	Show unstaged changes
<code>git restore filename.py</code>	Discard changes to a file
<code>git commit --amend -m "message"</code>	Update the most recent commit message
<code>git push</code>	Upload commits to GitHub
<code>git pull</code>	Download the latest changes from GitHub

Typical Session

<-> Python

```
git status
```

```
git add .
```

```
git commit -m "Add user authentication"
```

```
git push
```

Best Practices

- Run `git status` before staging changes.
 - Write commit messages that clearly describe the work completed.
 - Commit related changes together.
 - Review changes with `git diff` before committing when necessary.
 - Push completed work to GitHub regularly.
-

Developer's Note

Although Git offers many advanced commands, we relied on a small core set throughout the **Python Crash Course**. By consistently using `git status`, `git add`, `git commit`, and `git log`, version control quickly became a natural part of the development process rather than an afterthought.

Key Takeaways

- A small number of Git commands handle most daily tasks.
- `git status` should become part of every coding session.
- Make small, descriptive commits.
- Push completed work to GitHub regularly.
- Learn additional Git commands as your projects become more complex.

4.3 GitHub

GitHub is an online platform that hosts Git repositories. It provides a secure location to store your projects, back up your code, track development history, and collaborate with other developers.

For individual developers, GitHub also serves as a professional portfolio that showcases coding projects and commit history.

Why Use GitHub?

GitHub allows you to:

- Back up your projects online.
- Access your code from multiple computers.
- Share repositories with others.
- Track the history of every commit.
- Demonstrate your development experience to employers.

Typical Workflow

After committing changes locally:

<-> Python

`git push`

Git uploads your commits to the remote GitHub repository.

When working from another computer or after changes have been made remotely:

<-> Python

`git pull`

Git downloads the latest changes to your local repository.

Repository Essentials

Every GitHub repository should include:

- A meaningful repository name.
- A descriptive **README.md** file.
- A **.gitignore** file to exclude unnecessary files.
- Regular commits documenting project progress.

Best Practices

- Push completed work to GitHub regularly.
- Keep your README updated as the project evolves.
- Use meaningful repository names.
- Avoid committing temporary files, virtual environments, or sensitive information.
- Treat your GitHub profile as part of your professional résumé.

Developer's Note

As the Python Crash Course progressed, GitHub became more than a backup location—it became a record of growth as a developer. Each completed project, commit history, and repository demonstrated steady progress and provided tangible evidence of new skills. Building a consistent GitHub history proved just as valuable as completing the projects themselves.

Key Takeaways

- GitHub hosts your Git repositories online.
- `git push` uploads local commits.
- `git pull` downloads remote changes.
- A well-maintained GitHub profile showcases your work.
- Consistent commits tell the story of your development as a programmer.

4.4 Branching (*future*)

Overview

A branch is an independent line of development within a Git repository. Branches allow developers to work on new features, bug fixes, or experiments without affecting the main project.

While branching is essential for team development, it is not required for every personal project.

Why Branches Are Useful

Branches allow you to:

- Develop new features safely.
 - Fix bugs without disrupting stable code.
 - Experiment with new ideas.
 - Work on multiple features simultaneously.
 - Collaborate with other developers.
-

Our Approach

Throughout the **Python Crash Course**, we worked exclusively on the **main** branch.

This simplified the learning process by allowing us to focus on Python, Git fundamentals, and project development without introducing additional Git concepts.

As future projects become larger or involve multiple contributors, feature branches will become an important part of the development workflow.

Best Practices

- Keep the **main** branch stable.
 - Create feature branches for significant new work.
 - Merge completed features back into **main** after testing.
 - Delete branches that are no longer needed.
-

Developer's Note

Learning Git and Python at the same time provided plenty of new concepts to master. Staying on the **main** branch allowed us to build confidence with version control before introducing branching strategies. As future projects become more complex, branching will be a natural next step.

Key Takeaways

- Branches isolate new development from the main code base.
- Personal projects can often begin on the **main** branch.
- Feature branches become more valuable as projects grow.
- Branching is an important skill to learn when collaborating with others.

5 PYTHON CRASH COURSE PROJECTS

5.1 Alien Invasion

Purpose

The **Alien Invasion** project is the first large-scale application developed in *Python Crash Course*. Unlike the earlier chapters, which focus on individual programming concepts, this project demonstrates how those concepts work together to create a complete, interactive application.

Building the game introduces many of the core practices used in professional software development, including project organization, object-oriented programming, event-driven programming, game loops, collision detection, state management, and incremental feature development.

Although the final application is a simple arcade-style game, the architecture and development techniques extend well beyond game development and apply to desktop applications, simulations, automation tools, and many other Python projects.

Skills Demonstrated

Alien Invasion demonstrates the successful application of the following development skills:

- Organizing a multi-file Python project
 - Designing classes with single responsibilities
 - Handling keyboard events
 - Implementing a continuous game loop
 - Rendering graphics using the Pygame library
 - Detecting collisions between game objects
 - Managing collections of sprites
 - Tracking scores, levels, and game state
 - Building features incrementally while maintaining a working code base
-

Technologies Used

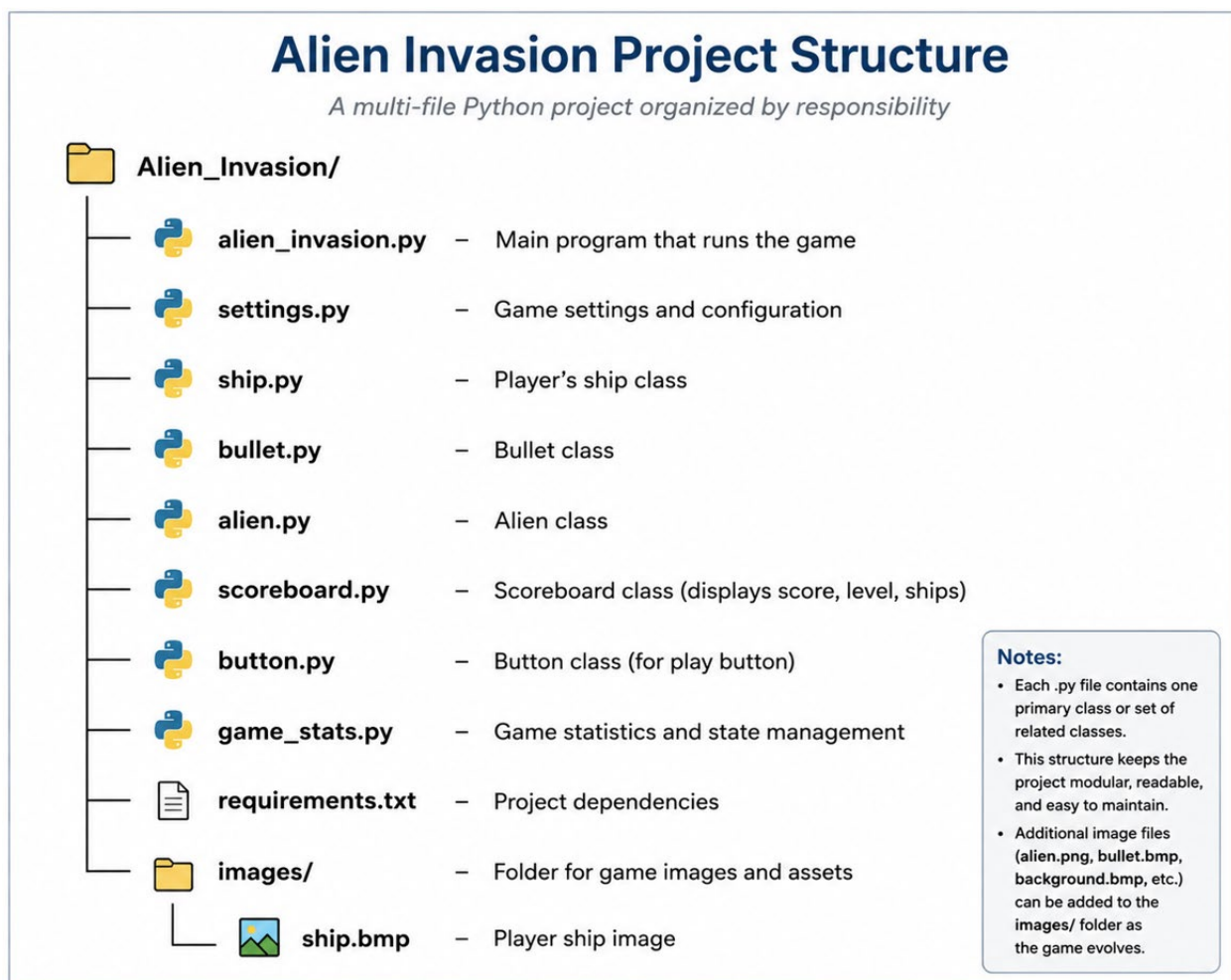
The project utilizes several development tools and libraries.

Component	Purpose
Python	Core programming language
Pygame	Graphics, sound, input handling, and game framework
VS Code	Development environment
Git	Version control throughout development

Project Structure

Alien Invasion is organized into multiple Python modules rather than a single source file. Each module has a specific responsibility, making the application easier to understand, maintain, debug, and expand.

Project Structure Diagram



Concepts Applied

Alien Invasion brings together many of the concepts introduced in earlier chapters.

- Classes and objects
- Methods and attributes
- Object composition
- Lists and sprite groups
- Loops
- Conditional logic
- Event handling
- Modular program design
- Incremental debugging
- Version control with Git

Rather than demonstrating these concepts individually, the project integrates them into a complete, working application.

Lessons Learned

Developing Alien Invasion reinforces several important software development principles.

- Build one feature at a time.
- Test frequently.
- Commit working code to Git regularly.
- Separate responsibilities into individual classes.
- Keep methods focused on a single task.
- Read error messages carefully before changing code.
- Small improvements accumulate into complete applications.

These practices become increasingly valuable as projects grow in size and complexity.

Developer's Note

Alien Invasion was the point where Python stopped feeling like a collection of individual exercises and started feeling like software development. Each chapter added another feature—movement, bullets, aliens, scoring, levels, and game state—until a complete application emerged.

Equally important, this project reinforced disciplined development practices such as making frequent Git commits, testing after every change, and organizing code into well-defined modules. Those habits have carried forward into every project that followed.

Key Takeaways

- Alien Invasion is the first complete application developed in this handbook.
- The project combines programming concepts learned in earlier chapters into a single application.
- Pygame provides the framework for graphics, animation, and user input.
- Object-oriented programming makes larger applications easier to organize and maintain.
- Incremental development and regular Git commits simplify debugging and improve code quality.
- The architectural patterns used throughout the project apply to many types of Python applications, not just games.

5.2 Data Visualization Projects

Purpose

The **Data Visualization** projects used Python to transform numerical data into clear, readable charts and graphs. These exercises demonstrated how visual presentation can reveal patterns, comparisons, and trends that are difficult to recognize in raw data alone.

The projects included plotted number sequences, random walks, and simulated dice rolls. Together, they introduced the fundamentals of creating, formatting, and interpreting visual output with Python.

Skills Demonstrated

The Data Visualization projects demonstrate the application of the following development skills:

- Generating and organizing numerical data
- Plotting data with Matplotlib
- Creating interactive charts with Plotly
- Customizing titles, labels, markers, and axis values
- Applying built-in color maps
- Simulating random events
- Comparing expected and observed results
- Saving visualizations as image files
- Separating data-generation logic from presentation logic

Technologies Used

Component	Purpose
Python	Core programming language
Matplotlib	Static charts and plotted visualizations
Plotly	Interactive charts and browser-based output
VS Code	Development environment
Git	Version control throughout development

Projects Completed

Plotting Squares

A sequence of square numbers was plotted using Matplotlib. The project demonstrated how to define data points, display them on a chart, customize the chart's appearance, and save the completed visualization.

Random Walk

The Random Walk project generated a path from a series of randomly selected movements. It demonstrated how repeated small decisions can produce complex and unpredictable patterns.

The project also reinforced the value of separating responsibilities by placing the data-generation logic in one class and the visualization logic in a separate program.

Dice Simulation

The Dice Simulation project used Python to model repeated dice rolls and calculate how often each possible result occurred.

Plotly was then used to display the results as an interactive bar chart. The simulation demonstrated how programming can test probability concepts by repeating an experiment thousands of times and analyzing the results.

Concepts Applied

- Lists and ranges
 - Loops and conditional logic
 - Classes and methods
 - Random number generation
 - Mathematical calculations
 - Data aggregation
 - Chart formatting
 - Color mapping
 - Interactive visualization
 - File output
 - Modular program design
-

Lessons Learned

Data visualization is not only about producing attractive charts. A useful visualization must present information clearly and help the reader understand what the data represents.

These projects reinforced the importance of:

- Selecting the appropriate chart for the data
 - Adding meaningful titles and axis labels
 - Keeping visual output readable
 - Checking the underlying data before plotting it
 - Testing visual changes incrementally
 - Separating data processing from visual presentation
-

Developer's Note

The Data Visualization projects showed that Python could do more than calculate and print results. It could turn information into something immediately understandable.

The Random Walk project was especially useful because each execution produced a different result while following the same rules. The dice simulations demonstrated the opposite principle: individual results were unpredictable, but repeated trials produced recognizable statistical patterns.

Key Takeaways

- Matplotlib is well suited for static charts and detailed formatting.
- Plotly provides interactive visualizations that can be viewed in a browser.
- Visualizations make patterns and comparisons easier to understand.
- Random simulations can be used to explore probability and statistical behavior.
- Clear labels and formatting are as important as the underlying calculations.
- Separating data generation from presentation produces cleaner, more maintainable code.

5.3 Downloading Data

Purpose

The **Downloading Data** projects introduced the process of obtaining real-world data from external sources and transforming it into meaningful visualizations. Rather than using manually created datasets, these projects worked with actual weather and earthquake data to demonstrate how Python can process and analyze information collected from outside systems.

These exercises emphasized the importance of data preparation before visualization, including reading files, validating data, and handling formatting differences.

Skills Demonstrated

The Downloading Data projects demonstrate the application of the following development skills:

- Reading CSV data files
- Parsing and formatting dates
- Extracting selected data fields
- Handling missing or inconsistent data
- Creating line charts with Matplotlib
- Visualizing geographic data
- Working with GeoJSON datasets
- Organizing projects into reusable modules
- Using Git throughout development

Technologies Used

Component	Purpose
Python	Core programming language
CSV Module	Reading comma-separated data files
Matplotlib	Weather data visualization
Plotly	Interactive earthquake maps
GeoJSON	Geographic data format
VS Code	Development environment
Git	Version control throughout development

Projects Completed

Weather Data Visualization

Weather data from Sitka, Alaska and Death Valley, California was imported from CSV files and displayed as line graphs. These projects demonstrated how climate data can be analyzed and compared visually while reinforcing the importance of handling dates and missing values correctly.

Global Earthquake Map

Earthquake information was downloaded in GeoJSON format from the United States Geological Survey (USGS). Plotly was then used to create an interactive world map that displayed earthquake locations and magnitudes.

The completed project demonstrated how large geographic datasets can be transformed into meaningful visual presentations.

Concepts Applied

- Reading external data files
 - CSV file processing
 - Date parsing
 - Data validation
 - Exception handling
 - Geographic datasets
 - Interactive mapping
 - Data visualization
 - Modular programming
 - File management
-

Lessons Learned

Working with real-world data is often more challenging than working with examples created specifically for learning. Data may contain missing values, unexpected formats, or inconsistent information that must be handled before it can be analyzed.

These projects reinforced the importance of:

- Validating imported data
 - Reading documentation carefully
 - Cleaning data before visualization
 - Handling exceptions gracefully
 - Testing visual output after each modification
 - Building visualizations incrementally
-

Developer's Note

These projects marked the transition from working with simulated data to working with information collected from real-world sources. The Weather Data projects demonstrated how raw data can reveal trends when presented visually, while the Global Earthquake Map highlighted the power of combining publicly available datasets with interactive visualization tools.

This chapter also reinforced that successful programming depends as much on understanding and preparing data as it does on writing code.

Key Takeaways

- Python can read and process real-world datasets from external sources.
- CSV and GeoJSON are common formats used for storing structured data.
- Data validation is an essential step before creating visualizations.
- Interactive maps provide an effective way to display geographic information.
- Well-prepared data produces clearer and more meaningful visualizations.
- Real-world projects often require data cleaning before analysis can begin.

5.4 Working with API's

Purpose

The **Working with APIs** projects introduced the process of retrieving live data from online services instead of reading information from local files. Application Programming Interfaces (APIs) allow software applications to exchange data, making them an essential part of modern software development.

These projects demonstrated how Python can request information from web services, process JSON data, and present the results in meaningful visual formats.

Skills Demonstrated

The Working with APIs projects demonstrate the application of the following development skills:

- Connecting to REST APIs
- Sending HTTP requests
- Processing JSON responses
- Extracting and filtering returned data
- Handling API response errors
- Creating interactive visualizations with Plotly
- Working with web-based data sources
- Using Git throughout development

Technologies Used

Component	Purpose
Python	Core programming language
Requests	Sending HTTP requests to APIs
JSON	Processing returned data
Plotly	Interactive data visualization
GitHub API	Repository statistics
Hacker News API	Top story retrieval
VS Code	Development environment
Git	Version control throughout development

Projects Completed

GitHub Repository Analysis

Repository information was retrieved from the GitHub API and displayed using interactive Plotly charts. The project demonstrated how API data can be filtered, organized, and presented visually while providing direct links to individual repositories.

Hacker News Top Stories

The Hacker News API was used to retrieve the most popular news stories from the platform. Story titles, links, and discussion statistics were processed and displayed using interactive visualizations, demonstrating how Python can work with frequently changing online data.

Concepts Applied

- REST APIs
 - HTTP requests
 - JSON processing
 - Data filtering
 - Lists and dictionaries
 - Error handling
 - Interactive visualization
 - External data integration
 - Modular programming
 - Version control with Git
-

Lessons Learned

APIs provide access to constantly changing information without requiring manual downloads. Learning to retrieve, process, and present this data is a valuable skill used in many software development projects.

These projects reinforced the importance of:

- Reading API documentation carefully
- Validating returned data
- Handling request failures gracefully
- Separating data retrieval from visualization
- Testing applications with live data
- Using version control throughout development

Developer's Note

Working with APIs demonstrated how Python applications can interact with services across the internet. Instead of relying on static datasets, the projects retrieved current information directly from online sources and transformed it into useful visualizations.

This chapter highlighted how APIs connect software systems together and provided a practical introduction to techniques used throughout modern web development and data analysis.

Key Takeaways

- APIs allow applications to communicate and exchange data.
- JSON is one of the most common formats used for API responses.
- Python's requests library simplifies communication with web services.
- Live data can be transformed into meaningful visualizations using Plotly.
- Proper error handling improves application reliability.
- API integration is a fundamental skill in modern software development.

5.5 Django - Learning Log

Purpose

The **Learning Log** project introduced web application development using the Django framework. Unlike previous projects, which focused on standalone Python programs, this project demonstrated how Python can be used to build dynamic, database-driven web applications.

The project combined many of the programming concepts learned throughout the book into a complete web application featuring user authentication, database management, templates, forms, and data relationships.

Skills Demonstrated

The Learning Log project demonstrates the application of the following development skills:

- Building web applications with Django
- Creating and configuring Django projects and apps
- Working with models and databases
- Managing data relationships
- Creating HTML templates
- Using forms for user input
- Implementing user authentication
- Protecting user data
- Using Git throughout development

Technologies Used

Component	Purpose
Python	Core programming language
Django	Web application framework
SQLite	Database management
HTML	Web page structure
Bootstrap	User interface styling
VS Code	Development environment
Git	Version control throughout development

Projects Completed

Learning Log Web Application

A complete Django web application was developed to allow users to create topics and record journal entries associated with each topic.

The application included user registration, login and logout functionality, secure database access, editable entries, and ownership protection so that each user could access only their own information.

Throughout development, the application was expanded incrementally while maintaining a fully functional code base through regular testing and Git commits.

Concepts Applied

- Django project structure
 - Applications (Apps)
 - Models
 - Views
 - URL routing
 - Templates
 - Forms
 - User authentication
 - Database migrations
 - CRUD operations
 - Version control with Git
-

Lessons Learned

Developing Learning Log demonstrated how modern web applications are built by combining multiple components into a cohesive system.

These projects reinforced the importance of:

- Planning application structure before coding
- Keeping applications modular
- Separating data, logic, and presentation
- Testing changes frequently
- Applying database migrations carefully
- Using version control throughout development

Developer's Note

Learning Log represented a significant milestone because it brought together nearly everything learned throughout *Python Crash Course*. Instead of writing individual scripts, the project demonstrated how multiple technologies work together to create a complete web application.

Completing this project also reinforced the value of disciplined development practices, including incremental feature development, regular testing, and consistent Git commits. These same practices continue to serve as the foundation for larger software projects.

Key Takeaways

- Django simplifies the development of database-driven web applications.
- Models, views, templates, and URLs work together to create dynamic websites.
- User authentication protects application data and personal information.
- Database migrations provide a controlled method for updating application data structures.
- Modular application design makes projects easier to maintain and expand.
- Learning Log demonstrated how individual Python concepts combine into a complete web application.

6 DATA VISUALIZATION

6.1 CSV Files

What is a CSV File?

A **CSV (Comma-Separated Values)** file is a plain text file used to store tabular data. Each line in the file represents one record, and each value is separated by a comma.

Think of it as a spreadsheet saved as plain text.

Example:

```
STATION,NAME,DATE,TAVG,TMAX,TMIN
USW00025333,Sitka,2021-07-01,57,61,53
USW00025333,Sitka,2021-07-02,56,60,52
```

6.2 Reading CSV Files

Import the Required Modules

<-> Python

```
from pathlib import Path
import csv
```

Why?

- Path is used to locate and read the file.
- csv understands how to split comma-separated values into usable data.
-

Open the File

<-> Python

```
path = Path('weather_data/sitka_weather_07-2021_simple.csv')
lines = path.read_text(encoding='utf-8').splitlines()
```

What happens?

- Create a path to the CSV file.
 - Read the file.
 - Split the file into individual lines.
 - Store those lines in a list.
-

Create a CSV Reader

<-> Python

```
reader = csv.reader(lines)
```

The `csv.reader()` creates an object that reads each line of the CSV file and separates the values into a list.

Read the Header Row

<-> Python

```
header_row = next(reader)
```

The first row usually contains the column names.

Example:

```
['STATION', 'NAME', 'DATE', 'TAVG', 'TMAX', 'TMIN']
```

After calling `next(reader)`, the reader automatically moves to the first row of actual data.

6.3 Preparing Data

Finding Column Numbers

<-> Python

```
for index, column_header in enumerate(header_row):  
    print(index, column_header)
```

Output

```
0 STATION  
1 NAME  
2 DATE  
3 TAVG  
4 TMAX  
5 TMIN
```

Using `enumerate()` lets you identify the index of each column instead of guessing.

Reading Values from a Column

<-> Python

```
highs = []  
  
for row in reader:  
    high = int(row[4])  
    highs.append(high)
```

What happens?

For each row in the CSV file:

- Read column **4** (TMAX).
 - Convert the value from a string to an integer.
 - Store the value in the highs list.
-

Result

<-> Python

```
print(highs)
```

Output

<-> Python

```
[61, 60, 66, 60, 65, 59, ...]
```

The list now contains all of the high temperatures from the CSV file.

Things to Remember:

- CSV files store data as **text**.
- `csv.reader()` converts each line into a list.
- The first row usually contains the headers.
- `next(reader)` reads the header and advances to the next row.
- Use `enumerate()` to locate column indexes.
- Values read from a CSV file are strings until converted with `int()` or `float()`.

6.4 Matplotlib

Purpose

Matplotlib is a Python library used to create static charts, graphs, and data visualizations. It provides detailed control over the appearance of a figure and is widely used in data analysis, scientific computing, engineering, and technical reporting.

Matplotlib was used throughout the Python Crash Course projects to visualize numerical and weather data, plot random walks, compare temperature ranges, and save completed charts as image files.

Basic Workflow

A Matplotlib visualization generally follows four steps:

1. Import the pyplot module.
2. prepare the data.
3. Create and customize the chart.
4. Display or save the finished figure.

<-> Python

```
import matplotlib.pyplot as plt
```

```
x_values = [1, 2, 3, 4, 5]  
y_values = [1, 4, 9, 16, 25]
```

```
plt.plot(x_values, y_values)  
plt.show()
```

The pyplot module is normally imported using the alias:

```
import matplotlib.pyplot as plt
```

This standard convention keeps Matplotlib commands short and recognizable.

Creating a Figure and Axes

A common Matplotlib pattern uses the `subplots()` function:

<-> Python

```
import matplotlib.pyplot as plt
```

```
x_values = [1, 2, 3, 4, 5]
```

```
y_values = [1, 4, 9, 16, 25]
```

```
fig, ax = plt.subplots()
```

```
ax.plot(x_values, y_values)
```

```
plt.show()
```

The two returned objects serve different purposes:

Object	Purpose
fig	Represents the entire figure or chart area
ax	Represents the individual plot where data is displayed

Most chart formatting is performed through the `ax` object.

Line Charts

Line charts display changes or trends across a sequence of values.

<-> Python

```
fig, ax = plt.subplots()
```

```
ax.plot(x_values, y_values)
```

```
ax.set_title("Square Numbers")
```

```
ax.set_xlabel("Value")
```

```
ax.set_ylabel("Square of Value")
```

```
plt.show()
```

Line charts are useful for:

- Weather data
 - Time-based measurements
 - Performance trends
 - Numerical sequences
 - Comparing related datasets
-

Scatter Plots

Scatter plots display individual data points without connecting them with lines.

<-> Python

```
fig, ax = plt.subplots()
```

```
ax.scatter(x_values, y_values)
```

```
ax.set_title("Square Numbers")
```

```
ax.set_xlabel("Value")
```

```
ax.set_ylabel("Square of Value")
```

```
plt.show()
```

Scatter plots are useful for identifying patterns, relationships, groupings, and unusual values within a dataset.

Plotting Multiple Data Series

More than one dataset can be displayed on the same axes.

<-> Python

```
highs = [58, 61, 64, 67, 70]
```

```
lows = [42, 44, 47, 49, 52]
```

```
fig, ax = plt.subplots()
```

```
ax.plot(highs, label="High Temperatures")
```

```
ax.plot(lows, label="Low Temperatures")
```

```
ax.set_title("Daily Temperatures")
```

```
ax.set_ylabel("Temperature")
```

```
ax.legend()
```

```
plt.show()
```

The label argument identifies each data series, while `ax.legend()` displays those labels on the chart.

Filling the Area Between Lines

The `fill_between()` method shades the area between two datasets.

<-> Python

```
fig, ax = plt.subplots()
```

```
ax.plot(highs)
```

```
ax.plot(lows)
```

```
ax.fill_between(  
    range(len(highs)),  
    highs,  
    lows,  
    alpha=0.1  
)
```

```
plt.show()
```

This technique was used in the weather visualization projects to emphasize the range between daily high and low temperatures.

The alpha value controls transparency.

Titles and Axis Labels

Clear titles and labels help the reader understand the chart.

<-> Python

```
ax.set_title("Daily High and Low Temperatures")
ax.set_xlabel("Date")
ax.set_ylabel("Temperature (°F)")
```

Common formatting methods include:

Method	Purpose
set_title()	Adds a chart title
set_xlabel()	Labels the horizontal axis
set_ylabel()	Labels the vertical axis
legend()	Displays data-series labels
tick_params()	Formats tick marks and labels

Styling Tick Labels

Tick labels can be adjusted for readability.

<-> Python

```
ax.tick_params(axis="both", labelsiz=12)
```

Date labels may also be rotated automatically:

<-> Python

```
fig.autofmt_xdate()
```

This is especially useful when charting dates that would otherwise overlap.

Applying Built-In Styles

Matplotlib includes built-in styles that modify the appearance of a chart.

To view the available styles:

<-> Python

```
import matplotlib.pyplot as plt

print(plt.style.available)
```

To apply a style:

<-> Python

```
plt.style.use("seaborn-v0_8")
```

The selected style should be applied before creating the figure.

Styles affect features such as:

- Background appearance
 - Grid lines
 - Fonts
 - Line presentation
 - Tick formatting
-

Using Color Maps

A color map assigns colors according to a range of values.

<-> Python

```
fig, ax = plt.subplots()
```

```
ax.scatter(  
    x_values,  
    y_values,  
    c=y_values,  
    cmap=plt.cm.Blues  
)
```

```
plt.show()
```

In this example, larger values appear with stronger color intensity.

Color maps are useful when color represents:

- Magnitude
- Frequency
- Temperature
- Elevation
- Density
- Relative value

Setting the Figure Size

The size of the chart can be controlled through figsize.

```
fig, ax = plt.subplots(figsize=(10, 6))
```

The dimensions are measured in inches:

```
figsize=(width, height)
```

Larger figures improve readability when charts contain long labels, dates, or many data points.

Saving a Chart

A chart can be saved instead of, or before, displaying it.

<-> Python

```
plt.savefig("square_numbers.png")
```

Additional options may be included:

```
plt.savefig(  
    "square_numbers.png",  
    bbox_inches="tight"  
)
```

The `bbox_inches="tight"` argument removes unnecessary whitespace around the saved image.

Saving the chart before calling `plt.show()` avoids the possibility of saving a blank figure in some environments.

Common Chart Types

Chart Type	Common Method	Typical Use
Line chart	<code>ax.plot()</code>	Trends and time-series data
Scatter plot	<code>ax.scatter()</code>	Relationships between values
Bar chart	<code>ax.bar()</code>	Comparing categories
Horizontal bar chart	<code>ax.barh()</code>	Comparing categories with long labels
Histogram	<code>ax.hist()</code>	Distribution of numerical values
Filled range	<code>ax.fill_between()</code>	Showing variation between datasets

Common Errors

Mismatched Data Lengths

The x-values and y-values must contain the same number of items.

<-> Python

```
x_values = [1, 2, 3]
y_values = [1, 4]
```

This results in an error because the two datasets cannot be paired correctly.

Calling show() Too Early

Commands placed after plt.show() may not affect the displayed figure.

Complete all chart formatting before calling:

<-> Python

```
plt.show()
```

Missing Labels

A chart may run correctly but still be difficult to understand if it lacks a title, axis labels, or a legend.

Visual accuracy includes clearly explaining what the data represents.

Incorrect Data Types

Values read from CSV files frequently begin as strings and must be converted before plotting.

<-> Python

```
temperature = int(row[5])
```

Matplotlib can only produce reliable results when the source data has been prepared correctly.

Best Practices

- Verify the data before creating the chart.
 - Select a chart type that matches the information being presented.
 - Add a meaningful title and axis labels.
 - Keep formatting clear and consistent.
 - Avoid unnecessary visual effects.
 - Use legends when displaying multiple datasets.
 - Rotate date labels when necessary.
 - Save important charts as image files.
 - Separate data preparation from visualization logic.
 - Test changes incrementally.
-

Developer's Note

Matplotlib was the first library that demonstrated how Python could turn raw numerical data into something immediately understandable. Plotting squares showed the basic workflow, while the weather projects demonstrated how real CSV data could be cleaned, converted, and displayed as a meaningful chart.

The ability to adjust titles, labels, dimensions, transparency, and saved output also showed that creating a visualization involves more than plotting values. The final presentation must communicate the information clearly.

Key Takeaways

- Matplotlib creates static charts and technical visualizations.
- `pyplot` is normally imported using the alias `plt`.
- `fig` represents the complete figure, while `ax` represents the plot.
- Line charts display trends, while scatter plots display individual relationships.
- Titles, labels, legends, and readable tick marks provide necessary context.
- `fill_between()` can highlight the range between two datasets.
- Color maps can represent differences in magnitude or value.
- Charts can be displayed with `plt.show()` or saved with `plt.savefig()`.
- Clear data preparation and thoughtful formatting are essential to producing useful visualizations.

6.5 Plotly

What is Plotly?

Plotly is an interactive data visualization library for Python. It can create charts, graphs, maps, and dashboards that users can zoom, pan, and interact with in a web browser.

Unlike Matplotlib, which creates static images, Plotly generates interactive visualizations that allow users to explore the data.

Why Use Plotly?

Plotly is particularly useful when working with large datasets or geographic information because it allows users to interact with the visualization instead of simply viewing a static chart.

Common uses include:

- Scatter plots
 - Line charts
 - Bar charts
 - Geographic maps
 - 3D charts
 - Interactive dashboards
-

Plotly Express vs. Graph Objects

Plotly provides two primary ways to create visualizations.

Plotly Express

The high-level interface used throughout **Python Crash Course**.

Advantages:

- Simple syntax
- Requires less code
- Ideal for most charts
- Great for rapid development

Example:

<-> Python

```
import plotly.express as px
```

Graph Objects

The lower-level interface that provides greater control over every aspect of a visualization.

Advantages:

- Highly customizable
- Supports advanced layouts
- Better suited for complex dashboards

Example:

<-> Python

```
import plotly.graph_objects as go
```

Basic Plotly Workflow

Most Plotly programs follow the same pattern:

1. Import Plotly.
 2. Read or retrieve the data.
 3. Organize the data.
 4. Create the figure.
 5. Customize the chart.
 6. Display or save the visualization.
-

Common Figure Methods

Method	Purpose
px.scatter()	Scatter plot
px.line()	Line chart
px.bar()	Bar chart
px.imshow()	Display image data
fig.update_layout()	Modify titles, axes, legend
fig.update_traces()	Modify markers and lines
fig.show()	Display the visualization

Common Plotly Parameters

Parameter	Purpose
title=	Chart title
labels=	Rename axis labels
x=	X-axis data
y=	Y-axis data
color=	Color categories
size=	Marker size
hover_name=	Hover label
hover_data=	Additional hover information
template=	Visual theme
color_continuous_scale=	Apply a color scale

Things to Remember:

- Plotly creates **interactive** visualizations.
- Plotly Express is easier to learn and is suitable for most projects.
- Graph Objects provide advanced customization.
- Most Plotly visualizations are displayed in a web browser.
- Nearly every chart begins by creating a **Figure** object.

Plotly Built in Color Scales

Plotly provides built-in color scales that can be applied to charts, maps, and other data visualizations. A color scale is a gradient that represents data values with color.

How to Use

Pass the color scale name to the appropriate Plotly parameter, such as:

<-> Python

```
color_continuous_scale='viridis'
```

Color Scale	Preview	Color Scale	Preview
viridis		reds	
plasma		greys	
inferno		rainbow	
magma		thermal	
cividis		ice	
turbo		earth	
blues		sunset	
greens			

Things to Remember

- Color scales are gradients used to represent data values.
- Different color scales can change how data patterns are perceived.
- Experiment with several color scales to find the one that best fits your data.
- Copy the color scale name and use it directly in your Plotly code.

All Plotly Built-in Color Scales (1 of 2)

Color Scale	Preview	Color Scale	Preview
aggrnyl		mint	
agsunset		orrd	
blackbody		oranges	
bluered		oryel	
blues		peach	
blugrn		pinkyl	
bluyl		plasma	
brwnyl		plotly3	
bugn		pubu	
bupu		pubugn	
burg		purd	
burgyl		purp	
cividis		purples	
darkmint		purpor	
electric		rainbow	
emrld		rdbu	
gnbu		rdpu	
greens		redor	
greys		reds	
hot		sunset	
inferno		sunsetdark	
jet		teal	
magenta		tealgrn	
magma			

All Plotly Built-in Color Scales (2 of 2)

Color Scale	Preview	Color Scale	Preview
turbo		prgn	
viridis		piyg	
ylgn		picnic	
ylgnbu		portland	
ylorbr		puor	
ylorrd		rdgy	
algae		rdylbu	
amp		rdylgn	
deep		spectral	
dense		tealrose	
gray		temps	
haline		tropic	
ice		balance	
matter		curl	
solar		delta	
speed		oxy	
tempo		edge	
thermal		hsv	
turbid		icefire	
armyrose		phase	
brbg		twilight	
earth		mrybm	
fall		mygbm	
geyser			

7 APIs

7.1 What is an API?

An Application Programming Interface (API) is a way for one program to request information or services from another program.

Instead of reading data from a local file, your Python program sends a request to another computer, called a server. The server processes the request and returns the requested data, which your program can then use.

The returned information can be processed, analyzed, stored, displayed, or used to generate visualizations. An API simply provides access to data or services; what your program does with that information depends on the purpose of your application.

How APIs Were Used

In Python Crash Course, APIs were used to automatically retrieve current information from websites instead of manually downloading data files.

Because the data comes directly from the source each time the program runs, the application always works with the most up-to-date information available. This makes APIs especially useful when working with information that changes frequently.

Examples include:

- Weather data
- Earthquake data
- GitHub repository information
- Hacker News articles

Why Use an API?

- Access current data automatically.
 - Eliminate the need to manually download data files.
 - Build applications that update themselves whenever new data becomes available.
 - Process, analyze, or store real-world data.
 - Use real-world data in visualizations and applications.
-

Best Practices

- Read the API documentation before writing code.
 - Verify HTTP status codes before processing returned data.
 - Handle request failures and exceptions gracefully.
 - Validate JSON data before using it.
 - Keep API requests and data processing in separate functions.
 - Protect API keys and credentials by storing them outside the source code.
 - Respect API rate limits and usage policies.
-

Developer's Note

Working with APIs was one of the first experiences that showed how Python applications can interact with systems across the internet. Rather than relying on static files, programs were able to retrieve live information from services such as GitHub, Hacker News, and the USGS Earthquake API.

Learning to request, validate, and process JSON data demonstrated that modern applications often spend as much time communicating with external services as they do processing local information. This chapter provided the foundation for integrating real-time data into future Python projects.

Related Topics

- **Chapter 6 – Data Visualization** — Visualizing data retrieved from APIs.
 - **2.1.2 Data Types** — Understanding JSON data types.
 - **2.1.3 Type Conversion** — Converting returned values into usable Python objects.
 - **2.6 Exceptions** — Handling connection and request errors.
 - **8.11 Views** — Using API data within Django applications.
-

Key Takeaways

- APIs allow applications to exchange information over a network.
- HTTP provides the communication protocol used by most web services.
- REST APIs commonly exchange data using JSON.
- The requests library simplifies communication with web APIs.
- Always verify responses before processing returned data.
- Proper error handling and validation improve application reliability.
- API integration is a fundamental skill for modern Python development.

7.2 What is HTTP?

Overview

HTTP (Hypertext Transfer Protocol) is the communication protocol used by web browsers, web servers, and most web APIs to exchange information over the internet.

Whenever your browser opens a web page or a Python program requests information from an API, it is using HTTP to communicate with a server.

Think of HTTP as the **language** that allows a client and a server to understand one another.

How HTTP Works

Every HTTP communication follows the same basic sequence:

1. A **client** (such as a web browser or Python program) sends a request.
2. The **server** receives the request and processes it.
3. The server sends a response back to the client.
4. The client displays or uses the returned information.

Example

When you enter the following address into your browser:

`https://api.github.com`

your browser sends an HTTP request to GitHub's server.

The server processes the request and returns an HTTP response containing the requested data.

The same process occurs when Python uses the requests library to retrieve data from an API.

7.2.1 HTTP Methods

What are HTTP Methods?

HTTP methods define the action a client wants a server to perform. They can be thought of as **verbs** because they describe what should happen when a request is made.

Method	Purpose
GET	Request information from the server.
POST	Send new information to the server.
PUT	Replace existing data.
DELETE	Remove existing data.

Think of It This Way

- **GET** → "Please send me this information."
- **POST** → "Here is some new information."
- **PUT** → "Replace the existing information with this."
- **DELETE** → "Remove this information."

Developer's Note: Throughout the *Python Crash Course*, the APIs used in the examples primarily relied on the **GET** method to retrieve current information such as GitHub repositories, Hacker News articles, and earthquake data.

7.2.2 HTTP Status Codes

What are HTTP Status Codes?

Every HTTP request returns a **status code** that tells your program whether the request was successful or whether an error occurred.

Think of the status code as the server's response to your request.

Code	Meaning
200	Request completed successfully.
301	The resource has moved permanently.
302	The resource has been temporarily redirected.
403	Access is forbidden.
404	The requested page or resource could not be found.
500	Internal server error. The problem exists on the server.

Things to Remember

- **200** usually means everything worked correctly.
- **404** is one of the most common errors and indicates the requested page doesn't exist.
- **500** indicates a problem on the server rather than in your request.
- Your Python program can check the status code before attempting to process returned data.

7.3 REST APIs

Purpose

A **REST API (Representational State Transfer Application Programming Interface)** is a standardized way for software applications to communicate over the web. Rather than exchanging files directly, applications send HTTP requests to specific endpoints and receive structured data in return, most commonly in JSON format.

REST APIs are the foundation of modern web services and are used by websites, mobile applications, cloud platforms, and countless online services.

Why It Matters

REST APIs allow applications to retrieve, update, create, and delete information without needing direct access to another application's database.

Python applications commonly use REST APIs to:

- Retrieve live data
- Submit information to web services
- Integrate with third-party applications
- Automate repetitive tasks
- Build cloud-connected applications

Understanding REST APIs is an essential skill for modern Python development.

REST Principles

Although implementations vary, most REST APIs follow several common principles.

Principle	Description
Client-Server	The client requests information from the server.
Stateless	Each request contains all information needed to process it.
Resource-Based	Everything is treated as a resource identified by a URL.
Standard HTTP Methods	Uses GET, POST, PUT/PATCH, and DELETE to interact with resources.
JSON Responses	Data is commonly exchanged using JSON.

Example Request

A Python application sends an HTTP request:

GET <https://api.example.com/users>

The server responds with JSON data:

<-> Python

```
[
    {
        "id": 1,
        "name": "Alice"
    },
    {
        "id": 2,
        "name": "Bob"
    }
]
```

Python can then process the returned data as native dictionaries and lists.

Common REST Operations

HTTP Method	Operation
GET	Retrieve information
POST	Create new information
PUT / PATCH	Update existing information
DELETE	Remove information

Best Practices

- Read the API documentation before writing code.
- Verify HTTP status codes before processing returned data.
- Validate JSON data before using it.
- Handle request failures gracefully.
- Keep API requests separate from business logic.
- Protect API keys and authentication credentials.

Developer's Note

Although REST APIs may appear complex initially, they all follow the same general pattern: send a request, receive a response, and process the returned data. Once this workflow becomes familiar, integrating external services into Python applications becomes much more straightforward.

Working with the GitHub API, Hacker News API, and USGS Earthquake API demonstrated that the same REST principles apply regardless of the service being accessed.

Related Topics

- **7.2** HTTP
 - **7.4** Requests
 - **7.5** JSON
 - **6.5** Plotly
 - **5.4** Working with APIs
-

Key Takeaways

- REST APIs provide a standardized method for applications to communicate.
- Resources are accessed through URLs using standard HTTP methods.
- Most REST APIs exchange information using JSON.
- Python commonly interacts with REST APIs through the requests library.
- Understanding REST APIs enables Python applications to integrate with modern web services.

7.4 Requests

Purpose

The **Requests** library is one of Python's most popular third-party packages for communicating with web services. It simplifies sending HTTP requests and receiving responses, making it much easier to work with REST APIs than using Python's built-in networking libraries.

Throughout the Python Crash Course projects, the Requests library was used to retrieve live data from GitHub, Hacker News, and the United States Geological Survey (USGS).

Why It Matters

Modern applications frequently obtain information from external services rather than local files. The Requests library provides a simple and consistent interface for communicating with these services.

Common uses include:

- Retrieving data from REST APIs
 - Downloading information from web services
 - Sending data to remote applications
 - Accessing cloud-based resources
 - Integrating third-party services
-

Installation

Install the Requests library using pip:

<-> Python

```
python -m pip install requests
```

Import the library into your program:

<-> Python

```
import requests
```

Basic Workflow

The typical Requests workflow consists of four steps:

1. Import the Requests library.
2. Send an HTTP request.
3. Verify the response.
4. Process the returned data.

Example:

<-> Python

```
import requests
```

```
url = "https://api.github.com"
```

```
response = requests.get(url)
```

```
print(response.status_code)
```

A successful request normally returns a status code of **200**.

Common Request Methods

Method	Purpose
requests.get()	Retrieve data
requests.post()	Submit new data
requests.put()	Replace existing data
requests.patch()	Update part of an existing resource
requests.delete()	Remove data

Working with JSON Responses

Many REST APIs return information in JSON format.

The Requests library can automatically convert the JSON response into Python objects.

<-> Python

```
response = requests.get(url)
data = response.json()
```

The returned data can then be processed using standard Python dictionaries and lists.

Best Practices

- Verify the HTTP status code before processing data.
- Handle connection errors gracefully.
- Read API documentation carefully.
- Store URLs in variables or constants.
- Separate data retrieval from data processing.
- Protect API keys and authentication credentials.

Developer's Note

The Requests library made working with APIs surprisingly straightforward. Rather than dealing with low-level networking details, it allowed the focus to remain on retrieving and processing useful information. Once the basic request-response workflow became familiar, accessing data from different web services followed the same predictable pattern.

Related Topics

- **7.2** HTTP
 - **7.3** REST APIs
 - **7.5** JSON
 - **5.4** Working with APIs
 - **6.5** Plotly
-

Key Takeaways

- Requests is the standard Python library for working with web APIs.
- HTTP requests can be sent using simple methods such as `get()` and `post()`.
- Always verify the response before processing returned data.
- JSON responses can be converted directly into Python objects.
- Requests provides a clean and consistent interface for communicating with web services.

7.5 JSON

Purpose

JSON (JavaScript Object Notation) is a lightweight, text-based format used to store and exchange structured data. Although it originated with JavaScript, JSON has become the standard format for data exchanged between web services and applications.

Python applications frequently receive JSON data from REST APIs and convert it into native Python objects for processing.

Why It Matters

Most modern APIs communicate using JSON because it is:

- Easy for humans to read
- Easy for computers to parse
- Lightweight and efficient
- Supported by virtually every programming language

Understanding JSON is essential when working with web services, cloud platforms, and modern application development.

JSON Structure

JSON stores information using two basic structures:

Structure	Python Equivalent
Object	Dictionary (dict)
Array	List (list)

Example JSON:

<-> Python

```
{  
    "name": "Skip",  
    "language": "Python",  
    "experience": 1,  
    "projects": [  
        "Alien Invasion",  
        "Learning Log"  
    ]  
}
```

When processed in Python, the JSON object becomes a dictionary containing strings, numbers, and lists.

Working with JSON in Python

The Requests library can automatically convert a JSON response into native Python objects.

<-> Python

```
import requests
```

```
url = "https://api.github.com"
```

```
response = requests.get(url)
```

```
data = response.json()
```

```
print(type(data))
```

After conversion, the returned data can be accessed using standard dictionary and list operations.

Common JSON Data Types

JSON Type	Python Type
String	str
Number	int or float
Boolean	bool
Array	list
Object	dict
Null	None

Best Practices

- Validate JSON data before using it.
 - Verify that expected keys exist before accessing them.
 - Handle missing or unexpected values gracefully.
 - Keep JSON parsing separate from application logic.
 - Read API documentation to understand the returned data structure.
-

Developer's Note

Working with JSON demonstrated that most web APIs follow a common pattern: retrieve structured data, convert it into Python objects, and process only the information needed. Once the relationship between JSON objects and Python dictionaries became clear, working with API responses became much more intuitive.

Related Topics

- **2.1.2** Data Types
 - **2.1.3** Type Conversion
 - **7.3** REST APIs
 - **7.4** Requests
 - **5.4** Working with APIs
-

Key Takeaways

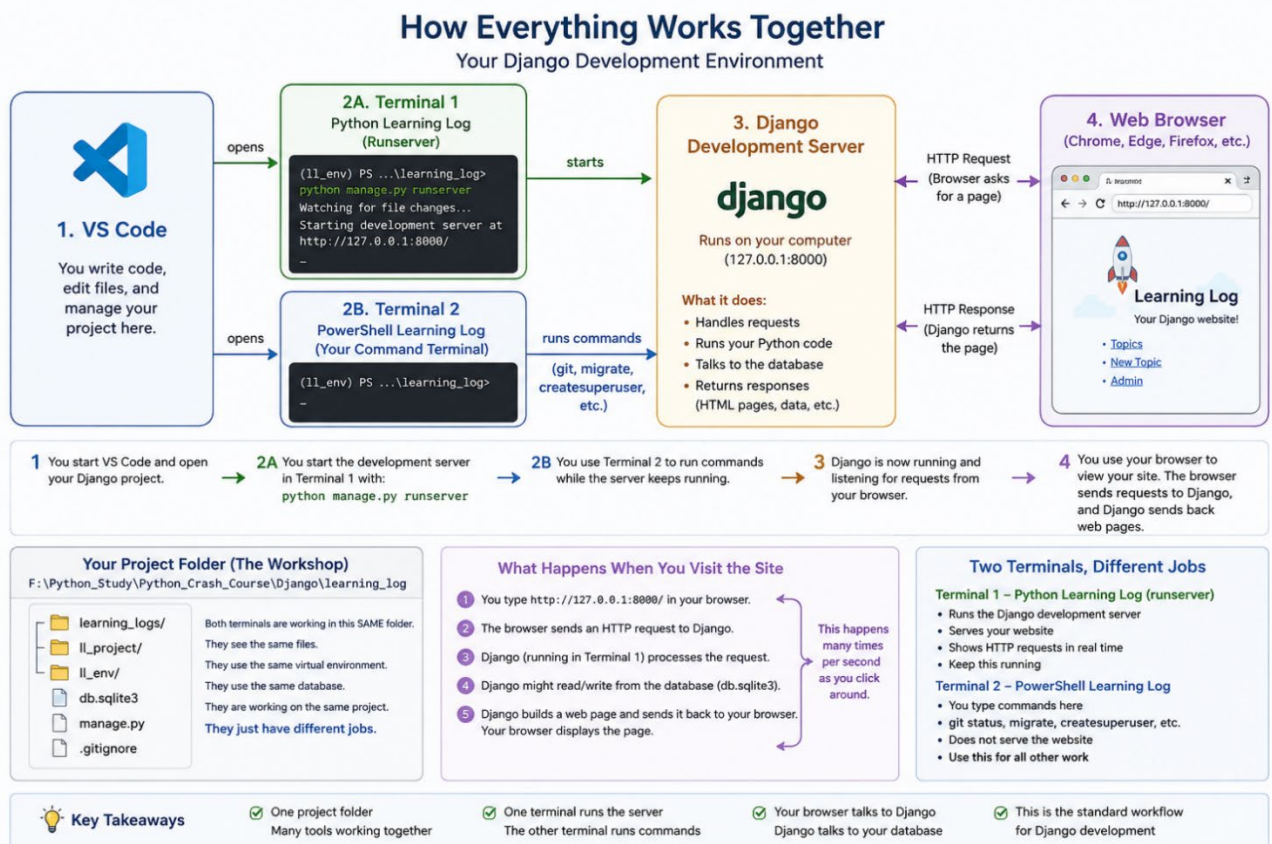
- JSON is the standard format for exchanging structured data over the web.
- JSON objects become Python dictionaries, and JSON arrays become Python lists.
- The Requests library can automatically convert JSON into Python objects.
- Understanding JSON is fundamental when working with REST APIs.
- Validating returned data improves the reliability of Python applications.

8 DJANGO WEB DEVELOPMENT

8.1 Introduction

Django is Python's most popular web framework; a set of tools designed for building interactive web applications.

A full spec details the project goals, describes the project's functionality, and discusses its appearance and user interface. Like any good project or business plan, a spec should keep you focused and help keep your project on track.



8.2 Django Starter Commands

COMMAND	PURPOSE
<code>python -m venv ll_env</code>	Create a virtual environment
<code>.\ll_env\Scripts\Activate.ps1</code>	Activate the virtual environment (PowerShell)
<code>python -m pip install --upgrade pip</code>	Update `pip` safely
<code>pip install django</code>	Install Django
<code>django-admin startproject ll_project .</code>	Create a new Django project in the current directory
<code>python manage.py migrate</code>	Create/update the project's database
<code>python manage.py runserver</code>	Start the local development server

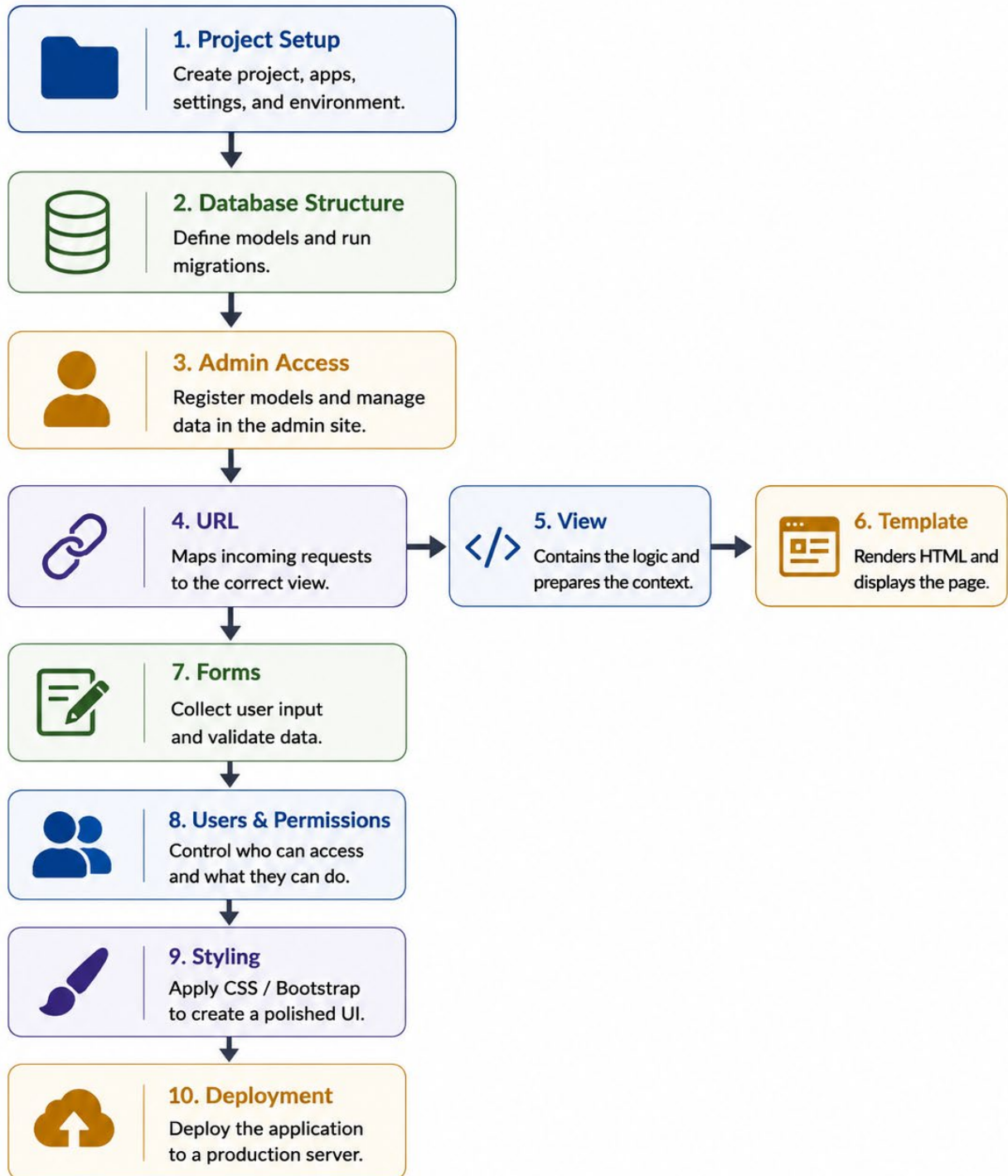
8.3 Project Build Map

1. Create project folder
2. Create virtual environment
3. Install Django
4. Start project
5. Start app
6. Start Git tracking
7. Register app in settings.py
8. Define models
9. Make migrations
10. Migrate database
11. Register models in admin.py
12. Create superuser
13. Define URLs
14. Create views
15. Create templates
16. Add forms
17. Add user accounts
18. Restrict data by user ownership
19. Add Bootstrap styling
20. Test site
21. Commit clean checkpoints with Git throughout
22. Deploy

8.4 The Big Mental Model

The Big Mental Model

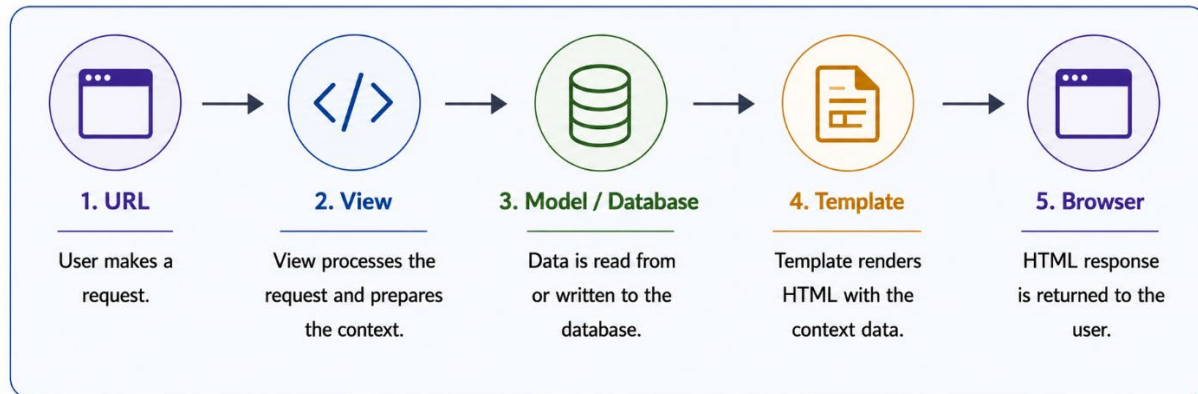
This is the complete path from project start to deployment.
Each layer builds on the previous one.



8.5 The Core Mental Loop

The Core Mental Loop

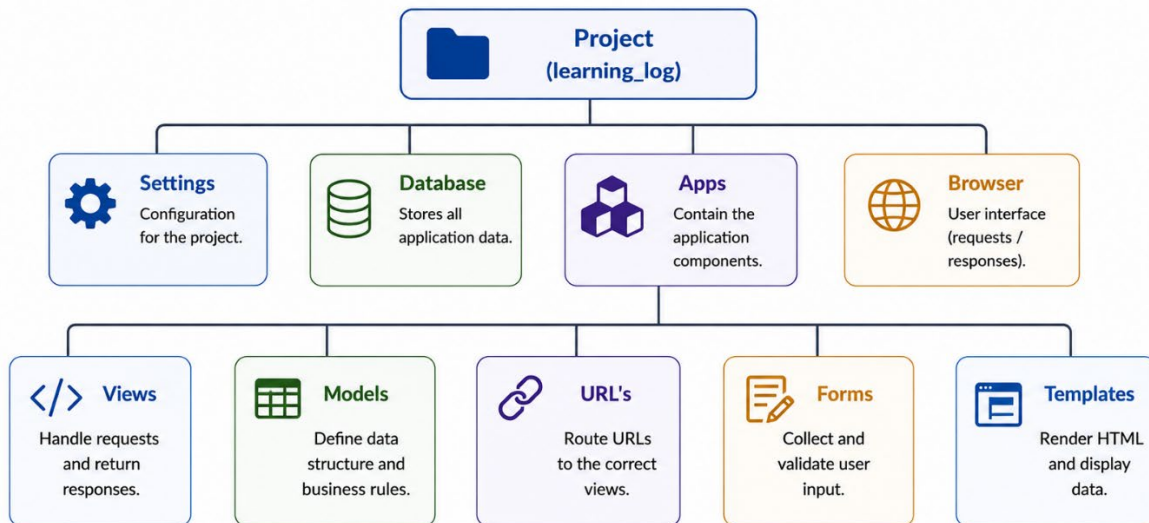
Every time a user interacts with the application, Django follows this loop.



8.6 Django Mental Model:

Django Mental Model

This diagram shows how the major pieces of a Django project fit together.



Part	Purpose
Project	Holds everything together
App	A self-contained feature of the project.
Model	Defines what data is stored.
Database	Stores the actual data.
View	Contains the logic for handling requests.
URL	Directs requests to the correct view.
Template	Controls what the user sees in the browser.
Form	Collects user input.
Admin	Lets administrators manage data.

8.7 Development Workflow

Purpose:

This workflow is the standard startup, development, testing, and shutdown procedure for any Django project developed in VS Code.

For this writing the path F:\Python_Study\Python_Crash_Course\Django\learning_log shown here is for a complete example and what to check for.

Choose the correct current path for the project

Things to check before running commands:

✓ Am I in the correct folder?

F:\Python_Study\Python_Crash_Course\Django\learning_log

✓ Is my virtual environment active?

Look for:

(ll_env) –

Note: the above environment is just an example of what should be seen. It will be the current project name chosen.

✓ Is the development server running?

Look for:

Watching for file changes...

If all three are true, you're ready to work

1. Open VS Code

Open the Django workspace or project.

For this initial project, the workspace contains:

```
F:\Python_Study\Python_Crash_Course
```

2. Open Terminal #1 (Development Server)

This terminal's only job is to run the Django development server.

Step 1 – Change to the project directory

```
cd F:\Python_Study\Python_Crash_Course\Django\learning_log
```

Verify that the prompt is:

```
PS F:\Python_Study\Python_Crash_Course\Django\learning_log>
```

Step 2 – Activate the virtual environment

If you already see:

```
(ll_env) in front of the path the virtual environment is already active.
```

If you do not see `(ll\_env)`, activate it:

```
powershell
```

```
.\ll_env\Scripts\Activate.ps1
```

Your prompt should now become:

```
(ll_env) PS F:\Python_Study\Python_Crash_Course\Django\learning_log>
```

Step 3 – Start the Django server

powershell

```
python manage.py runserver
```

The terminal will display something similar to:

Watching for file changes...

Starting development server at

http://127.0.0.1:8000/

Leave this terminal running.

Do not close it.

3. Open Terminal #2 (Working Terminal)

This terminal is where you work.

Open:

Terminal → New Terminal

If necessary, move to the project folder:

powershell

```
cd F:\Python_Study\Python_Crash_Course\Django\learning_log
```

Again, check the prompt.

If you already see:

```
(ll_env)
```

you're ready to go.

If not, activate the virtual environment:

powershell

```
.\ll_env\Scripts\Activate.ps1
```

4. Common Django Commands

Examples include:

Create an app:

powershell

```
python manage.py startapp learning_logs
```

Create migrations:

powershell

```
python manage.py makemigrations
```

Apply migrations:

powershell

```
python manage.py migrate
```

Create a superuser:

powershell

```
python manage.py createsuperuser
```

Run Server

powershell

```
python manage.py runserver
```

5. Edit Your Code

Use the VS Code Explorer to open files such as:

- * `models.py`

- * `views.py`

- * `urls.py`

- * `admin.py`

Save your work with:

Ctrl + S

Test Your Changes

Open your browser and visit:

`http://127.0.0.1:8000/`

or

`http://localhost:8000/`

Refresh the page after making changes.

Because the server is running, Django automatically reloads most code changes.

6. Stop the Server

When you're finished coding:

Go to the ****Server Terminal**** and press:

Ctrl + C

The prompt returns:

```
(11_env) PS F:\Python_Study\Python_Crash_Course\Django\learning_log>
```

The server is now stopped.

7. Commit Your Work

From the repository root:

enter these commands:

```
git status  
git add .  
git commit -m "Describe what you accomplished"  
git status
```

look for the response:

On branch main

nothing to commit, working tree clean

8.8 App Anatomy

At first glance it looks like Django dumps a bunch of random files.

It doesn't.

Everyone has a purpose.

models.py

★ Probably the most important file.

This is where you define the data.

For the Learning Log project I eventually created things like:

Topic

Entry

Think of it as defining the tables in your database.

views.py

This answers the question:

"What should happen when someone visits this page?"

Example:

User visits

/topics/

views.py decides

what data to retrieve

what page to display

admin.py

Django comes with an administrator website.

This file tells Django

"Allow me to manage these models from the admin interface."

This will be used surprisingly often.

tests.py

Professional software is tested.

This file is where automated tests go.

The book didn't spend much time here, but it's an important part of larger projects.

apps.py

Configuration information for this app.

It is rarely edited.

It simply tells Django:

"This application exists."

migrations

This folder keeps track of database changes.

Every time a model is added or modified

Django records the change here.

Think of it as a history of your database's evolution.

__init__.py

This file tells Python

"Treat this directory as a package."

Most of the time it stays empty.

8.9 Request Cycle

Purpose

The **Django Request Cycle** describes the sequence of events that occurs every time a user interacts with a Django application. Whether a user enters a URL, clicks a link, or submits a form, Django follows the same workflow to process the request and return a response.

Understanding this cycle is one of the most important concepts in Django because it explains how the major components of a project work together.

Why It Matters

Every Django application follows the same request-response pattern.

Once this workflow becomes familiar, it becomes much easier to understand:

- URL routing
- Views
- Models
- Templates
- Forms
- Authentication

Rather than thinking of these as separate topics, the request cycle shows how they interact to produce a web page.

The Request Cycle

1. The browser sends a request to the Django application.
2. Django examines the requested URL.
3. The URL configuration selects the appropriate view.
4. The view performs the application logic.
5. If required, the view retrieves or updates data using one or more models.
6. The view passes the resulting data to a template.
7. The template renders the HTML page.
8. Django returns the completed response to the browser.

*(Insert **The Core Mental Loop** diagram here.)*

Mental Model

Think of Django as a receptionist in a busy office.

A visitor arrives and asks a question.

The receptionist:

- Determines where the request should go.
- Directs it to the appropriate person.
- Retrieves any required information.
- Prepares the response.
- Returns the completed answer to the visitor.

Every request follows this same sequence, regardless of the complexity of the application.

Best Practices

- Keep views focused on a single responsibility.
 - Place business logic in models whenever appropriate.
 - Use templates only for presentation.
 - Let URL patterns handle request routing.
 - Think of every page as completing one request cycle.
-

Developer's Note

The request cycle was the point where Django finally "clicked" for me. Earlier topics—URLs, views, models, templates, and forms—felt like individual pieces. Once I understood how a request moved through the application, those pieces became parts of a single workflow. From that point forward, debugging became much easier because I could identify where in the request cycle a problem was occurring.

Related Topics

- **8.8** App Anatomy
 - **8.10** Models
 - **8.11** Views
 - **8.12** Templates
 - **8.13** Forms
 - **8.14** Authentication
-

Key Takeaways

- Every interaction with a Django application follows the same request-response cycle.
- URLs determine which view processes a request.
- Views contain the application logic.
- Models manage application data.
- Templates generate the HTML returned to the browser.
- Understanding the request cycle makes the overall Django architecture much easier to understand.

8.10 Models

Purpose

A **Model** defines the structure of the data used by a Django application. Models describe what information is stored, how different pieces of data relate to one another, and how that data is managed within the database.

Each model is represented as a Python class that inherits from Django's Model class.

Why It Matters

Models serve as the foundation of every Django application. Rather than writing SQL statements directly, developers define data structures using Python classes, and Django automatically creates and manages the corresponding database tables.

Models provide a consistent and secure way to store, retrieve, update, and delete application data.

Basic Structure

A Django model consists of:

- A class that inherits from `models.Model`
- One or more fields that define the stored data
- Optional methods that provide additional functionality

<-> Python

```
from django.db import models
```

```
class Topic(models.Model):  
    text = models.CharField(max_length=200)  
    date_added = models.DateTimeField(auto_now_add=True)
```

After defining a model, Django uses migrations to create the corresponding database table.

Common Field Types

Field	Purpose
CharField	Short text
TextField	Long text
IntegerField	Whole numbers
FloatField	Decimal numbers
BooleanField	True or False values
DateField	Dates
DateTimeField	Date and time
ForeignKey	Relationship between models

Model Relationships

Models often reference one another.

For example, in the **Learning Log** project:

- A **Topic** can contain many **Entries**.
- Each **Entry** belongs to one **Topic**.

This relationship is created using a ForeignKey.

Best Practices

- Create one model for each type of application data.
- Choose field types that accurately represent the stored information.
- Keep business logic within models whenever appropriate.
- Use meaningful model and field names.
- Apply migrations whenever models are modified.

Developer's Note

Models were one of the most powerful concepts introduced in Django. Instead of designing database tables manually, defining a simple Python class allowed Django to build and manage the database automatically. During the Learning Log project, this made it much easier to focus on the application's behavior rather than the underlying database implementation.

Related Topics

- **8.9** Request Cycle
- **8.11** Views
- **8.12** Templates
- **8.13** Forms
- **8.14** Authentication

Key Takeaways

- Models define the structure of application data.
- Django automatically converts models into database tables.
- Field types determine how information is stored.
- Relationships connect models together.
- Migrations keep the database synchronized with the application's models.
- Models form the foundation of every Django application.

8.11 Views

Purpose

A **View** contains the application logic that processes a user's request and determines what information is returned to the browser. When a URL is requested, Django calls the appropriate view, which retrieves any required data, performs the necessary processing, and returns an HTTP response.

Views act as the connection between URLs, models, and templates.

Why It Matters

Views control how an application responds to user requests. They determine:

- What data is retrieved
- What business logic is executed
- Which template is displayed
- What information is passed to the template

Well-designed views keep applications organized by separating processing logic from presentation.

Basic Structure

A Django view is simply a Python function (or class) that accepts a request and returns a response.

<-> Python

```
from django.shortcuts import render
```

```
def index(request):  
    return render(request, "learning_logs/index.html")
```

The `render()` function combines a template with any required data and returns the completed HTML page to the browser.

Responsibilities

A typical view performs the following tasks:

- Receives the user's request.
- Retrieves or updates data using models.
- Performs any required application logic.
- Passes data to a template.
- Returns an HTTP response.

Views should focus on processing requests rather than storing data or generating HTML.

Best Practices

- Keep views focused on a single responsibility.
 - Retrieve only the data required for the request.
 - Place business logic in models whenever appropriate.
 - Keep presentation logic within templates.
 - Write clear, descriptive view names.
-

Developer's Note

Views became much easier to understand once I realized they function as the decision-making component of a Django application. Every request eventually reaches a view, where the application determines what needs to happen before returning a response. During the Learning Log project, recognizing that every page had its own view made the overall application flow much easier to follow and debug.

Related Topics

- **8.9** Request Cycle
 - **8.10** Models
 - **8.12** Templates
 - **8.13** Forms
 - **8.14** Authentication
-

Key Takeaways

- Views process incoming requests and return responses.
- A view connects URLs, models, and templates.
- Views retrieve data and perform application logic.
- The render() function combines templates with data to generate HTML.
- Keeping views simple and focused improves readability and maintainability.

8.12 Templates

Purpose

A **Template** defines how information is presented to the user. Templates combine HTML with Django's template language to display dynamic content while keeping presentation separate from application logic.

Templates allow the same page layout to display different information depending on the data provided by a view.

Why It Matters

Templates separate **what the user sees** from **how the application works**.

This separation makes applications easier to:

- Develop
- Maintain
- Modify
- Reuse

A designer can update the appearance of a page without changing the application's business logic, while a developer can modify the logic without affecting the page layout.

Basic Structure

Templates are HTML files that contain Django template tags and variables.

<-> Python

```
<h1>{{ topic.text }}</h1>
```

```
<ul>
```

```
{% for entry in entries %}
```

```
    <li>{{ entry.text }}</li>
```

```
{% endfor %}
```

```
</ul>
```

When the template is rendered, Django replaces the template variables with actual data supplied by the view.

Django Template Syntax

Django templates use two primary types of delimiters:

Syntax	Purpose	Think of It As
{% ... %}	Performs an action or executes template logic.	"Django, do something."
{{ ... }}	Displays the value of a variable or expression.	"Django, show something."

Examples:

<-> Python

```
{% for topic in topics %}
    <li>{{ topic.text }}</li>
{% endfor %}
```

Reading the template in plain English:

- {% for topic in topics %} → Django, do something.
- {{ topic.text }} → Django, show something.
- {% endfor %} → Django, stop doing that thing.

Common Template Features

Feature	Purpose
{{ variable }}	Display a variable
{% for %}	Loop through a collection
{% if %}	Display content conditionally
{% extends %}	Inherit a base template
{% block %}	Define replaceable content sections
{% url %}	Generate application URLs

Template Inheritance

One of Django's most powerful features is **template inheritance**.

A base template defines the common page structure, while individual templates provide only the content unique to that page.

This approach:

- Reduces duplicated code.
 - Maintains a consistent appearance.
 - Simplifies future updates.
-

Best Practices

- Keep business logic out of templates.
 - Use templates only for presentation.
 - Create a base template for shared layouts.
 - Use meaningful template names.
 - Keep templates simple and easy to read.
-

Developer's Note

Template inheritance was one of the features that made Django feel both organized and efficient. Instead of repeating navigation menus, page headers, and footers throughout the application, a single base template provided the common layout while individual pages supplied only their unique content. This significantly reduced duplication and made the Learning Log application much easier to maintain.

Related Topics

- **8.9** Request Cycle
 - **8.10** Models
 - **8.11** Views
 - **8.13** Forms
 - **8.14** Authentication
-

Key Takeaways

- Templates control how information is displayed to users.
- Django templates combine HTML with dynamic content.
- Template inheritance promotes consistency and reduces duplicated code.
- Views provide data, while templates present it.
- Keeping presentation separate from application logic improves maintainability.

8.13 Forms

Purpose

Forms provide the primary method for users to enter and submit information to a Django application. They simplify the process of collecting input, validating data, displaying errors, and saving information to the database.

Rather than manually processing user input, Django forms provide a structured and secure approach to handling data.

Why It Matters

Nearly every web application relies on forms.

Common examples include:

- User registration
- Login pages
- Contact forms
- Search boxes
- Creating new records
- Editing existing records

Django forms help ensure that submitted information is accurate, complete, and safe before it is processed.

Basic Structure

A Django form is defined as a Python class that specifies the fields a user can complete.

<-> Python

```
from django import forms
```

```
class TopicForm(forms.Form):
```

```
    text = forms.CharField(max_length=200)
```

When submitted, Django validates the entered data before making it available to the application.

Form Processing Workflow

A typical Django form follows the same sequence:

- Display an empty form.
- The user enters information.
- The browser submits the form.
- Django validates the data.
- If valid, the data is processed or saved.
- If invalid, the form is redisplayed with error messages.

This process ensures that only valid information reaches the application.

Form Validation

One of Django's greatest strengths is automatic validation.

Validation can verify:

- Required fields
- Maximum field length
- Numeric values
- Date formats
- Email addresses
- Custom business rules

If validation fails, Django automatically displays appropriate error messages.

Best Practices

- Validate all user input.
 - Keep forms focused on a single task.
 - Display clear validation messages.
 - Never trust user input without validation.
 - Use Django's built-in form classes whenever possible.
-

Developer's Note

Forms transformed the Learning Log project from a read-only application into one that users could actively interact with. Creating topics, adding entries, and editing existing information all relied on Django forms. The built-in validation greatly reduced the amount of code required while helping prevent many common user input errors.

Related Topics

- **8.10** Models
 - **8.11** Views
 - **8.12** Templates
 - **8.14** Authentication
 - **8.15** Deployment
-

Key Takeaways

- Forms provide a structured method for collecting user input.
- Django automatically validates submitted data.
- Validation helps ensure data accuracy and application security.
- Forms work closely with views, templates, and models.
- Built-in form classes simplify the development of interactive web applications.

8.14 Authentication

Purpose

Authentication is the process of verifying a user's identity before allowing access to protected areas of a Django application. Django includes a complete authentication system that manages user accounts, login, logout, passwords, permissions, and session management.

Using Django's built-in authentication system eliminates the need to develop these security features from scratch.

Why It Matters

Most web applications contain information that should only be accessible to authorized users.

Authentication helps protect:

- Personal information
- User accounts
- Private data
- Administrative functions
- Database records

In the Learning Log project, authentication ensured that each user could view and modify only their own topics and entries.

Authentication Workflow

A typical authentication process follows these steps:

- A user registers an account.
- The user logs in with valid credentials.
- Django verifies the username and password.
- A secure session is created.
- The user accesses protected pages.
- The user logs out, ending the session.

This workflow provides secure access while maintaining a smooth user experience.

Key Components

Component	Purpose
User	Stores account information
Login	Verifies user credentials
Logout	Ends the user's session
Session	Maintains authentication between requests
Permissions	Controls access to application features
@login_required	Restricts access to authenticated users

Protecting Views

Django provides the `@login_required` decorator to restrict access to authenticated users.

<-> Python

```
from django.contrib.auth.decorators import login_required
```

```
@login_required
def topics(request):
    ...
```

If an unauthenticated user attempts to access the view, Django automatically redirects them to the login page.

Best Practices

- Use Django's built-in authentication system whenever possible.
- Protect private views with `@login_required`.
- Never store passwords in plain text.
- Limit access to only the data each user owns.
- Test authentication and authorization throughout the application.

Developer's Note

Authentication was one of the most rewarding additions to the Learning Log project because it transformed the application from a personal notebook into a multi-user web application. Once user accounts and ownership were implemented, each person could safely manage their own data without accessing anyone else's information. It also demonstrated how much security Django provides with relatively little code.

Related Topics

- **8.9** Request Cycle
 - **8.10** Models
 - **8.11** Views
 - **8.12** Templates
 - **8.13** Forms
 - **8.15** Deployment
-

Key Takeaways

- Authentication verifies a user's identity.
- Django includes a complete, built-in authentication framework.
- User sessions allow authenticated users to navigate securely.
- The `@login_required` decorator protects private views.
- Proper authentication is essential for securing web applications and protecting user data.

8.15 Deployment

Purpose

Deployment is the process of making a Django application available to users outside the development environment. During development, an application runs on a local computer using Django's built-in development server. Deployment moves the application to a production environment where it can be accessed over the internet.

Deployment represents the final stage of the software development lifecycle.

Why It Matters

An application provides value only when users can access it.

Deployment allows a Django application to:

- Be accessed from any web browser
- Serve multiple users simultaneously
- Store production data securely
- Operate continuously on a web server
- Become a real-world application

Although the Learning Log project was developed locally, it was structured so it could later be deployed to a production environment.

Typical Deployment Process

A typical Django deployment includes the following steps:

- Prepare the application for production.
- Configure environment variables and secret keys.
- Select a hosting provider.
- Configure the production database.
- Collect static files.
- Apply database migrations.
- Launch the application.
- Test the deployed site.

Each hosting platform provides its own deployment workflow, but these steps remain largely the same.

Common Deployment Components

Component	Purpose
Web Server	Serves the application to users
Application Server	Runs the Django application
Database	Stores application data
Static Files	CSS, JavaScript, and images
Environment Variables	Protect sensitive configuration values
HTTPS	Encrypts communication between users and the server

Development vs. Production

Development	Production
Local computer	Public server
SQLite commonly used	Production database
Debug mode enabled	Debug mode disabled
Development server	Production web server
Single developer	Multiple users

Best Practices

- Never deploy with DEBUG = True.
 - Protect secret keys using environment variables.
 - Apply migrations before launching the application.
 - Test the deployed application thoroughly.
 - Keep dependencies documented in requirements.txt.
 - Maintain backups of production data.
-

Developer's Note

Although the Learning Log project remained a local development project, completing it provided a clear understanding of what is required before an application can be shared with others. The project emphasized that deployment is more than copying files to a server—it involves security, configuration, databases, and careful testing to ensure the application operates reliably in a production environment.

Related Topics

- **1.8** Managing Dependencies with requirements.txt
- **8.10** Models
- **8.11** Views
- **8.14** Authentication
- **9.3** Learning Log

Key Takeaways

- Deployment makes a Django application available to users.
- Production environments require additional security and configuration.
- Environment variables protect sensitive information.
- Database migrations and static files are part of the deployment process.
- Careful testing is essential before releasing an application.
- Deployment is the final step in delivering a completed Django project.

8.16 Django Markers

Purpose

Throughout this handbook, **Django Markers** are used to identify commonly encountered files, functions, decorators, commands, and template tags. Rather than repeatedly explaining these items in multiple chapters, this section serves as a quick-reference guide for the markers most frequently used during Django development.

These markers are intended to refresh your memory while reading code or troubleshooting an application.

Python Modules

Marker	Purpose
models.py	Defines the application's data models.
views.py	Contains the application logic that processes requests.
urls.py	Maps URLs to views.
forms.py	Defines forms used for user input.
admin.py	Registers models with the Django administration site.
settings.py	Stores project configuration settings.
manage.py	Command-line utility for managing the Django project.

Common Commands

Command	Purpose
python manage.py runserver	Starts the development server.
python manage.py makemigrations	Creates migration files.
python manage.py migrate	Applies database migrations.
python manage.py createsuperuser	Creates an administrative user.
python manage.py shell	Opens the Django interactive shell.
python manage.py collectstatic	Collects static files for deployment.

Frequently Used Decorators

Decorator	Purpose
@login_required	Restricts access to authenticated users.

Common Template Tags

Tag	Purpose
{% extends %}	Inherits a base template.
{% block %}	Defines replaceable content areas.
{% url %}	Generates application URLs.
{% if %}	Performs conditional rendering.
{% for %}	Loops through a collection of data.
{% csrf_token %}	Protects forms against Cross-Site Request Forgery attacks.
{{ variable }}	Displays the value of a variable.

Common Model Fields

Field	Purpose
CharField	Short text values.
TextField	Long text values.
DateTimeField	Date and time information.
BooleanField	True or False values.
ForeignKey	Creates relationships between models.

Best Practices

- Learn the purpose of each Django file.
 - Use Django's built-in features whenever possible.
 - Keep application logic organized by responsibility.
 - Refer to this section as a quick reference rather than memorizing every command.
 - Rely on Django's conventions to maintain consistency across projects.
-

Developer's Note

As the Learning Log project grew, I found myself repeatedly returning to the same files, commands, and template tags. Having a quick-reference section for these common markers made it much easier to navigate the project without searching through documentation. Over time, many of these became second nature, but this reference remains a useful reminder when starting a new Django project or returning to one after a break.

Related Topics

- **8.2** Django Starter Commands
 - **8.8** App Anatomy
 - **8.10** Models
 - **8.11** Views
 - **8.12** Templates
 - **8.13** Forms
 - **8.14** Authentication
-

Key Takeaways

- Django organizes applications into well-defined files with specific responsibilities.
- Built-in commands simplify project management and database maintenance.
- Template tags provide dynamic content within HTML pages.
- Model fields define how application data is stored.
- This section serves as a quick-reference guide for the Django features used most frequently.

9 PROJECTS

9.1 Time Logger

Purpose

The **Time Logger** project was developed to record time worked on individual tasks, calculate earnings based on hourly rates, and maintain a running history of completed work. It served as both a practical Python application and a productivity tool for managing freelance and contract work.

Unlike the tutorial projects completed during Python Crash Course, this was a self-directed project designed to solve a real-world problem.

Project Objectives

The primary objectives of the Time Logger project were to:

- Record time worked on individual tasks.
- Calculate earnings automatically using hourly rates.
- Maintain a permanent work history.
- Store data in Microsoft Excel format.
- Provide an easy-to-use interface for adding, editing, and deleting entries.
- Practice designing a complete Python application from concept to implementation.

Technologies Used

Component	Purpose
Python	Core programming language
openpyxl	Reading and writing Excel workbooks
VS Code	Development environment
Git	Version control
Excel	Time log storage and reporting

Major Features

The completed application includes:

- Task-based time tracking.
 - Automatic earnings calculations.
 - Editable work records.
 - Running totals for hours worked.
 - Running totals for earnings.
 - Excel workbook integration.
 - Summary reporting.
-

Lessons Learned

Developing the Time Logger reinforced several important software development principles.

- Build software to solve real problems.
 - Design applications one feature at a time.
 - Test new functionality before adding more complexity.
 - Separate user input, calculations, and data storage into logical components.
 - Version control makes experimentation much safer.
 - Small improvements can significantly enhance usability.
-

Developer's Note

The Time Logger was the first application I designed specifically to support my own workflow rather than to complete a programming exercise. Building an application that I actually used every day changed the way I viewed software development. Instead of simply writing code to learn Python, I was writing software to solve a practical problem. That shift in perspective made the project one of the most rewarding experiences in the handbook.

Key Takeaways

- The Time Logger was the first self-directed Python application developed for everyday use.
- The project combined file handling, calculations, and user interaction into a practical business tool.
- Excel provided a familiar and effective method for storing and reviewing work records.
- Incremental development and frequent testing simplified debugging.
- Real-world projects reinforce programming concepts far more effectively than isolated exercises.

9.2 Alien Invasion

Purpose

Alien Invasion was the first large-scale application developed during the Python learning journey. Using the Pygame library, the project combined many of the concepts introduced earlier in the handbook into a complete, interactive arcade-style game.

The project demonstrated how individual programming concepts work together to build a real software application.

Project Objectives

The primary objectives of the Alien Invasion project were to:

- Develop a complete object-oriented application.
- Organize code across multiple Python modules.
- Handle keyboard input and user interaction.
- Implement animation using a game loop.
- Detect collisions between game objects.
- Track scores, levels, and player lives.
- Practice incremental development using Git.

Technologies Used

Component	Purpose
Python	Core programming language
Pygame	Graphics, animation, and input handling
VS Code	Development environment
Git	Version control

Major Features

The completed game includes:

- Player-controlled spaceship.
 - Animated alien fleet.
 - Bullet firing and collision detection.
 - Increasing game difficulty.
 - Score and high-score tracking.
 - Multiple player lives.
 - Start and restart functionality.
-

Lessons Learned

Developing Alien Invasion reinforced several important software development principles.

- Break large projects into small, manageable tasks.
 - Assign a single responsibility to each class.
 - Test frequently after every new feature.
 - Commit working code to Git regularly.
 - Read error messages carefully before making changes.
 - Well-organized code is easier to debug and extend.
-

Developer's Note

Alien Invasion was the project where Python began to feel like real software development rather than individual programming exercises. Each completed feature built upon the previous one until a fully functional game emerged. Just as importantly, the project established development habits—incremental coding, frequent Git commits, and structured debugging—that continued throughout every project that followed.

Key Takeaways

- Alien Invasion was the first complete Python application developed in this handbook.
- Object-oriented programming simplified the organization of a multi-file project.
- Pygame provided a practical introduction to graphics, animation, and event handling.
- Incremental development reduced debugging time and improved code quality.
- The development practices learned during this project apply to far more than game programming.

9.3 Learning Log

Purpose

Learning Log was the capstone project completed during the Django portion of *Python Crash Course*. The application demonstrated how a database-driven web application is built by combining models, views, templates, forms, authentication, and URL routing into a cohesive project.

Learning Log represented the transition from writing standalone Python programs to developing full-stack web applications.

Project Objectives

The primary objectives of the Learning Log project were to:

- Build a complete Django web application.
- Design and manage a relational database.
- Create dynamic web pages using templates.
- Process user input with Django forms.
- Implement user authentication and authorization.
- Organize a project using Django's application structure.
- Practice version control throughout development.

Technologies Used

Component	Purpose
Python	Core programming language
Django	Web application framework
SQLite	Development database
HTML	Page structure
Bootstrap	User interface styling
VS Code	Development environment
Git	Version control

Major Features

The completed application includes:

- User registration and authentication.
 - Secure login and logout.
 - Topic and entry management.
 - User-specific data ownership.
 - Dynamic page generation.
 - Form validation.
 - Database integration using Django models.
-

Lessons Learned

Developing Learning Log reinforced several important software development principles.

- Organize applications using Django's project structure.
 - Separate responsibilities between models, views, templates, and forms.
 - Protect user data through authentication and authorization.
 - Build applications one feature at a time.
 - Test frequently after each enhancement.
 - Maintain version control throughout development.
-

Developer's Note

Learning Log was the project where everything came together. Concepts that initially seemed independent—URLs, views, models, templates, forms, and authentication—became parts of a single request-response workflow. Completing the application provided a practical understanding of how modern web applications are designed and reinforced the importance of clean architecture, incremental development, and disciplined debugging.

Key Takeaways

- Learning Log was the first complete Django web application developed in this handbook.
- Django's architecture separates data, logic, and presentation into well-defined components.
- Authentication enables secure, multi-user applications.
- Models, views, templates, and forms work together to process every request.
- The project provided a solid foundation for developing future Python web applications.

9.4 Future Projects

Purpose

This section identifies projects that will be completed as Python knowledge and professional experience continue to grow. Each new project provides an opportunity to reinforce existing skills while introducing new concepts, libraries, and development techniques.

The goal is to continue building practical applications that solve real-world problems rather than simply completing programming exercises.

Planned Projects

Future projects may include:

- AI-assisted development tools.
- Data analysis and reporting applications.
- Automation and scripting utilities.
- Database-driven business applications.
- REST API development and integration.
- Desktop applications.
- Cloud-based services.
- Additional Django web applications.
- Open-source contributions.

As each project is completed, it will be documented using the same format established throughout this handbook.

Development Philosophy

Future projects will continue to follow the principles used throughout this handbook:

- Solve real-world problems whenever possible.
 - Build one feature at a time.
 - Test frequently.
 - Commit working code to Git regularly.
 - Keep applications organized and maintainable.
 - Document lessons learned for future reference.
-

Developer's Note

Completing the projects in this handbook marks the beginning rather than the end of the journey. Every new project introduces unfamiliar problems, and solving those problems is how developers continue to improve. The most valuable lesson learned throughout this handbook is that confidence comes from building software, making mistakes, debugging patiently, and steadily refining both the code and the development process.

Key Takeaways

- Learning continues through practical application.
- Real-world projects strengthen programming skills.
- Every completed project becomes a reference for future work.
- Consistent development practices improve both code quality and productivity.
- This handbook will continue to evolve as new projects and technologies are explored.

10 APPENDICES

10.1 Appendix A – Python Quick Reference

10.1.1 Purpose

This appendix provides a consolidated reference to the Python syntax, commands, and language features used throughout this handbook. It is intended as a quick lookup rather than a detailed tutorial.

Includes

- Variables
 - Data Types
 - Operators
 - Conditional Statements
 - Loops
 - Functions
 - Classes
 - File Operations
 - Exceptions
 - Modules
 - Common Built-in Functions
-

10.2 Appendix B – Git Quick Reference

10.2.1 Purpose

This appendix summarizes the Git commands used during daily development.

10.2.2 Common Commands

Command	Purpose
git status	Show repository status
git add .	Stage all changes
git commit -m "message"	Commit staged changes
git log --oneline	Display commit history
git restore filename	Restore a file
git branch	List branches
git checkout branch	Switch branches
git pull	Retrieve updates
git push	Upload commits

10.3 Appendix C – Django Quick Reference

10.3.1 Purpose

A consolidated reference to the Django commands, files, and template syntax used throughout the Learning Log project.

Includes

- Starter commands
 - Common manage.py commands
 - Project structure
 - Template tags
 - Model fields
 - Request cycle
 - Authentication decorators
-

10.4 Appendix D – Resources

10.4.1 Purpose

A collection of references, documentation, and learning resources used throughout the development of this handbook.

Official Documentation

- [Python Documentation](#)
- [Django Documentation](#)
- [Pygame Documentation](#)
- [Plotly Documentation](#)
- [Matplotlib Documentation](#)
- [Git Documentation](#)

Books

- *Python Crash Course* — Eric Matthes

Development Tools

- [Visual Studio Code](#)
 - [Git](#)
 - [GitHub](#)
-

Final Thoughts

This handbook began as a collection of notes while learning Python. Through continuous development, project work, and real-world experimentation, it evolved into a comprehensive reference manual documenting the concepts, workflows, and practices that proved most valuable throughout the learning process.

Although no handbook can replace experience, experience becomes far more valuable when it is documented, organized, and available for future reference.

This manual is therefore not intended to be finished. As new projects are completed, new technologies are explored, and new lessons are learned, it will continue to grow alongside my development journey.

"The best way to learn software development is to build software."